

REAL ANALYSIS

SECTION -01

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CHAPTER 1

ELEMENTARY SET THEORY AND COUNTABILITY

1.1. Definition

1.1.1. Set

A set is a well defined collection of distinct objects. By well defined we mean there is no confusion regarding inclusion or exclusion of objects. Sets are conventionally denoted with capital letters and small letters will represent the members of the set.

Note: This definition is widely accepted but not absolute.

Following notations will be used for some of the specific sets that are commonly used.

$\mathbb{N}$ = The set of all natural numbers or all positive integers.

$\mathbb{Z}$ = The set of all integers.

$\mathbb{Q}$ = The set of all rational numbers.

$\mathbb{Q}^c$ = The set of all Irrational numbers.

Note: No set is member of itself.

- i.e. $A \notin A$ for every set A .

Russell's Paradox: There is no set of all sets.

Reason: Let if possible, X be the set of all sets

i.e. $X = \{A \mid A \text{ is set}\}$

By definition of set, X is set.

And By definition of X

$X \in X$

Which is a contradiction.

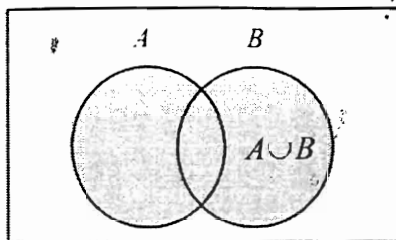
1.1.2. Operations on Sets

Let A and B be two sets. Then

1. **Union:** The union of A and B , denoted by $A \cup B$, is defined as the set $\{x \mid x \in A \text{ or } x \in B\}$.

While writing $A \cup B$, elements common to A and B should be taken only once in the union.

Also note that $A \cup A = A$ and if $A \subset B$, then $A \cup B = B$



Put Your Own Notes

Example:

(a) Take $A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

Then $A \cup B = \{1, 2, 3, 4, 5, 6\}$

(b) If $A = \mathbb{Q}$

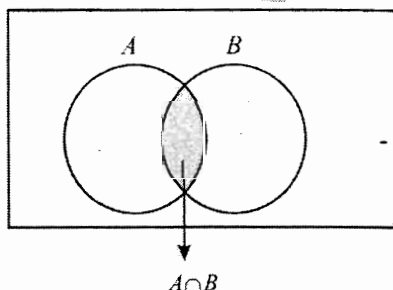
$B = \mathbb{Q}^c$

Then $A \cup B = \mathbb{R}$

2. **Intersection:** The intersection of A and B , denoted by $A \cap B$, is defined as the set $\{x | x \in A \text{ and } B \text{ both}\}$.

Thus $A \cap B$ consists of all elements common to A and B . Also note that $A \cap A = A$ and if $A \subset B$ then $A \cap B = A$.

If $A \cap B = \phi$, then the sets A and B are called **Disjoint**.



Example:

(a) Let $A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

Then $A \cap B = \{3, 4\}$

(b) If $A = \left\{ \frac{1}{n}, n \in \mathbb{N} \right\}$ and $B = \left\{ -\frac{1}{n}, n \in \mathbb{N} \right\}$

Then $A \cap B = \phi$

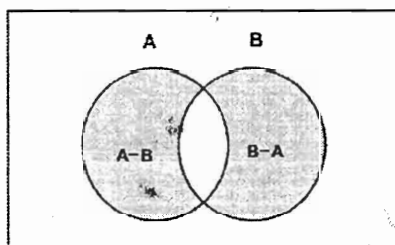
3. **Difference:** The difference of A and B , denoted by $A - B$, is defined as the set $\{x | x \in A \text{ and } x \notin B\}$.

Thus $A - B$ is the set of all elements of A not belonging to B .

Similarly, we can define $B - A$

Note:

- (a) $A - B \neq B - A$
- (b) $A - B \subset A, B - A \subset B$
- (c) $(A - B) \cap (B - A) = \phi$



Example:

- (a) If $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{4, 5, 6, 7, 8\}$

$$\text{Then } A - B = \{1, 2, 3\}$$

- (b) If $A = \mathbb{R}$

$$B = \mathbb{Q}$$

$$\text{Then } A - B = \mathbb{R} - \mathbb{Q} = \mathbb{Q}^c$$

4. **Complement:** The complement of A in the set U (universal set), denoted by A^c , is defined as the set $\{x | x \in U \text{ such that } x \notin A\}$. In other notation we can see it as $U - A$.

From this definition, it is easy to verify that $U = A \cup A^c$, $A \cap A^c = \phi$, $(A^c)^c = A$ and if $A, B \subset U$ and $A \subset B$ then $B^c \subset A^c$.

Example:

- (a) If $A = \{1, 2, 3, 4, 5\}$ and

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$\text{Then } A^c = \{6, 7, 8, 9, 10\}$$

- (b) If $A = \mathbb{Q}$

$$U = \mathbb{R}$$

$$\text{Then } A^c = \mathbb{Q}^c$$

5. **Power Set:** The power set of A , denoted by $P(A)$, is defined as a set $\{X | X \subseteq A\}$. Thus, $P(A)$ is collection of all possible subsets of A .

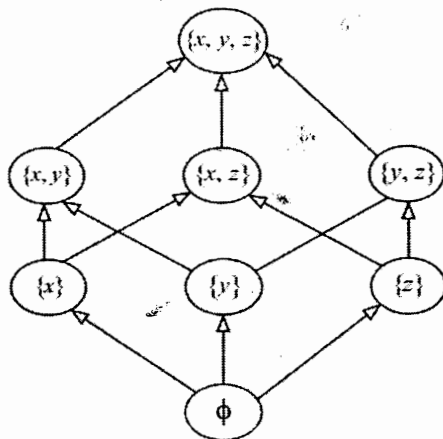
$$\text{If } |A| = n, \text{ Then } |P(A)| = 2^n$$

(where, $|A|$ denotes the number of elements in the set A . i.e, cardinality of the set A)

If $A = \{x, y, z\}$

Then

$$P(A) = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}\}$$



The cardinality of the set $\{x, y, z\}$, is three, while there are eight elements in its power set.

6. **Cartesian Product:** If A and B are nonempty sets, then the Cartesian product $A \times B$ of A and B is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. That is,

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

Thus if $A = \{1, 2, 3\}$ and $B = \{1, 5\}$, then the set $A \times B$ is the set whose elements are the ordered pairs

$$(1, 1), (1, 5), (2, 1), (2, 5), (3, 1), (3, 5)$$

We may visualize the set $A \times B$ as the set of six points in the plane with the coordinates that we have just listed.

Note: $A \times B \neq B \times A$ when $A \neq B$ and also if any of A and B is empty then $A \times B = \emptyset$.

Relation: Any subset of $A \times B$ defines a relation from A to B .

Example: Let $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ define

$$R = \{(x, y) : |x - y| \text{ is even}\}$$

$$= \{(1, 5), (2, 4), (3, 5)\}$$

Put Your Own Notes

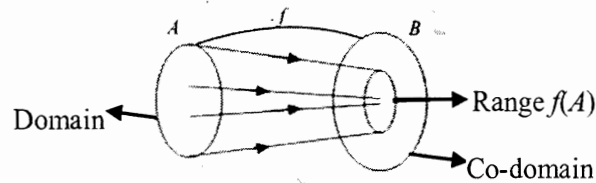
1.1.3. Functions And Their Properties

Put Your Own Notes

1. **Function on a set:** A relation f from a non-empty set A to another non empty set B is said to be a function, if each element $x \in A$ corresponds to a unique element $y \in B$ and the y which corresponds in this way to a given x is denoted by $f(x)$, and is called the value of x under f .

Note: If f is a function from a set A to set a B then we write it as $f : A \rightarrow B$.

2. **Graph:** If f is a function from A to B , then set of all ordered pairs $(x, f(x))$ is called the graph of the function f .
3. **Domain, Co-domain & Range of a Function:** For a function f from A to B , the set A is called the domain of f and the set of values of f is called the range of f and written as $f(A)$, whereas B is called the Co-domain of f . Clearly $f(A) \subseteq B$. All these sets can be seen in following figure easily.



Equality of Two Functions: Two functions f and g are said to be equal if

- (a) Domain of f = Domain of g
- (b) $\forall x, f(x) = g(x)$

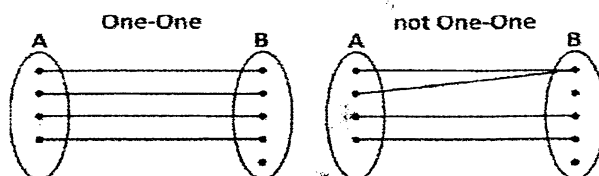
Note: The terms 'function' and 'mapping' are synonyms.

4. **Counting of Functions:** Let A and B be two finite sets having m and n elements respectively. Then each element of A can be associated to any one of n elements of set B . So, total number of functions from set A to set B is equal to ways of having m job where each job can be done in n ways. Hence total number of function from A to B is n^m or $|B|^{|A|}$

Example: Let A be the set containing 10 elements and B be the set containing 5 elements. Then total number of functions from A to B is 5^{10} .

1.1.4. Classification of Functions

- One-One (Injective) Function:** We say a function $f: A \rightarrow B$ is one-one if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \quad \forall x_1, x_2 \in A$ i.e. no two elements in the domain correspond to the same element in the range.



Algorithm to Check Injectivity of Function

Step I: Take two arbitrary elements x_1 and x_2 in the domain

Step II: Put $f(x_1) = f(x_2)$

Step III: Solve $f(x_1) = f(x_2)$. If $f(x_1) = f(x_2)$ gives $x_1 = x_2$ only, then $f: A \rightarrow B$ is a one-one function otherwise not.

Example: $f: [0,1] \rightarrow \mathbb{R}$ defined by

$f(x) = x^2$ is a one-one function.

$$\because f(x) = f(y)$$

$$\Rightarrow x^2 = y^2$$

$$\Rightarrow x = y \quad (\because x, y \geq 0) \quad \forall x, y \in [0,1]$$

Remark:

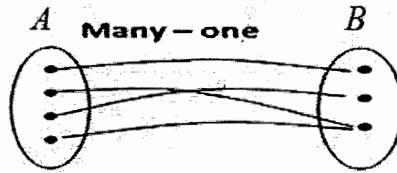
- Injectivity of a function can also be checked from its graph. If any straight line parallel to x -axis intersects the curve $y = f(x)$ at most at one point, then the function $f(x)$ is one-one or an injection otherwise it is not.
- If $f: A \rightarrow B$ then $x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$ is true for all $x_1, x_2 \in A$. However $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ for all $x_1, x_2 \in A$ is true only when f is an injective map.
- If A and B are finite sets having m and n elements respectively then

number of one-one functions from A to $B = \begin{cases} {}^n P_m, & \text{if } n \geq m \\ 0 & \text{if } n < m \end{cases}$ where

$${}^n P_m = \frac{n!}{(n-m)!}$$

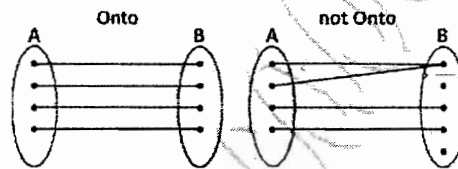
Put Your Own Notes

2. **Many-One Function:** A function $f: A \rightarrow B$ is said to be many-one function if two or more elements of set A (if exists) have the same image in B .



Example: Consider the function $f: [-1, 1] \rightarrow \mathbb{R}$, defined by $f(x) = x^2$, then f is many to one function.

3. **Onto (Surjective) Function:** A function $f: A \rightarrow B$ is said to be an onto function or a surjective function if every element of B is the f -image of some element of A i.e. if $f(A) = B$ or range of f is the co-domain of f .



Remark: It depends upon domain and co-domain respectively for a function to be one-one and onto.

Example: The function $f: [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is a one-one function where as function $f: [-1, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is not a one-one function. ($\because f(-1) = f(1) = 1$)

Also function $f: [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is not an onto function, but $f: [0, 1] \rightarrow [0, 1]$ defined by $f(x) = x^2$ is an onto function.

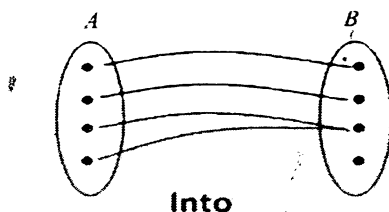
Remark: If A and B are two sets having m and n elements respectively and let $1 \leq n \leq m$, then number of onto functions from A to B is

$$\left[n^m - \sum_{r=1}^n (-1)^{r+1} n_{c_r} (n-r)^m \right] \text{ where } n_{c_r} = \frac{n!}{r!(n-r)!}.$$

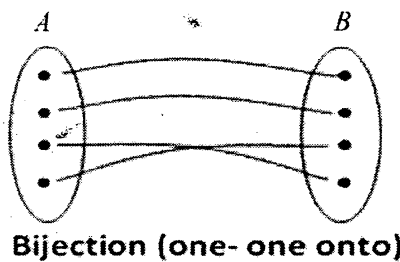
Example: If $|A| = 4, |B| = 3$ then there are 36 onto maps from A to B .

4. **Into Function:** A function of $f: A \rightarrow B$ is an into function if there exists an element in B having no pre-image in A . In other words, $f: A \rightarrow B$ is an into function if $f(A) \subset B$.

Put Your Own Notes



5. **One-One Onto (Bijective) Function:** The map $f : A \rightarrow B$ is bijective if it is one-one as well as onto.



Example: The map $f : \mathbb{N} \rightarrow \mathbb{Z}$ defined as $f(n) = (-1)^n \left\lfloor \frac{n}{2} \right\rfloor$

(Where $\left\lfloor \frac{n}{2} \right\rfloor$ denotes the greatest integer function) is bijective function

Cantor's Theorem: For any set A , there is no map from A onto the set $P(A)$, the power set of A .

Reason: Let $\phi : A \rightarrow P(A)$ is onto.

Since for every $x \in A$, $\phi(x)$ is a subset of A . Then either $x \in \phi(x)$ or it does not. Now consider $X = \{x \in A : x \notin \phi(x)\}$

Since X is a subset of A and ϕ is onto, then $X = \phi(t)$ for some $t \in A$, so either $t \in X$ or $t \notin X$. If $t \in X$ and since $X = \phi(t)$. We must have $t \in \phi(t)$. Which is contrary to the definition of X .

Similarly, If $t \notin X$ then $t \notin \phi(t)$ so that $t \in X$ which is also a contradiction.

So ϕ cannot be onto.

Remark: There is always exist a one-one map from A to $P(A)$.

1.2. Countable and Uncountable Sets

Put Your Own Notes

1. **Similar/Equivalent/Equipotential Sets:** Two set A and B are called Similar if and only if there exists an bijective function between them and then we write $A \sim B$
2. **Finite Set:** Consider $S^{(n)} = \{1, 2, 3, 4, \dots, n\}$. Then a non-empty set A is said to be finite iff there exist $n \in \mathbb{N}$ and a bijective function f from $S^{(n)}$ to A . (n gives the number of elements in the set A)
 If no such n exist, then set A is said to be infinite.

Example: Set of all vowels $A = \{a, e, i, o, u\}$ is a finite set, since there exist $f: A \rightarrow S^{(5)}$ which is a bijection.

Note:

- (a) Two finite sets are similar if their cardinality are equal.
 - (b) Every set A is equivalent to itself as there exists an identity map on A .
 - (c) If $A \sim B$ and $B \sim C$ then $A \sim C$.
 - (d) The main difference between a finite and an infinite set is that every infinite set is equivalent to at least one of its proper subset whereas a finite set can never be equivalent to any of its proper subset.
 #It is so important that some of the authors use this note as the definition of finite and infinite sets.
 - (e) Subset of a finite set is a finite set.
 - (f) Empty set is considered as a finite set.
3. **Countably Infinite Set:** A set S is said to be countably infinite if it is similar to set of natural numbers i.e. if $S \sim \mathbb{N}$.

Example: $\mathbb{Z}$ is a countably infinite set.

Reason: Define $\phi: \mathbb{N} \rightarrow \mathbb{Z}$ as

$$\phi(n) = (-1)^n \left\lfloor \frac{n}{2} \right\rfloor$$

Clearly ϕ is bijective function from $\mathbb{N}$ to $\mathbb{Z}$

$\therefore \mathbb{Z}$ is a countably infinite set.

In Abstract: We say a set can be put under counting system or process if a rule can be imposed over that set by which successor is defined.

4. **Countable Set:** A set S is called countable if it is either finite or countably infinite. In case of countably infinite there is function f which establishes a bijection between $\mathbb{N}$ and the elements of S . Therefore the set S can be displayed as $S = \{f(1), f(2), \dots, f(n), \dots\}$. If we write $\{a_1, a_2, \dots, a_n, \dots\}$ in place of $\{f(1), f(2), \dots, f(n), \dots\}$ then $S = \{a_1, a_2, \dots, a_n, \dots\}$
5. **Uncountable Set:** A set which is not a countable set, called uncountable set.

Note: If every map from A to B fails to be onto. Then we say B is larger than A in the sense of cardinality. i.e. the co-domain has more potential than the domain.

Or

If the domain has less potential then every map from A to B fails to be onto.

6. Let A and B are finite set and $|A| < |B|$. Then A is similar to a subset of B and B is not similar to any of the subset of A .

Hence $|A| = |B|$ iff $A \sim X \subseteq B$ and $B \sim Y \subseteq A$

Note: If we define cardinality of infinite sets no matter how it is defined but these cardinality will follow the above rules.

Cardinality of infinite set: For infinite sets we use symbols to denote their cardinality

$$|\mathbb{N}| = \aleph_0 \text{ (Aleph naught)}$$

$$|\mathbb{R}| = c \text{ (Continuum)}$$

These symbols which are used to denote cardinality of infinite sets are known as transfinite numbers.

7. **Arithmetic of Cardinal Numbers:** $\aleph_0$ denotes the cardinality of $\mathbb{N}$ and c denotes the cardinality of $\mathbb{R}$.

$$(a) \underbrace{\aleph_0 + \aleph_0 + \dots + \aleph_0}_{n \text{ times}} = \aleph_0$$

$$(b) n + \aleph_0 = \aleph_0, n \in \mathbb{N}$$

$$(c) \aleph_0 + c = c$$

$$(d) \underbrace{c + c + \dots + c}_{n \text{ times}} = c$$

$$(e) n \cdot \aleph_0 = \aleph_0, n \in \mathbb{N}$$

$$(f) c \cdot c \cdot c \cdot \dots \cdot c = c$$

$$(g) \aleph_0 \cdot c = c$$

$$(h) 2^{\aleph_0} = c > \aleph_0$$

$$(i) \aleph_0^{\aleph_0} = 2^{\aleph_0} = c$$

$$(j) \aleph_0^c = 2^c$$

$$(k) \aleph_0 < c$$

$$(l) n < \aleph_0$$

For any transfinite number α, β, γ

$$(m) \alpha^\beta \cdot \alpha^\gamma = \alpha^{\beta+\gamma}$$

Put Your Own Notes

- (n) $(\alpha^\beta)^\gamma = (\alpha)^{\beta\gamma}$
- (o) $(\alpha\beta)^\gamma = \alpha^\gamma \cdot \beta^\gamma$
- (p) $\alpha^\gamma \leq \beta^\gamma$ if $\alpha \leq \beta$
- (q) $\gamma^\alpha \leq \gamma^\beta$ if $\alpha \leq \beta$
8. There is no transfinite number between $\aleph_0$ and c .
9. If A and B are two sets with cardinality α and β respectively and $F = \{f \mid f: A \rightarrow B\}$ then $\text{card } |F| = \beta^\alpha$
10. $F_1 = \{f \mid f: \mathbb{N} \rightarrow \mathbb{N}\}$, then $\text{card } |F_1| = \aleph_0^{\aleph_0} = 2^{\aleph_0} = c \Rightarrow F_1$ is uncountable.
11. $F_2 = \{f \mid f: [0,1] \rightarrow [0,1]\}$ Then $\text{card } |F_2| = c^c = 2^c \Rightarrow F_2$ is uncountable
12. The set of all equivalence classes of an equivalence relation on a set is called Quotient set.
- (a) Quotient set of countable set is always countable.
- (b) Quotient set of uncountable set may be countable or uncountable.

1.3. Properties of Sets

- Two finite set are similar iff they have same number of elements.
- Every countable set is equivalent to a subset of natural numbers.

Reason: Let A be any countable set. Now if A is finite then it is similar to $S^{(n)}$ for some n .

Else if A is countably infinite then A can be written as

$$A = \{a_1, a_2, a_3, \dots\}$$

Now define $\phi: A \rightarrow \mathbb{N}$ as $\phi(a_k) = k \quad \forall k \in \mathbb{N}$

This ϕ is a bijective map and thus A is equivalent to $\mathbb{N}$.

- Every set equivalent to finite/infinite set is finite/infinite set.
Reader should prove it themselves.
- If A is similar to a subset of B and B is similar to subset of A , then $A \sim B$

Reason: As A is similar to a subset of B

$\Rightarrow A$ has less potential than B ... (1)

Also B is similar to a subset of A

$\Rightarrow B$ has less potential than A ... (2)

By (1) and (2)

$$A \sim B$$

5. A set is infinite iff it contains a countably infinite subset.

Reason: Let X be any infinite set.

Take $x \in X$ and name it as a_1 and define $A_1 = \{a_1\}$

Again take another $y \in X - A_1$ and name it as a_2

Define $A_2 = \{a_2\} \cup A_1$

$\vdots$

Proceeding in same way, we have

$$A_n = A_{n-1} \cup \{a_n\} \quad \forall n \in \mathbb{N}$$

$$\text{Let } A = \bigcup_{n \in \mathbb{N}} A_n = A_1 \cup A_2 \cup \dots = \{a_1, a_2, \dots, a_n, a_{n+1}, \dots\}$$

Clearly $A \subset X$

This shows that every infinite set has countable infinite subset

Converse is obvious.

Hence a set X is infinite iff it contains a countably infinite subset.

6. Every infinite set is equivalent to at least one of its proper subset.

Reason: As done in above property construct $A = \bigcup_{n \in \mathbb{N}} A_n$

Now take $B = A - \{a_1\}$ and define $Y = (X - A) \cup B$

Clearly $a_1 \notin Y \subset X$ thus Y is proper subset of X

Define $g : X \rightarrow Y$ such that

$$g(x) = \begin{cases} a_{i+1} & x = a_i \\ x & \text{else} \end{cases}$$

Clearly g is one-one and onto.

This shows that if we remove one element from an infinite set. Then its cardinality does not change.

By the same procedure if we remove finite number of elements from an infinite set then its cardinality remains same.

7. A finite set is not equivalent to any of its proper subset.

Reason: Let A is any finite set and $B \subset A$ then clearly B has less number of elements than A and two finite set are equivalent iff they have same number of elements so A and B can-not be equivalent. Thus no finite set can be equivalent to its proper subset.

8. Countable union of countable set is countable.

9. If B is an uncountable set and A is a countable set then $(B - A)$ is an uncountable set.

Reason: Suppose that $(B - A)$ is countable set and also $B = (B - A) \cup A$ (being a union of two countable set) is countable thus B is a countable set. But this is not possible as B is an uncountable set. Thus $B - A$ is uncountable set.

10. Family of all finite subsets of countable set is countable.

Reason: Let A be any countable set.

If A is a finite set then nothing to prove. Now, Suppose A is countably infinite set.

Then $A \sim \mathbb{N}$. If we show that collection of all finite subsets of $\mathbb{N}$ is countable then we are done.

Now define $A_1 =$ union of all subsets of $\mathbb{N}$ having single element i.e.

$$A_1 = \{1\} \cup \{2\} \cup \{3\} \cup \{4\} \cup \dots$$

Thus, A_1 is countable.

Now, Let $A_2 =$ Union of all subset of $\mathbb{N}$ having 2 elements. Then A_2 is countable.

In this manner A_n is countable $\forall n \in \mathbb{N}$

Thus if $\mathcal{F}$ denotes the collection of all finite subset of $\mathbb{N}$ then $\mathcal{F}$

$$= \bigcup_{n=1}^{\infty} A_n \text{ which is countable (being countable union of countable sets).}$$

11. The family of all infinite subsets of an infinite set X is uncountable.

Reason: Let $\mathcal{B}$ denote the family of all subsets of the set X and $\mathcal{A}$ denote the family of all finite subsets of X . Then family of all infinite subset of the set X is equal to $\mathcal{B} - \mathcal{A}$. Which is uncountable.

(By Property 9)

Remark:

(a) The family of all subsets of countably infinite set is uncountable.

(b) $P(\mathbb{N}), P(\mathbb{Q}), P(\mathbb{Z})$ all are uncountable set.

12. Finite Cartesian product of countable set is countable.

Application: $\mathbb{N} \times \mathbb{N} \times \dots \times \mathbb{N}$ is countable set.

13. The set of all positive rational numbers is countable.

Reason: $\mathbb{Q}^+ = \left\{ \frac{p}{q}; p, q \text{ are co-prime positive integers} \right\}$

Let $A = \{(p, q); p, q \text{ are co-prime positive integers}\}$

Clearly the elements of $\mathbb{Q}^+$ and A are in one-one correspondence and therefore $\mathbb{Q}^+$ is countable iff A is countable. Since $A \subset \mathbb{N} \times \mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$ is countable, therefore, A is countable. (being subset of countable set)

Hence $\mathbb{Q}^-$ is also countable

Note:

(a) $\mathbb{Q} = \mathbb{Q}^+ \cup \mathbb{Q}^- \cup \{0\}$ is countable

(b) By view of 12 & 13, we can say $\mathbb{Z} \times \mathbb{Z} \times \dots \times \mathbb{Z}$, $\mathbb{Q} \times \mathbb{Q} \times \dots \times \mathbb{Q}$ all are countable sets.

14. Infinite Cartesian product of countable sets $A_1, A_2, A_3, \dots, A_n, \dots$ (where $|A_i| \geq 2$, for infinitely many values of i) is uncountable.

15. Set of all polynomial with rational coefficients is a countable set.

Reason: Let $P_m = \{a_0 + a_1x + \dots + a_mx^m : a_i \in \mathbb{Q}\}$

Then P_m is countable $\forall m \in \mathbb{N} \cup \{0\}$ as there is a bijection $\phi: P_m \rightarrow \mathbb{Q} \times \mathbb{Q} \times \dots \times \mathbb{Q}$ defined as

$$\phi(a_0 + a_1x + \dots + a_mx^m) = (a_0, a_1, \dots, a_m)$$

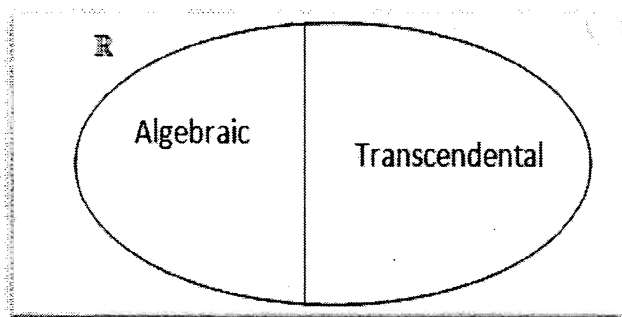
Now if P denote the class of all polynomials over $\mathbb{Q}$ then clearly

$$P = \bigcup_{n=1}^{\infty} P_n \text{ which (being countable union of countable set) is countable.}$$

16. Set of all circles whose Centre and radius are rational number is countable.

Reason: Let $X = \{c \mid c \equiv (x-a)^2 + (y-b)^2 = r^2 \text{ where } a, b, r \in \mathbb{Q}\}$

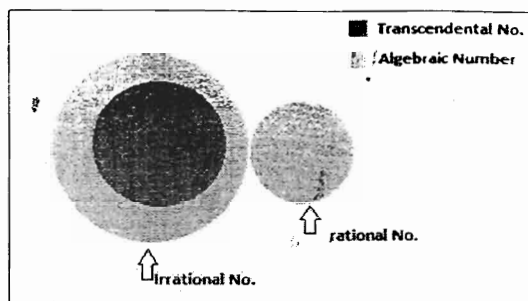
Define $\phi: X \rightarrow \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}$ as $\phi(c) = (a, b, r)$. This ϕ is a bijective map and thus X a countable set.



Algebraic number: A real number is called an algebraic number, if it is a root of a polynomial $P(x) = a_0 + a_1x + \dots + a_nx^n = 0$ ($a_n \neq 0$) with rational coefficient.

Transcendental number: A real number which is not Algebraic is called a transcendental number.

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17. Set of all algebraic numbers and set of all transcendental numbers forms a partitions of the set $\mathbb{R}$.
18. Every rational number is algebraic.
19. Every transcendental number is irrational.
20. Irrational number may be algebraic or transcendental.

Example:

$\sqrt{2}$ is an algebraic number and π is a transcendental number.

21. Set of all algebraic number is countable.

Reason: For a fixed $m \in \mathbb{N} \cup \{0\}$ define

$P_m = \{a_0 + a_1x + a_2x^2 + \dots + a_mx^m \mid a_i \in \mathbb{Q}\}$. This P_m is countable.

Now take $P = \bigcup_{m \in \mathbb{N}} P_m$ and being countable union of countable sets it is also countable. So, P is countable.

Now if α is any algebraic number then there is a polynomial in P such that whose root is α .

Thus the set of all algebraic numbers is contained in the set of zeros of polynomials in P . Now all we need to show is that the set of zeros of these polynomials is countable.

Since every polynomial can have at most k real roots where k is degree of that polynomial. Now we denote the set A_m as the set of zeros of polynomials in P_m . Since P_m has countable number of polynomial and further all of those polynomial have finite number of zeros so A_m will have countable number of elements.

Now define $A = \bigcup_{m \in \mathbb{N}} A_m$, the set of zeros of polynomials in P , being countable union of countable set is countable and set of algebraic numbers being subset of this set.

$\therefore$ Set of algebraic numbers is countable.

22. Set of all transcendental numbers is uncountable.

Reason: Since $\mathbb{R} = \mathcal{A} \cup \mathcal{T}$ where $\mathcal{A}$ is set of all algebraic numbers and $\mathcal{T}$ is set of all transcendental numbers. Now since $\mathcal{A}$ is countable and $\mathbb{R}$ is uncountable

$\therefore \mathcal{T} = \mathbb{R} - \mathcal{A}$ Is uncountable

(By using the result 9)

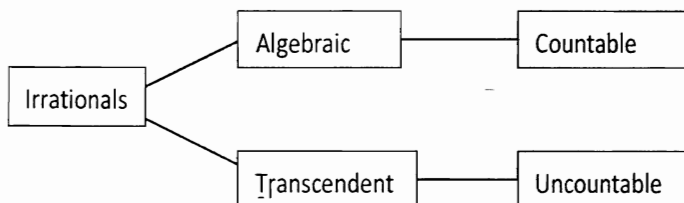
23. Arbitrary collection of disjoint open intervals is countable.

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Reason: consider $\mathcal{I}$ to be the collection of disjoint open intervals. Now if I is any interval then we can find a rational number which is not in any other interval we denote that by a_1 if we take any other interval there we can find any other rational number a_2 in this way we can find a sequence of rational numbers $a_1, a_2, a_3, \dots, a_n, \dots$ where each a_i is from different interval and thus we can generate a set of rational numbers having one-one correspondence with the set $\mathcal{I}$ and since that set of rational number is countable so $\mathcal{I}$ is also countable.

24. There does not exist any onto function from a countable set to an uncountable set.
25. There does not exist any one-one function from an uncountable set to a countable set.
26. **Some special Sets**

Sets	Countable / Uncountable
Natural Numbers ($\mathbb{N}$)	Countable
Integers ($\mathbb{Z}$)	Countable
Rational Numbers ($\mathbb{Q}$)	Countable
Non Empty , non-singleton Intervals (Open, closed, semi-open, semi-closed)	Uncountable
Irrationals ($\mathbb{Q}^c$)	Uncountable
Real numbers ($\mathbb{R}$)	Uncountable
Algebraic numbers ($\mathcal{A}$)	Countable
Transcendental Numbers ($\mathcal{T}$)	Uncountable



27. There exist intervals which are countable e.g. $[a, a] \quad \forall a \in \mathbb{R}$
28. Every countably infinite set has cardinality $\aleph_0$ (**Aleph Naught**)
29. ' c ' is the smallest cardinality of any uncountable set.
30. The unit interval $[0, 1]$ is uncountable set

Reason: Let us assume that $[0,1]$ is countable.

$\Rightarrow$ either $[0,1]$ is finite or countably infinite.

Since every non empty, non singleton interval is an infinite set, $[0,1]$ is Countably infinite.

$\Rightarrow$ There is an enumeration $x_1, x_2, x_3, \dots$ of real numbers in $[0,1]$.

Expanding each x_i in the form of an infinite decimal, we have

$$\begin{aligned} x_1 &= 0.a_{11}a_{12}a_{13}a_{14}\dots a_{1n}\dots \\ x_2 &= 0.a_{21}a_{22}a_{23}a_{24}\dots a_{2n}\dots \\ &\dots\dots\dots \\ x_n &= 0.a_{n1}a_{n2}a_{n3}a_{n4}\dots a_{nn}\dots \\ &\dots\dots\dots \end{aligned}$$

where each $a_{ij} \in \{0,1,2,3,4,5,6,7,8,9\}$.

Consider the number b with decimal representation $b = 0.b_1b_2b_3\dots b_n\dots$

where b_1 is any integer from 0 to 9 such that $b_1 \neq a_{11}$

b_2 is any integer from 0 to 9 such that $b_2 \neq a_{22}$

b_n is any integer from 0 to 9 such that $b_n \neq a_{nn}$ and so on.

Clearly, $b \in [0,1]$ and $b \neq x_n \forall n$ since the decimal representation of b is different from the decimal representation of x_n as $b_n \neq a_{nn}$. Thus b escapes enumeration and we arrive at a contradiction.

Hence $[0,1]$ is not countable.

31. $(0,1)$ is uncountable set. (By property 6)
32. Any open interval (a,b) is uncountable where $a \neq b$
33. Any open interval (a,b) is equivalent to any other open interval (c,d) where $a \neq b$ & $c \neq d$

Reason: Let $x \in (a,b)$. Consider a function $f : (a,b) \rightarrow (c,d)$ given by

$$f(x) = c + \frac{d-c}{b-a}(x-a).$$

It is easy to verify that f is one-one and onto. Hence the result follows.

34. Any closed interval $[a,b]$ is equivalent to any other closed interval $[c,d]$ where $a \neq b$ & $c \neq d$
35. Any two non-trivial intervals are equivalent (non-empty, non-singleton intervals are non-trivial)

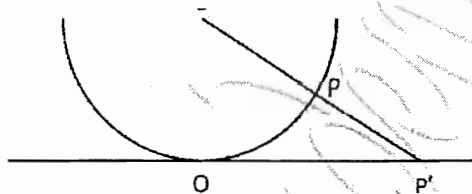
36. The set $\mathbb{R}$ of all real numbers is uncountable.

Reason: To prove this, it is sufficient to show that $(0,1) \sim \mathbb{R}$. Consider the function $f : (0,1) \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} \frac{2x-1}{x} & \text{if } 0 < x < \frac{1}{2} \\ \frac{2x-1}{1-x} & \text{if } \frac{1}{2} \leq x < 1 \end{cases}$$

Clearly, f define a one-to-one correspondence between $(0,1)$ and $\mathbb{R}$. This proves that $(0,1) \sim \mathbb{R}$.

Geometrical Reason: Let (a,b) be an open interval. We bend (a,b) into a semi-circle and rest this semi-circle tangentially on the real line $\mathbb{R}$ as shown in figure.



Let P be a point of (a,b) . Then, by joining P with the centre of the semi-circle and producing it to meet the real line at P' , we can locate a point on the real line corresponding to each point of (a,b) ; and conversely. This establishes a one-to-one correspondence between the sets (a,b) and $\mathbb{R}$. Hence $\mathbb{R}$ is uncountable.

37. Any uncountable subset of $\mathbb{R}$ is similar to $\mathbb{R}$.

38. $\mathbb{R} \sim (a, b)$ Or $[a, b]$ or $(a, b]$ or $[a, b)$ Also $\mathbb{R} \sim \mathbb{R}^n$ Implies $\mathbb{R} \sim \mathbb{C} \sim \mathbb{R}^2 \Rightarrow$ Set of all points of the plane is uncountable where $a \neq b$. Also $(0, 1) \times (0, 1) \sim (0, 1)$.

39. Let A be any set with $\text{card} A = n$ then $|P(A)| = 2^n \Rightarrow |P(\mathbb{N})| = 2^{\aleph_0} = c$, $|P(\mathbb{R})| = 2^c$.

40. Set of all functions having domain A (countably infinite) to the co-domain B (containing at least two points) is uncountable set.

1.4. Cantor Set

For describing the Cantor set, we shall give the following preliminaries on the ternary expansions of the real numbers in $[0,1]$ which is needed for the arithmetical characterization of the Cantor set.

In the ternary expansion of a real number x , we use the digits 0, 1 and 2. Thus in the ternary scale, $x = 0.a_1a_2a_3\dots$ means that $x = \frac{a_1}{3} + \frac{a_2}{3^2} + \frac{a_3}{3^3} + \dots$ where $a_1, a_2, a_3, \dots$, will take any one of the values 0, 1 and 2. Except for numbers like $\frac{1}{3}, \frac{2}{3}$, etc. The ternary expansion of every real number is unique.

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In the case of $\frac{1}{3}$, we have the following two expansions:

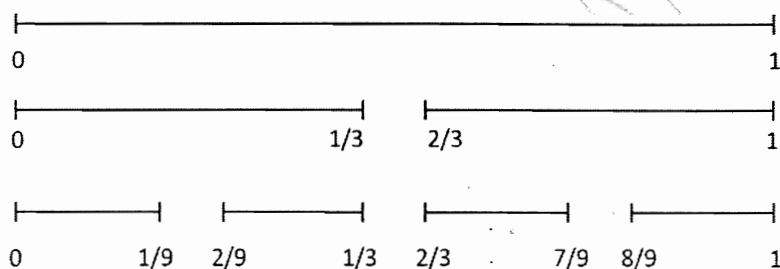
$$\frac{1}{3} = 0.1000 \dots, \quad \frac{1}{3} = 0.0222 \dots$$

In the case of $\frac{2}{3}$, we have the following two expansions

$$\frac{2}{3} = 0.1222 \dots \text{ and } \frac{2}{3} = 0.2000 \dots$$

The numbers $\frac{1}{2}$ and $\frac{1}{4}$ have the following expansion,

$$\frac{1}{2} = 0.1111 \dots \text{ and } \frac{1}{4} = 0.020202 \dots$$



Now consider the closed interval $[0, 1]$. From $[0, 1]$, let us remove $\left(\frac{1}{3}, \frac{2}{3}\right)$ which the open middle third part is of $[0, 1]$. Then we obtain the intervals $\left[0, \frac{1}{3}\right]$ and $\left[\frac{2}{3}, 1\right]$. Now we shall remove the open middle thirds of each one

of these intervals. That is we take away $\left(\frac{1}{9}, \frac{2}{9}\right)$ in $\left[0, \frac{1}{3}\right]$ and $\left(\frac{7}{9}, \frac{8}{9}\right)$ in $\left[\frac{2}{3}, 1\right]$ and get these intervals. $\left[0, \frac{1}{9}\right]$, $\left[\frac{2}{9}, \frac{1}{3}\right]$, $\left[\frac{2}{3}, \frac{7}{9}\right]$, and $\left[\frac{8}{9}, 1\right]$.

At each one of the succeeding stage, we shall remove the open middle third of each open of the remaining close intervals. If this process is carried for a countable number of times, removing the middle thirds of the left out intervals at each stage, the set of points left out in $[0, 1]$ is called the Cantor ternary set or Cantor middle third.

First note that any number in $\left(\frac{1}{3}, \frac{2}{3}\right)$ removed in the first stage has the expansion of the form $0.1a_2a_3a_4\dots$. For all numbers removed in the second state in $\left[\frac{1}{9}, \frac{2}{9}\right]$ and $\left[\frac{7}{9}, \frac{8}{9}\right]$, 1 will occur in the second place in the ternary scale on the assumption $\frac{2}{3}$ can be represented as $0.11111\dots$. In a similar manner, all the numbers removed in the n th stage will have 1 in the n th position in the ternary scale. All the left out numbers will not have one in their expansions. Hence the Cantor set K is the set of all numbers x in $[0, 1]$ which have a ternary expansion without digit 1. Where we assume that if the

end points like $\frac{1}{3}$ and $\frac{2}{3}$ have two expansions, we shall take that expansion without one's.

Note: From the construction, it may appear that only the end points $\frac{1}{3}, \frac{2}{3}, \frac{1}{9}, \frac{2}{9}, \frac{7}{9}, \frac{8}{9}, \dots$ are left alone in the Cantor set. But this is not so. For example $0.020202\dots$ in the scale of 3 is $\frac{1}{4}$ and it is in the Cantor set, since 1 does not appear in the expansion.

Further it is interesting to note the sum of the lengths of the intervals removed in $[0, 1]$ is one as seen below.

$$\frac{1}{3} + \left(\frac{1}{9} + \frac{1}{9}\right) + \left(\frac{1}{27} + \frac{1}{27} + \frac{1}{27} + \frac{1}{27}\right) + \dots = \frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \dots = \sum_{n=0}^{\infty} \frac{1}{3} \cdot \left(\frac{2}{3}\right)^n$$

The sum of this geometric series is $\frac{1}{3} \cdot \frac{1}{1 - \frac{2}{3}} = 1$

Results on Cantor's Set:

1. $A = \left\{ x \mid x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}, a_n = 0 \text{ or } 2 \right\}$ is another expression of Cantor set.
2. Cantor set contains no interior point.
3. Cantor set doesn't contain any open interval.
4. It is closed and bounded.
5. Each point of cantor set A is limit point of A .
6. Cantor set is perfect.

Note: For definition of interior point, limit point, perfect set, closed and bounded set See Chapter 2.

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CHAPTER 2

POINT SET TOPOLOGY

2.1. Definitions and Properties

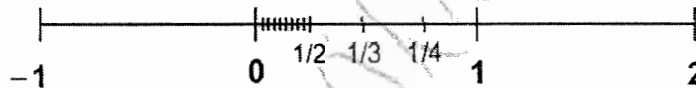
I Archimedean Property Of Real Numbers

1. If x, y are two positive real numbers then exist a positive integer n such that $ny > x$.
2. If x is any positive real number then there exist a positive integer n such that $n > x$.
3. For any real number x , there exist an integer n such that $\frac{1}{n} < x$

II Important Definitions

1. **Upper bound:** Let S be any subset of $\mathbb{R}$ then a real number M is called an **upper bound** of S if every element of S is less than or equal to M i.e. $x \leq M \quad \forall x \in S$.

Example: $S = \left\{ \frac{1}{n}; n \in \mathbb{N} \right\}$

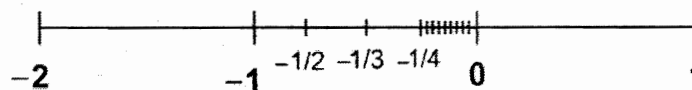


Then $x \leq 1 \quad \forall x \in S$

$\therefore$ upper bound of S is 1

2. **Lower bound:** Let S be any subset of $\mathbb{R}$ then a real number m is called **lower bound** of S if every element of S is greater than or equal to m i.e. $m \leq x \quad \forall x \in S$.

Example: $S = \left\{ -\frac{1}{n}; n \in \mathbb{N} \right\}$

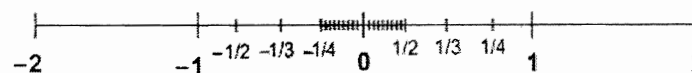


Then $-1 \leq x \quad \forall x \in S$

$\therefore$ lower bound of S is -1

3. **Bounded Set:** A set $S \subseteq \mathbb{R}$ is called bounded if $\exists$ two real numbers m and M such that $m \leq x \leq M \quad \forall x \in S$

Example: $S = \left\{ \frac{1}{n}; n \in \mathbb{N} \right\} \cup \left\{ -\frac{1}{n}; n \in \mathbb{N} \right\}$



Then $-1 \leq x \leq 1 \quad \forall x \in S$

$\therefore S$ is bounded.

4. **Greatest Element:** If an upper bound of a set belongs to the set then it is called greatest element of the set.

Examples:

(a) $S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$, 1 is the Greatest element of S .

(b) $S = (0, 2]$, 2 is the Greatest element of S .

5. **Smallest Element:** If lower bound of a set belongs to the set then it is called smallest element of the set.

Examples:

(a) $S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} \cup \{0\}$, 0 is the least element of S .

(b) $S = [2, 3]$, 2 is the least element of S .

6. **Supremum of a set:** Let $S \subseteq \mathbb{R}$, then smallest of all the upper bounds of a set S is called the **Supremum** of the set S or the **least upper bound (l.u.b.)** of the set S .

Examples:

(a) $S = \left\{ 1, 1 + \frac{1}{2}, 1 + \frac{1}{2} + \frac{1}{2^2}, 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \right\}$

$\sup S = 2$ & $2 \notin S$

(b) $S = \left\{ \frac{4n+3}{n} : n \in \mathbb{N} \right\}$

$\sup S = 7$ & $7 \in S$.

7. **Infimum of the set:** Let $S \subseteq \mathbb{R}$, Then greatest of all the lower bounds of a set S is called the **Infimum** of the set S or the **greatest lower bound (g.l.b.)** of the set S .

Examples:

(a) $S = \left\{ \frac{4n+2}{n} : n \in \mathbb{N} \right\}$

$\inf S = 4 \notin S$

(b) $S = \left\{ 1, 1 + \frac{1}{2}, 1 + \frac{1}{2} + \frac{1}{2^2}, \dots, 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \dots \right\}$

$\inf S = 1 \in S$

8. **Interval:** Let S be a subset of $\mathbb{R}$ then we say S is an interval if whenever $a, b \in S$ such that $a \neq b$. Then $\forall a < x < b \Rightarrow x \in S$.

Note:

- (a) S is not an interval if there exist $a, b \in S, a \neq b$ and there exist t such that $a < t < b$ but $t \notin S$.

- (b) Empty Set and singleton Sets are intervals.

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- (c) If S is an interval and $\sup S = \beta$, $\inf S = \alpha$ then S is denoted by $(\alpha, \beta), [\alpha, \beta], [\alpha, \beta), (\alpha, \beta]$ (depending whether they are member of S or not.)
- (d) Length of interval = $\sup S - \inf S$.
- (e) An interval S is called an open interval if $\sup S$ and $\inf S$ both does not belong to S .
- (f) An interval S is called closed interval if $\sup S$ and $\inf S$ both belong to S .
9. **Order Completeness Property of $\mathbb{R}$ (O.C.P.):** Every non empty bounded Subset of $\mathbb{R}$ has Supremum and Infimum in $\mathbb{R}$.
10. **Neighbourhood of a Point:** Let $S \subseteq \mathbb{R}, a \in \mathbb{R}$, we say S is a Neighbourhood of a , if there exist an open interval I such that $a \in I$ and $I \subseteq S$.

Examples:

- (a) $\mathbb{R}$ is a neighbourhood of every real number.
- (b) $\mathbb{N}$ is not a neighbourhood of any real number.
11. **Adherent point:** Let $a \in \mathbb{R}$ and $S \subseteq \mathbb{R}$, then a is called Adherent point of S , If for every $\delta > 0, (a - \delta, a + \delta) \cap S \neq \emptyset$.

Examples:

- (a) $S = \mathbb{N}$, Then Set of all adherent points of S is S itself.
- (b) $S = (0, 1]$, Then Set of all adherent points of S is $[0, 1]$.
12. **Isolated Point:** Let $a \in S$ and $S \subseteq \mathbb{R}$, then a is called Isolated point of S , if there exist $\delta > 0$, such that $(a - \delta, a + \delta) \cap S = \{a\}$

Examples:

- (a) $S = \mathbb{N}$, then every $a \in S$ is an Isolated point of S .
- (b) $S = \mathbb{R}$, then no point of $\mathbb{R}$ is Isolated point of S .
13. **Limit Point:** Let $a \in \mathbb{R}$ and $S \subseteq \mathbb{R}$, then a is called Limit point of S , if for every $\delta > 0, (a - \delta, a + \delta) \cap S - \{a\} \neq \emptyset$

Examples:

- (a) $S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$ then 0 is only limit point of S .
- (b) $S = [0, 1)$ then set of all limit points of S is $[0, 1]$
14. **Interior Point:** Let $S \subseteq \mathbb{R}$ and $a \in \mathbb{R}$ we say a is an interior point of S if there exist $\delta > 0$ such that $(a - \delta, a + \delta) \subset S$

Set of all interior points of S is denoted by S°

Examples:

- (a) $S = \mathbb{N}$, then no real number is an Interior point of S .
- (b) $S = \mathbb{R}$ or $S = (0, 1)$ then every member of S is an interior point of S .

15. **Exterior Point:** Let $a \in \mathbb{R}$ and $S \subseteq \mathbb{R}$, we say a is an exterior point of S if it is an interior point of S^c in $\mathbb{R}$ (where S^c means complement of S in $\mathbb{R}$). Set of all exterior point of S is denoted by $\text{Ext } S$.

Examples:

- (a) $S = \mathbb{Q}$ then no real number which is Exterior point of $\mathbb{Q}$
(b) $S = [0, 1]$ then set of all exterior Point of S is $(-\infty, 0) \cup (1, \infty)$

16. **Condensation Point:** Let $a \in \mathbb{R}$ and $S \subseteq \mathbb{R}$, then a is called Condensation point of S if for every $\delta > 0$, $(a - \delta, a + \delta)$ must contain uncountable members of S . Hence if S is not countable, then there exists a point in S such that x is condensation point of S .

Example:

- (a) $S = \mathbb{Q}^c$ then every Real number is a Condensation Point of S
(b) $S = \mathbb{Q}$ then no Real number is a Condensation Point of S .

17. **Frontier Point:** $a \in \mathbb{R}$ is said to be a frontier Point of $S \subseteq \mathbb{R}$ if it is neither an interior Point of S nor an exterior Point of S . Set of all frontier point of S is denoted by $\text{Fr } S$.

Example:

- (a) $S = \mathbb{R}$

$$S^o = \mathbb{R} \Rightarrow (S^o)^c = \phi$$

$$\text{Ext } S = \phi \Rightarrow (\text{Ext } S)^c = \mathbb{R}$$

$$\text{Now, Fr}(S) = (S^o)^c \cap (\text{Ext } S)^c = \phi \cap \mathbb{R} = \phi$$

- (b) $S = \mathbb{N}$

$$S^o = \phi \Rightarrow (S^o)^c = \mathbb{R}$$

$$\text{Ext } S = \mathbb{R} - \mathbb{N} \Rightarrow (\text{Ext } S)^c = \mathbb{N}$$

$$\text{Fr } S = \mathbb{R} \cap \mathbb{N} = \mathbb{N}$$

18. **Boundary Point:** Frontier Point of S which are member of S are referred as boundary point. Set of all boundary point of S is denoted by ∂S .

Example:

- (a) $S = \mathbb{R}$

$$\text{Then } \partial S = \text{Fr } S \cap S = \phi \cap S = \phi$$

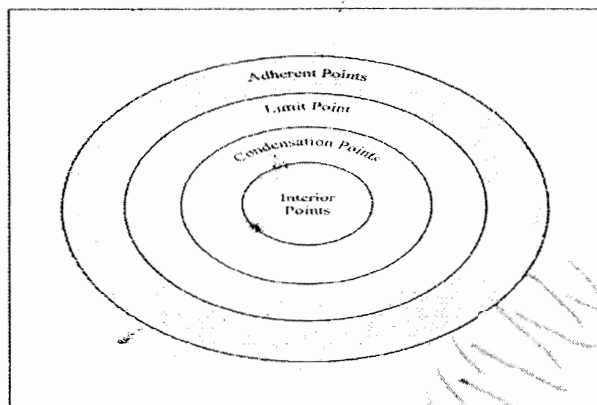
- (b) $S = \mathbb{N}$

$$\partial S = \text{Fr } S \cap S = \mathbb{N} \cap \mathbb{N} = \mathbb{N}$$

19. **Open Set:** A set S is called open, if it is neighbourhood of each of its point.

20. **Closed Set:** A set S is called closed, if it contains all its limit point.

21. **Derived Set:** Collection of all limit point of a given set S is called derived set of S and it is denoted by S' .
22. **Perfect Set:** If $S = S'$ then S is called perfect set.
23. **Dense Set:** A set S is said to be Dense in any set X iff $\bar{S} = X$.



2.2. Bolzano Weierstrass Theorem

Every infinite bounded subset of $\mathbb{R}$ has a limit point.

Note:

- (a) This is only a sufficient condition for a set to have a limit point i.e. unbounded infinite sets may have limit point

Example: $S = \mathbb{R}$

- (b) Condition of infiniteness cannot be dropped.

Example: $S = \{0, 1\}$ does not have any limit point.

- (c) Condition of boundedness cannot be relaxed.

Example: $S = \mathbb{Z}$ does not have any limit point.

2.3. Properties

1. A set may or may not possess upper and/or lower bounds.

Example:

- (a) $S = (-\infty, 0)$ then upper bound of S is 0 and lower bound of S does not exist

- (b) $S = [0, \infty)$ then upper bound of S does not exist and Lower bound of S is 0

- (c) $S = [0, 1]$ then upper bound of S is 1 and lower bound of S is 0

2. Bounds of a set may or may not belong to the set.
3. If a set has one upper bound / lower bound then it has an infinite number of upper bounds/lower bounds.
4. If S is a bounded non-empty subset of $\mathbb{R}$. Then $\inf S \leq \sup S$

5. If $S \subseteq T \subseteq \mathbb{R}$ where $S \neq \emptyset$ then

(a) If T is a bounded above then $\sup S \leq \sup T$ (existence of $\sup$ is followed by O.C.P.)

(b) If T is bounded below then $\inf S \geq \inf T$.

6. For any two non-empty bounded subsets S, T of $\mathbb{R}$ such that $S \subseteq T$;
 $\inf T \leq \inf S \leq \sup S \leq \sup T$

7. For any two non-empty subsets S, T of $\mathbb{R}$ such that $s \leq t \forall s \in S, t \in T$ then S is bounded above and T is bounded below and $\sup(S) \leq \inf T$.

8. If S and T be arbitrary non-empty bounded subsets of $\mathbb{R}$ then

(a) $\sup(S \cup T) = \max\{\sup S, \sup T\}$

(b) $\inf(S \cup T) = \min\{\inf S, \inf T\}$

9. If S and T are two non-empty subset of $\mathbb{R}$ then $\sup(S \cap T) \neq \min\{\sup S, \sup T\}$

Example:

$S = [0, 1] \cup \{4\}$ and $T = [0, 2]$ then $\sup S = 4$ and $\sup T = 2$ and $\min\{\sup S, \sup T\} = 2$

$S \cap T = [0, 1]$ and $\sup(S \cap T) = 1$. Thus $\sup(S \cap T) \neq \min\{\sup S, \sup T\}$

10. Let S be any non-empty subset of $\mathbb{R}$ having Supremum and let $T = \{\lambda x : x \in S\}$ then T is bounded above $\sup T = \lambda \sup S \forall \lambda \geq 0$

11. Let S be a non-empty set of $\mathbb{R}$ which has Supremum and let $T = \{\lambda x : x \in S\}$ then T is bounded below and $\inf T = \lambda \sup S \forall \lambda \leq 0$

12. $\sup\{x \in \mathbb{Q} : x < a\} = a$ for each $a \in \mathbb{R}$.

13. Finite union and arbitrary intersection of bounded subsets of $\mathbb{R}$ is bounded.

14. A subset of bounded set is bounded.

15. If S and T are two bounded sets Then

(a) $\sup\{x + y : x \in S, y \in T\} = \sup S + \sup T$

(b) $\inf\{x + y : x \in S, y \in T\} = \inf S + \inf T$

(c) $\sup\{x - y : x \in S, y \in T\} = \sup S - \inf T$

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- (d) $\inf\{x - y : x \in S, y \in T\} = \inf S - \sup T$
- (e) $\sup\{xy : x \in S, y \in T\} = \sup S \sup T$ (where S and T are bounded subset of positive real numbers)
- (f) $\inf\{xy : x \in S, y \in T\} = \inf S \inf T$ (where S and T are bounded subset of positive real numbers)
16. Let S be a non-empty subset of $\mathbb{R}$ having supremum and $T = \{x \in \mathbb{R} : x - a \in S\}$, $C = \left\{x \in \mathbb{R} : \frac{x+a}{2} \in S\right\} \forall a \in \mathbb{R}$ then $\sup T = a + \sup S$ and $\sup C = 2\sup S - a$

Example:

Let $S = \{1, 2, 3, 4, 5\}$ then $\sup S = 5$ and

$$T = \{x \in \mathbb{R} : x - 5 \in S\} = \{6, 7, 8, 9, 10\}$$

Clearly $\sup T = 10 = 5 + \sup S$

Define

$$C = \left\{x \in \mathbb{R} : \frac{x+5}{2} \in S\right\} = \{5, 3, 1, -1, -3\}$$

Clearly $\sup C = 5 = 2 \sup S - 5$

17. Every bounded subset of $\mathbb{R}$ is contained in an interval of finite length.
18. Countably infinite set is not a neighbourhood of any of its points.
19. Non-empty finite set is not a neighbourhood of any of its points.
20. Empty set ϕ is neighbourhood of each of its points.
21. Every neighbourhood of a point is uncountable.
22. Uncountability is necessary condition for a set to be neighbourhood of a point but not sufficient.

Example: $S = \mathbb{Q}^c$ then S is an uncountable set but not neighbourhood of any of its points.

23. Superset of a neighbourhood is also neighbourhood of same point.
24. Subset of a neighbourhood may or may not be a neighbourhood of same point.

Example: $S = [-1, 1]$ is neighbourhood of 0 but it's subset

$$T = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \text{ is not.}$$

25. Arbitrary union of family of neighbourhood is also neighbourhood.
26. If S and T are two neighbourhoods of point $p \in \mathbb{R}$, then $S \cap T$ is also a neighbourhood of p .
27. Finite intersection of neighbourhoods of a point is also neighbourhood of that point.
28. Arbitrary intersection of neighbourhoods need not be a neighbourhood of a Point.

Example: $S_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$

Then each S_n is a neighbourhood of 0.

But $\bigcap_{n=1}^{\infty} S_n = \{0\}$ is not a neighbourhood of 0.

29. There exist a countable infinite set $S \subseteq \mathbb{R}$, such that $\sup S \in S^c$ and $\inf S \in S^c$.

Example: $S = (\sqrt{2}, \sqrt{3}) \cap \mathbb{Q}$ then $\sup S = \sqrt{3} \in S^c$ and $\inf S = \sqrt{2} \in S^c$.

30. There exist a uncountable set $S \subseteq \mathbb{R}$, such that $\sup S \in S^c$ and $\inf S \in S^c$.

Example: $S = (0, 1)$ then $\sup S = 1 \in S^c$ and $\inf S = 0 \in S^c$.

31. There exist countably infinite set $S \subseteq \mathbb{R}$, such that $\sup S \in S$ and $\inf S \in S^c$.

Example: $S = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$ then $\sup S = 1 \in S$ and $\inf S = 0 \in S^c$.

32. There exist uncountable set $S \subseteq \mathbb{R}$, such that $\sup S \in S$ and $\inf S \in S^c$.

Example: $S = (0, 1]$ then $\sup S = 1 \in S$ and $\inf S = 0 \in S^c$.

33. Intersection of the family of all neighbourhoods of a point x is singleton set $\{x\}$.

34. A finite set has no limit point.

35. Limit point of a set may or may not belong to the set.

Example: $S = (0, 1]$ then 0 is a limit point of S . But $0 \notin S$ also 1 is a limit point of S and $1 \in S$.

36. Infiniteness is the necessary condition for a set to have a limit point but not sufficient.

i.e. Infinite set may not have a limit point.

Example: $S = \mathbb{N}$ (Set of all Natural number), does not have a limit point.

Put Your Own Notes

37. Uncountable set must have limit point.
38. Set of Isolated points of a set is always countable.
39. Every limit point of a set is an adherent point of that set but converse may not be true.

Example: $S = \mathbb{N}$ (Set of all Natural number)

Then every member of S is an Adherent Point of S . But no member of S is a limit point of S

40. Limit point of a set S cannot be interior point of S^c

Reason: Let $p \in \mathbb{R}$ be a limit point of the set $S \subseteq \mathbb{R}$ and p is also an interior point of S^c

Then $\exists \varepsilon > 0$ such that $(p - \varepsilon, p + \varepsilon) \subset S^c$ then $(p - \varepsilon, p + \varepsilon) \cap S = \phi$ thus p is not a limit point of S which contradicts the hypothesis.

41. Every interior point of a set is limit point of that set. Converse may not be true.

Example: $S = (0, 1)$ then 0 is a limit point of the sets but not an interior point.

42. Every interior point of set is adherent point of a set. Converse may not be true

Example: $S = (0, 1]$ then 0 is an adherent point of S but not an interior point of S .

43. Interior point of S^c cannot be limit point of S .
44. Supremum of a set S (if exist) is either greatest member of S or a limit point of S
45. Infimum of a set S (if exist) is either least member of S or a limit point of S
46. For every set S , S° is an open set.
47. S° is the largest open set contained in S
48. S and T are two set. Then
 - (i) $S \subseteq T \Rightarrow S^\circ \subseteq T^\circ$
 - (ii) $(S \cap T)^\circ = S^\circ \cap T^\circ$

(iii) $(S \cup T)^\circ \supset S^\circ \cup T^\circ$

49. For any subset S of $\mathbb{R}$, $S^\circ \subset S$.

50. S is open iff $S = S^\circ$, for any subset S of $\mathbb{R}$.

51. For any set S ; $(S^\circ)^\circ = S^\circ$.

52. S° is equal to the union of all open subsets of S

53. A set is closed iff its complement is an open set.

54. The intersection of arbitrary family of closed sets is closed

55. Finite union of closed set is closed.

56. Arbitrary union of closed sets may not be closed.

Example: $S_n = \left[\frac{1}{n}, 1 - \frac{1}{n}\right] \forall n \geq 2$ then each S_n is closed set but

$\bigcup_{n=1}^{\infty} S_n = (0, 1)$ is an open set.

57. Every finite set is a closed set.

58. Derived set is always closed set

59. Derived set of a bounded set is bounded set.

60. If derived set (say) S' is finite then S is countable set.

61. Let S and T are two subset of real numbers. Then

(i) $S \subset T \Rightarrow S' \subset T'$

(ii) $(S \cup T)' = S' \cup T'$

(iii) $(S \cap T)' \subset S' \cap T'$ (where S', T' denote the derived set of S and T respectively)

62. Non-empty countable set cannot be open

63. If a set S is open then $S \subseteq S'$ converse need not be true

Example: $S = (0, 1]$ then $S \subseteq S' = [0, 1]$ but S is not an open set.

64. If S is an open set and T is closed set. Then $S - T$ is open and $T - S$ is closed set.

65. Arbitrary intersection of open sets may not be open.

Example: $S_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$ then each S_n is an open set. But $\bigcap_{n=1}^{\infty} S_n = \{0\}$

which is not an open set.

66. Every non-empty open set is union of open intervals.

67. Closure of a set S is denote by $\bar{S}$ and $\bar{S} = S \cup S'$ i.e. closure is union of S and set of limit points of S

Some Examples:

Put Your Own Notes

Set	Supremum	Infimum	Set of adherent points
$\mathbb{N}$	does not exist	1	$\mathbb{N}$
$\mathbb{Z}$	does not exist	does not exist	$\mathbb{Z}$
$\mathbb{Q}$	does not exist	does not exist	$\mathbb{R}$
$\mathbb{Q}^C$	does not exist	does not exist	$\mathbb{R}$
$\mathbb{R}$	does not exist	does not exist	$\mathbb{R}$
$\left\{\frac{1}{n}; n \in \mathbb{N}\right\}$	1	0	$\left\{\frac{1}{n}; n \in \mathbb{N}\right\} \cup \{0\}$
$[a, b]; a \neq b$	b	a	$[a, b]$
$\left\{x + \frac{1}{n}; n \in \mathbb{N}\right\}$	$x+1$	x	$\left\{x + \frac{1}{n}; n \in \mathbb{N}\right\}$
$\{\cos n; n \in \mathbb{N}\}$	1	-1	$[-1, 1]$

Set	Set of limit points	Frontier points	Boundary points	Nbd. of
$\mathbb{N}$	ϕ	$\mathbb{N}$	$\mathbb{N}$	None of its Points
$\mathbb{Z}$	ϕ	$\mathbb{Z}$	$\mathbb{Z}$	None of its Points
$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{Q}$	None of its Points
$\mathbb{Q}^C$	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{Q}^C$	None of its Points
$\mathbb{R}$	$\mathbb{R}$	ϕ	ϕ	Each of its Points
$\left\{\frac{1}{n}; n \in \mathbb{N}\right\}$	$\{0\}$	$\left\{\frac{1}{n}; n \in \mathbb{N}\right\}$	$\left\{\frac{1}{n}; n \in \mathbb{N}\right\}$	None of its Points
$[a, b]; a \neq b$	$[a, b]$	$\{a, b\}$	$\{a\}$	(a, b)
$\left\{x + \frac{1}{n}; n \in \mathbb{N}\right\}$	$\{x\}$	$\left\{x + \frac{1}{n}; n \in \mathbb{N}\right\}$	$\left\{x + \frac{1}{n}; n \in \mathbb{N}\right\}$	None of its Points
$\{\cos n; n \in \mathbb{N}\}$	$[-1, 1]$	$\{\cos n; n \in \mathbb{N}\}$	$\{\cos n; n \in \mathbb{N}\}$	None of its Points

****Students are advised to extend this table by themselves****

Put Your Own Notes

CHAPTER 3

SEQUENCES OF REAL NUMBERS

3.1. Definition

A map from $\mathbb{N}$ to $\mathbb{R}$ defines a sequence of real numbers when the images are arranged in the natural order of natural number i.e., starting with images of n is followed by that of $(n+c)$ i.e., $f: \mathbb{N} \rightarrow \mathbb{R}$ is a sequence of real numbers and $\langle f(n) \rangle = \langle f(1), f(2), \dots \rangle$ we write $\langle f(n) \rangle = \langle a_n \rangle, \langle b_n \rangle, \langle u_n \rangle, \langle v_n \rangle$ etc. $\forall n \in \mathbb{N}$, $\langle a_n \rangle = \langle a_1, a_2, a_3, \dots \rangle$.

When $f: \mathbb{N} \rightarrow \mathbb{C}$, then this function defines a sequence of complex numbers.

Note: A map $f: S_{(n)} \rightarrow \mathbb{R}$ (1). Define a finite sequence of length n called as string ($S_{(n)}$ denote the set of first n natural numbers). Generally sequence is always infinite (i.e., non-terminating) but if defined on finite set then it is given by (1).

3.1.1. Range set of a Sequence

The range set of a sequence is the set consisting of all distinct elements of a sequence and without regard to the position term. Thus the range may be finite or infinite set i.e., the range set of sequence $\langle u_n \rangle$ is given by $\{u_n : n \in \mathbb{N}\}$ or simply by $R(u)$.

Some Important Example of Sequence

1. $\langle a_n \rangle = \langle n \rangle$
2. $\langle a_n \rangle = \langle -n \rangle$
3. $\langle a_n \rangle = (-1)^n n$
4. $\langle a_n \rangle = (-1)^n$
5. $\langle a_n \rangle = \frac{1}{n}$
6. $a_n = 1 - \frac{(-1)^n}{n}$
7. $a_n = \begin{cases} 2 & n = 1 \text{ or prime} \\ n & \text{else} \end{cases}$
8. $a_n = \begin{cases} 2 & n = 1 \text{ or prime} \\ p & p/n \text{ \& p is the least prime} \end{cases}$

9. $a_{n+2} = a_n + a_{n+1}, \quad a_1 = 1, \quad a_2 = 1$
10. $a_{n+2} = \frac{1}{2}(a_n + a_{n+1})$ where a_1 & a_2 given
11. $a_{n+1} = \sqrt{2 + a_n}$ where $a_1 = 0$
12. $\langle a_n \rangle = 2012 \quad \forall n \in \mathbb{N}$
13. $\langle a_n \rangle = (-1)^n + \sin\left(\frac{n\pi}{6}\right)$
14. $\langle a_n \rangle = \begin{cases} -1 - \frac{1}{n} & n \text{ is even} \\ 1 + \frac{1}{n} & n \text{ is odd} \end{cases}$
15. $\langle a_n \rangle = \cos n$
16. $\langle a_n \rangle = \sin\left(\frac{1}{n}\right)$
17. $\langle a_n \rangle = \text{Last digit of } 3^n$
18. $\langle a_n \rangle = \{-2, -3, -1, 0, 1, 2, 1, 0, 1, 0, \dots\}$
19. $a_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \log n$
20. $a_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$

3.1.2. Bounded Sequence

A sequence $\langle u_n \rangle$ is defined as bounded if its range set is bounded. Hence $\langle u_n \rangle$ is bounded if there exist real numbers k' and k such that $k' \leq u_n \leq k \quad \forall n \in \mathbb{N}$.

OR

If there exist $K \geq 0$ such that $|u_n| \leq K \quad \forall n \in \mathbb{N}$.

Note:

- (a) Supremum and infimum of the range set is the supremum and infimum of the sequence.
- (b) If the range of the sequence $\langle a_n \rangle$ is finite then there exist $\alpha \in \mathbb{R}$ such that $a_n = \alpha$ for infinitely many values of n . however converse need not be true.

Example: $a_n = \begin{cases} 4 & n = 1 \text{ or prime} \\ n & \text{else} \end{cases}$

Then $a_n = 4$ for infinitely many values of n but range of the sequence $\langle a_n \rangle$ is not finite

3.1.3. Monotonic Sequence

Let $\langle a_n \rangle$ be a sequence of real numbers $\langle a_n \rangle$ is monotonic if either $a_n \leq a_{n+1} \forall n \in \mathbb{N}$ (Then the sequence is called monotonically increasing or non-decreasing).

$a_n \geq a_{n+1} \forall n \in \mathbb{N}$ (Then the sequence is called monotonically decreasing or non-increasing).

If $a_n < a_{n+1} \forall n \in \mathbb{N}$, then $\langle a_n \rangle$ is called strictly increasing or increasing sequence.

If $a_n > a_{n+1} \forall n \in \mathbb{N}$, then $\langle a_n \rangle$ is called strictly decreasing or decreasing sequence.

Examples:

(a) $\langle a_n \rangle = \langle n \rangle$ is increasing sequence.

(b) $\langle a_n \rangle = \langle -n \rangle$ is decreasing sequence.

(c) $a_{n+1} = \sqrt{2 + a_n} \forall n \in \mathbb{N}$ is monotonically increasing sequence.

3.1.4. Eventual Nature of a Sequence

1. A sequence $\langle a_n \rangle$ is said to be eventually monotonically increasing if $\exists m \in \mathbb{N}$ such that $a_{n+1} \geq a_n \forall n \geq m$.

Example: $\langle 0, 2, -5, 3, 7, 8, 9, 10, 11, 12, \dots \rangle$

2. A sequence $\langle a_n \rangle$ is said to be eventually monotonically decreasing if $\exists m \in \mathbb{N}$ such that $a_{n+1} \leq a_n \forall n \geq m$.

Example: $\left\langle 0, 2, 0, 2, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\rangle$

3. A sequence $\langle a_n \rangle$ is eventually constant sequence if $\exists m \in \mathbb{N}$ such that $a_n = c \forall n \geq m$ where $c \in \mathbb{R}$.

Example: $\langle 0, 2, 0, 2, 0, 3, 8, 5, 7, 7, 7, 7, \dots \rangle$

3.2. Limit Point of a Sequence

A real number p is said to be a **limit point** or a **cluster point** of a sequence if every neighbourhood of p contains an infinite number of elements of the given sequence.

In other words, a real number p is a **limit point** of a sequence $\langle a_n \rangle$ if for any $\varepsilon > 0$, $a_n \in (p - \varepsilon, p + \varepsilon)$ for infinitely many values of n .

Remark: A real number p is **not a limit point** of a sequence $\langle a_n \rangle$, if there exists at least one neighbourhood of p which contains only **finite** number of elements of $\langle a_n \rangle$.

Note:

- (a) Limit point of a sequence need not be a limit point of the range set.

Example: $\langle a_n \rangle = (-1)^n$

Then $\{1, -1\}$ is a set of limit points of $\langle a_n \rangle$ but neither 1 nor -1 limit point of its range set.

- (b) Limit point of the sequence can be obtained in three ways and they are as follows :

(a) Representation on real line.

(b) If the range set has a limit point, then sequence has limit point.

(c) If the range set is finite.

$\Rightarrow a_n = \alpha$ for infinitely many values of n where $\alpha \in \mathbb{R}$.

$\Rightarrow \alpha$ is a limit point of the sequence.

3.2.1. Existence of A Limit Point

Bolzano Weierstrass Theorem: Every bounded sequence has a limit point.

Reason: Since the sequence $\langle a_n \rangle$ say is bounded thus Range set of $\langle a_n \rangle$ is bounded.

Case I: If range set is infinite then by Bolzano Weierstrass Theorem for set, range set must have limit point thus sequence $\langle a_n \rangle$ has a limit point.

Case II: If range set is finite then $\exists \alpha \in \mathbb{R}$ such that $a_n = \alpha$ for infinitely many values of n . Evidently, α is a limit point of the sequence $\langle a_n \rangle$.

Hence, every bounded sequence has a limit point.

Application:

1. $\langle a_n \rangle = (-1)^n$ has limit points 1, -1

2. $\langle b_n \rangle = \langle \frac{1}{n} \rangle$ has limit point 0

3. $\langle c_n \rangle = \left(1 + \frac{1}{n}\right)^n$ has limit point e

3.3. Subsequence

Subsequence of a sequence is defined with the help of sequence of natural number. Consider a map $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(k) = n_k$ and let $\langle n_k \rangle$ be a strictly increasing sequence of natural numbers. Then for any sequence $\langle a_n \rangle$, the sequence $\langle a_{n_k} \rangle$ is defined as a subsequence of $\langle a_n \rangle$.

Example: Let $\langle a_n \rangle$ be any sequence of real number. Then $\langle a_{2k} \rangle, \langle a_{2k-1} \rangle, \langle a_{2^k} \rangle, \langle a_{3^k} \rangle$ all are subsequences of $\langle a_n \rangle$ where $\langle a_{2k} \rangle = \langle b_n \rangle = b_1, b_2, b_3, \dots = a_2, a_4, a_6, a_8, \dots$

Similarly $\langle a_{2^k} \rangle = \langle c_n \rangle = c_1, c_2, c_3, c_4, \dots = a_2, a_4, a_8, a_{16}, \dots$

Thus $\langle b_n \rangle$ is a sub sequence of $\langle a_n \rangle$ and $\langle c_n \rangle$ is a subsequence of $\langle b_n \rangle$.

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3.3.1. Complementary Subsequences

Let $\langle a_{n_k} \rangle$ and $\langle a_{n_{k'}} \rangle$ are subsequence of $\langle a_n \rangle$ then define $S_1 = \{n_k \mid n_k \in \mathbb{N}\}$ and $S_2 = \{n_{k'} \mid n_{k'} \in \mathbb{N}\}$, then $\langle a_{n_k} \rangle$ and $\langle a_{n_{k'}} \rangle$ are complementary subsequences if

1. $S_1 \cup S_2 = \mathbb{N}$
2. $S_1 \cap S_2 = \emptyset$

($S_1 \cup S_2$ may not be complete $\mathbb{N}$ but except finite subset of $\mathbb{N}$ & $S_1 \cap S_2$ may be a finite subset of $\mathbb{N}$).

Example: If $\langle a_n \rangle$ is a sequence of real numbers. Then $\langle a_{2n} \rangle$ and $\langle a_{2n-1} \rangle$ are complementary subsequence of $\langle a_n \rangle$.

Remark: Unbounded sequence may have limit point.

3.4. Limit of a Sequence

Let $\langle a_n \rangle$ be a sequence of real number and l be a real number, ($l \in \mathbb{R}$) then we say l is the limit of the sequence $\langle a_n \rangle$ if for any $\epsilon > 0 \exists m \in \mathbb{N}$ such that $|a_n - l| < \epsilon \forall n \geq m$.

Symbolically, we write it as $\lim_{n \rightarrow \infty} a_n = l$ or $a_n \rightarrow l$ as $n \rightarrow \infty$.

Algorithm: To claim l is limit of a sequence $\langle a_n \rangle$

1. Simplify $|a_n - l|$ to get $|a_n - l| \leq \phi(n)$, where $\phi(n)$ is function of n .
2. Solve the inequality $\phi(n) < \epsilon$ to get $n > \psi(\epsilon)$.
3. If $\psi(\epsilon)$ is defined $\forall \epsilon > 0$ then set $m = [\psi(\epsilon)] + 1$ (where $[\psi(\epsilon)]$ denote the ceiling function)
 $\Rightarrow |a_n - l| < \epsilon \forall n \geq m$.

Result Based on the Definition:

Let $\lim_{n \rightarrow \infty} a_n = a$ and $\lim_{n \rightarrow \infty} b_n = b$ ($a, b \in \mathbb{R}$), then

1. $\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n = a + b$
2. $\lim_{n \rightarrow \infty} (a_n - b_n) = \lim_{n \rightarrow \infty} a_n - \lim_{n \rightarrow \infty} b_n = a - b$
3. $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = \lim_{n \rightarrow \infty} (a_n) \lim_{n \rightarrow \infty} (b_n) = a \cdot b$
4. $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{\left(\lim_{n \rightarrow \infty} a_n \right)}{\left(\lim_{n \rightarrow \infty} b_n \right)} = \frac{a}{b}$, provided $b_n \neq 0 \forall n \in \mathbb{N}$ and $b \neq 0$.
5. $\lim_{n \rightarrow \infty} a_n = a \Leftrightarrow \lim_{n \rightarrow \infty} (ka_n) = k \lim_{n \rightarrow \infty} (a_n) = k \cdot a \quad \forall k \in \mathbb{R} - \{0\}$

Remarks:

- (a) In view the results 1–5 we can say that the operation of limit is compatible with addition subtraction multiplication division and scalar multiplication.
- (b) Converse of the above results 1–4 is not true.

Examples:

- (a) Take $\langle a_n \rangle = \langle n \rangle$ and $\langle b_n \rangle = \langle -n \rangle$, then $\langle a_n \rangle, \langle b_n \rangle$ are not convergent
- (i) $\langle a_n + b_n \rangle = \langle 0 \rangle$, which is convergent.
- (ii) $\frac{\langle a_n \rangle}{\langle b_n \rangle} = \langle -1 \rangle$, which is convergent.
- (b) Take $\langle a_n \rangle = \langle (-1)^n \rangle, \langle b_n \rangle = \langle (-1)^{n+1} \rangle$, then $\langle a_n \rangle, \langle b_n \rangle$ both are not convergent
- (i) $\langle a_n \cdot b_n \rangle = \langle (-1)^{2n+1} \rangle = \langle -1 \rangle$ Which is convergent.
- (ii) $\langle a_n - b_n \rangle = \langle 0 \rangle$ Which is convergent.
- (c) Every convergent sequence is bounded; however converse need not be true.

Reason: Let $\lim_{n \rightarrow \infty} a_n = a \quad (a \in \mathbb{R})$

Then for given $\epsilon > 0$, $\exists k \in \mathbb{N}$ such that

$$|a_n - a| < \epsilon \quad \forall n \geq k \Rightarrow a - \epsilon < a_n < a + \epsilon \quad \forall n \geq k$$

Take $M = \max\{a_1, a_2, \dots, a_{k-1}, a + \epsilon\}$ and $m = \min\{a_1, a_2, \dots, a_{k-1}, a - \epsilon\}$

Then $m \leq a_n \leq M \quad \forall n \in \mathbb{N}$. This implies $\langle a_n \rangle$ is bounded.

Note: Converse need not be true.

Example: Take $\langle a_n \rangle = \langle (-1)^n \rangle$, then $\langle a_n \rangle$ is bounded but not convergent.

- (c) Limit of a convergent sequence is unique.

Reason: Let $\langle a_n \rangle$ be a sequence of real number such that $\lim_{n \rightarrow \infty} a_n = a$

Let if possible, $\langle a_n \rangle$ also converges to a' ($a' \in \mathbb{R}$)

Then for given $\epsilon > 0$, $\exists m_1, m_2 \in \mathbb{N}$ such that

$$|a_n - a| < \epsilon/2 \quad \forall n \geq m_1$$

$$|a_n - a'| < \epsilon/2 \quad \forall n \geq m_2$$

Take $M = \max\{m_1, m_2\}$

$$\text{Now, } |a - a'| = |-a_n + a + a_n - a'|$$

$$\Rightarrow |a - a'| \leq |a_n - a| + |a_n - a'| \quad \forall n \geq m$$

$$< \epsilon/2 + \epsilon/2 = \epsilon \quad \forall n \geq m$$

$$\Rightarrow |a - a'| < \epsilon$$

Put Your Own Notes

Since ϵ was arbitrary small. So $\epsilon \rightarrow 0, a = a'$

Hence, limit of sequence $\langle a_n \rangle$ is unique.

Note:

- (i) l is a limit $\Leftrightarrow$ every Neighbourhood of l leaves out only finite members of the sequence.
- (ii) Limit of a sequence is limit point but converse need not be true.
- (d) If $\lim_{n \rightarrow \infty} u_n = l \Rightarrow \lim_{n \rightarrow \infty} |u_n| = |l|$ Converse need not be true.

Example: Take $\langle u_n \rangle = (-1)^n$ then $\langle |u_n| \rangle = \langle 1 \rangle$, is convergent but $\langle u_n \rangle$ is not convergent.

- (e) $\lim_{n \rightarrow \infty} |u_n - l| = 0 \Leftrightarrow \lim_{n \rightarrow \infty} u_n = l$.
- (f) If $|u_n| \leq |v_n| \forall n \geq m$ and $\lim_{n \rightarrow \infty} |v_n| = 0$ thus $\lim_{n \rightarrow \infty} |u_n| = 0$ (where $|u_n|$ denotes the absolute value of u_n).
- (g) If $\lim_{n \rightarrow \infty} a_n = a$ and $a_n \geq b \forall n \geq m, m \in \mathbb{N}$. Then $a \geq b$ ($a, b \in \mathbb{R}$).
- (h) A bounded sequence $\langle a_n \rangle$ is convergent $\Leftrightarrow$ it has unique limit point.

Note:

- (i) The above statement of the result provides a second definition for a sequence to be convergent i.e., a sequence is convergent if it is bounded and has a unique limit point.
- (ii) Condition of the above result cannot be relaxed i.e., if a bounded sequence have more than one limit point then it cannot be convergent.

Example: A sequence $\langle a_n \rangle = (-1)^n$, is bounded having two limit points 1 and -1 thus it is not convergent sequence.

Application:

1. If a sequence $\langle a_n \rangle$ have more than one limit point then $\langle a_n \rangle$ cannot be convergent.

Example:

- (a) $\langle a_n \rangle = (-1)^n$ is not convergent as 1 and -1 are two limit points of $\langle a_n \rangle$.
- (b) $\langle b_n \rangle = \langle (-1)^n + \frac{1}{n} \rangle$ is not convergent as 1 and -1 are two limit points of $\langle b_n \rangle$.

2. An unbounded sequence cannot converge to a real number.

Example:

- (a) $\langle a_n \rangle = \langle n \rangle$ is not convergent.
- (b) $\langle b_n \rangle = \langle (-1)^n + n \rangle$ is not convergent.

Put Your Own Notes

3. If a sequence has limit point, then it may not be convergent.

Example:

Consider a sequence $\langle a_n \rangle = \langle 1, 0, 2, 0, 3, 0, 4, 0, \dots \rangle$ then 0 is a limit point of $\langle a_n \rangle$ but $\langle a_n \rangle$ is not convergent as it is unbounded.

- (i) A monotonic sequence $\langle a_n \rangle$ is convergent $\Leftrightarrow$ it is bounded.

Remark:

- (i) Convergent sequence may not be monotonic.

Example:

$$a_n = 1 - \frac{(-1)^n}{n}$$

- (ii) A bounded and monotonically increasing sequence converges to its supremum.

- (iii) A bounded and monotonically decreasing sequence converges to its infimum.

- (iv) An eventually monotonic sequence is convergent $\Rightarrow$ it is bounded

- (v) If $\langle a_n \rangle, \langle b_n \rangle$ be two sequences of real numbers such that

$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n \quad \forall n \geq m \quad \text{and} \quad \lim_{n \rightarrow \infty} (a_n - b_n) = 0 \quad \text{Then}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n.$$

- (vi) Monotonic sequence can have at most one limit point.

Applications:

- (a) The sequence $\langle a_n \rangle = \langle \frac{1}{n} \rangle$ converges to 0 as it is monotonically decreasing sequence and 0 is its infimum.

- (b) The sequence $\langle a_n \rangle = \langle n \rangle$ is not convergent as it is monotonically increasing sequence which is not bounded above.

- (c) If $a_1 = \sqrt{3}$ and $a_{n+1} = \sqrt{3 + \sqrt{a_n}} \quad \forall n \in \mathbb{N}$, then $\langle a_n \rangle$ converges.

- (d) The sequence $\langle S_n \rangle$ defined by the recursion formula :

$$S_{n+1} = \sqrt{7 + S_n}, S_1 = \sqrt{7} \text{ converges to the positive root of } x^2 - x - 7 = 0$$

Solution: We have $S_1 = \sqrt{7}$ and $S_2 = \sqrt{7 + \sqrt{7}}$.

Then $S_2 > S_1$. Let us suppose that $S_{k+1} > S_k$.

$$\text{Then } 7 + S_{k+1} > 7 + S_k \Rightarrow \sqrt{7 + S_{k+1}} > \sqrt{7 + S_k} \Rightarrow S_{k+2} > S_{k+1}.$$

Thus, by the principle of mathematical induction, $S_{n+1} > S_n \quad \forall n \in \mathbb{N}$, and so $\langle S_n \rangle$ is monotonically increasing.

Again $S_1 = \sqrt{7} < 7$. Let us suppose that

$$S_k < 7 \Rightarrow \sqrt{7 + S_k} < \sqrt{7 + 7} < 7 \Rightarrow S_{k+1} < 7.$$

Thus, by the principle of mathematical induction, $S_n < 7 \quad \forall n$, and so

$\langle S_n \rangle$ is bounded above.

Put Your Own Notes

Put Your Own Notes

Since $\langle S_n \rangle$ is monotonically increasing and bounded above, so $\langle S_n \rangle$ is convergent.

Let $\lim_{n \rightarrow \infty} S_n = l$, so that $\lim_{n \rightarrow \infty} S_{n+1} = l$ (1)

Clearly $l > 0$.

Now $S_{n+1} = \sqrt{7 + S_n} \Rightarrow S_{n+1}^2 = 7 + S_n$.

Taking the limit as $n \rightarrow \infty$ and using (1), we get $l^2 = 7 + l \Rightarrow l^2 - l - 7 = 0$.

Hence l is a positive root of $x^2 - x - 7 = 0$.

(e) Let $\langle a_n \rangle$ be a sequence defined as $a_1 > 0$, $a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right)$, $n \geq 1$.

Then $\langle a_n \rangle$ is a bounded monotonic sequence.

Solution: From $a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right)$ we obtain $a_n^2 - 2a_n a_{n+1} + 2 = 0$, which is quadratic in a_n . Since this equation has real roots, therefore $a_{n+1}^2 - 2 \geq 0 \Rightarrow a_{n+1} \geq \sqrt{2} \forall n \in \mathbb{N}$.

Thus the sequence $\langle a_n \rangle$ is bounded below by $\sqrt{2}$.

Now

$$a_n - a_{n+1} = a_n - \frac{1}{2} \left(a_n + \frac{2}{a_n} \right) = \frac{1}{2} \left(a_n - \frac{2}{a_n} \right) = \frac{1}{2} \left(\frac{a_n^2 - 2}{a_n} \right) \geq 0 \forall n$$

$$\therefore a_n \geq a_{n+1} \text{ or } a_{n+1} \leq a_n \forall n$$

$\Rightarrow \langle a_n \rangle$ is a monotonically decreasing sequence.

Since $\langle a_n \rangle$ is a monotonically decreasing and bounded below sequence, therefore, $\langle a_n \rangle$ is convergent.

Let $\lim_{n \rightarrow \infty} a_n = l$, so that $\lim_{n \rightarrow \infty} a_{n+1} = l$. Using in $a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right)$, we get

$$\lim_{n \rightarrow \infty} a_{n+1} = \frac{1}{2} \left(\lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} \frac{2}{a_n} \right)$$

$$\Rightarrow l = \frac{1}{2} \left(l + \frac{2}{l} \right) \Rightarrow l^2 = 2 \Rightarrow l = \sqrt{2}. \quad (\because l > 0)$$

Hence $\lim_{n \rightarrow \infty} a_n = \sqrt{2}$.

(f) Let $\langle s_n \rangle$ be a sequence defined by $s_1 = 1$, $s_{n+1} = \left(\frac{3 + s_n^2}{2} \right)^{1/2}$

$\forall n \in \mathbb{N}$, then $\langle s_n \rangle$ is a bounded and monotonically increasing sequence and converges to $\sqrt{3}$.

(g) Let x_1, y_1 , are two positive unequal numbers and $x_n = \frac{1}{2}(x_{n-1} + y_{n-1})$ and $y_n = \sqrt{x_{n-1}y_{n-1}} \forall n \geq 2$ Then the sequence $\langle x_n \rangle$ and $\langle y_n \rangle$ are monotonic and they converge to the same limit.

Solution: Since x_1 and y_1 are unequal positive numbers, let $x_1 > y_1$

$$\Rightarrow \frac{1}{2}(x_1 + y_1) > \sqrt{x_1 y_1} \quad (\because \text{A.M.} > \text{G.M.})$$

$\Rightarrow x_2 > y_2$. Similarly, $x_3 > y_3$ and so on.

$$\text{Thus } x_n > y_n \quad \forall n \quad (2)$$

$$\text{Also } x_{n+1} = \frac{1}{2}(x_n + y_n) < \frac{1}{2}(x_n + x_n) = x_n, \text{ by (2)}$$

$$\therefore x_{n+1} < x_n \quad \forall n \quad (3)$$

Thus $\langle x_n \rangle$ is a monotonically decreasing sequence.

$$\text{Again } y_{n+1} = \sqrt{x_n y_n} > \sqrt{y_n y_n} = y_n, \text{ by (2).}$$

$$\therefore y_{n+1} > y_n \quad \forall n \quad (4)$$

Thus $\langle y_n \rangle$ is a monotonically increasing sequence.

$$\text{Now } x_n = \frac{1}{2}(x_{n-1} + y_{n-1}) > \frac{1}{2}(y_{n-1} + y_{n-1}) = y_{n-1} \Rightarrow x_n > y_{n-1}, \text{ by (2)}$$

$$\Rightarrow y_n < x_{n-1} < x_{n-2} < \dots < x_1, \text{ using (3)}$$

$$\Rightarrow y_n < x_1 \quad \forall n \text{ and so } \langle y_n \rangle \text{ is bounded above.}$$

Since $\langle x_n \rangle$ is monotonically decreasing and bounded below, therefore, $\langle x_n \rangle$ is convergent.

$$\text{Let } \lim_{n \rightarrow \infty} x_n = l \quad (5)$$

Since $\langle y_n \rangle$ is monotonically increasing and bounded above, therefore $\langle y_n \rangle$ is convergent.

$$\text{Let } \lim_{n \rightarrow \infty} y_n = l' \quad (6)$$

Now we shall show that $l = l'$.

$$\text{We have } x_n = \frac{1}{2}(x_{n-1} + y_{n-1}) \Rightarrow 2x_n = x_{n-1} + y_{n-1}$$

$$\Rightarrow 2 \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_{n-1} + \lim_{n \rightarrow \infty} y_{n-1}$$

$$\Rightarrow 2l = l + l' \Rightarrow l = l', \text{ by (5) and (6).}$$

Hence the sequences $\langle x_n \rangle$ and $\langle y_n \rangle$ are monotonic and converge to the same limit.

(j) Every sequence $\langle a_n \rangle$ has a monotonic subsequence $\langle a_{n_k} \rangle$.

(k) Let $\langle a_n \rangle$ be a sequence converging to l . Then every sub sequence of $\langle a_n \rangle$ also converges to l .

Reason: Let $\langle a_n \rangle$ be a sequence of real number which converge to l .

Let $\langle a_{n_k} \rangle$ be a subsequence of $\langle a_n \rangle$

Since $\langle a_n \rangle \rightarrow l$

So for given $\epsilon > 0$, $\exists m \in \mathbb{N}$ such that

$$|a_n - l| < \epsilon \quad \forall n \geq m$$

$$\Rightarrow |a_{n_k} - l| < \epsilon \quad \forall n_k \geq m$$

$$\Rightarrow \langle a_{n_k} \rangle \rightarrow l.$$

Applications:

(a) Let $\langle a_n \rangle = \langle \frac{1}{n} \rangle$ be a sequence then $\lim_{n \rightarrow \infty} a_n = 0$. Now

$\langle b_n \rangle = \langle \frac{1}{2n} \rangle, \langle c_n \rangle = \langle \frac{1}{3n} \rangle$ are subsequences of $\langle a_n \rangle$ so

$\lim_{n \rightarrow \infty} b_n = 0, \lim_{n \rightarrow \infty} c_n = 0$. (By above result)

(b) Let $\langle a_n \rangle = \langle \sin\left(\frac{1}{n}\right) \rangle$ be a sequence then $\lim_{n \rightarrow \infty} a_n = 0$. Now

$\langle b_n \rangle = \langle \sin\left(\frac{1}{2n+1}\right) \rangle$ is a subsequence

$\langle a_n \rangle$ So by above result, $\lim_{n \rightarrow \infty} b_n = 0$.

(1) A real number l is a limit point of a sequence $\langle a_n \rangle \Leftrightarrow$ there exist a subsequence $\langle a_{n_k} \rangle$ of $\langle a_n \rangle$ converges to l .

Remark:

- (a) Every bounded sequence has a convergent subsequence.
- (b) If each of the two subsequence $\langle a_{2n-1} \rangle$ and $\langle a_{2n} \rangle$ of a sequence $\langle a_n \rangle$ converges to l . Then $\langle a_n \rangle$ also converges to l . ($\because \langle a_{2n-1} \rangle$ and $\langle a_{2n} \rangle$ are complementary subsequences of $\langle a_n \rangle$)
- (c) If $\langle a_{2n-1} \rangle$ and $\langle a_{2n} \rangle$ converges to different limit then $\langle a_n \rangle$ cannot converge.

Example:

(a) Let $\langle a_n \rangle = \langle (-1)^n \rangle$.

Since $\langle a_n \rangle$ has two limit point -1 and 1 .

$\therefore$ There must exist two different subsequences of $\langle a_n \rangle$

Say $\langle b_n \rangle$ and $\langle c_n \rangle$ such that $\lim_{n \rightarrow \infty} b_n = 1$ and $\lim_{n \rightarrow \infty} c_n = -1$.

Hence $\langle a_n \rangle$ is not convergent.

(b) Let $\langle a_n \rangle = \left\langle (-1)^n + \frac{1}{n} \right\rangle$

Since $\langle b_n \rangle = \left\langle (-1)^{2n} + \frac{1}{2n} \right\rangle$ is a subsequence of $\langle a_n \rangle$ and $\langle b_n \rangle \rightarrow 1$.

$\therefore 1$ is a limit point of $\langle a_n \rangle$

Similarly, $\langle c_n \rangle = \left\langle (-1)^{2n+1} + \frac{1}{(2n+1)} \right\rangle$ is a subsequence of $\langle a_n \rangle$

and $\langle c_n \rangle \rightarrow -1$

$\Rightarrow -1$ is a limit point of $\langle a_n \rangle$

Hence $\langle a_n \rangle$ is not convergent.

(c) Let $\langle a_n \rangle$ be a sequence of positive real numbers such then

$$a_n = \sqrt{a_{n-1} a_{n-2}} \text{ for } n > 2 \text{ then } \langle a_n \rangle \rightarrow (a_1 a_2^2)^{1/3}.$$

(d) Let $a_1 > 0, a_2 > 0$ and define $a_n = \frac{2a_{n-1}a_{n-2}}{a_{n-1} + a_{n-2}}, n > 2$, then $\langle a_n \rangle$

converges to $\frac{3a_1a_2}{2a_1 + a_2}$.

(e) The sequence $\langle a_n \rangle$, where $\langle a_n \rangle$ is defined as

$$a_n = 1 - \frac{1}{2} + \frac{1}{3} - \dots + (-1)^{n+1} \frac{1}{n} \text{ is convergent.}$$

Reason: We have $a_n = 1 - \frac{1}{2} + \frac{1}{3} - \dots + (-1)^{n+1} \frac{1}{n}$

$$a_{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \dots + (-1)^{2n+1} \frac{1}{2n}$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{(-1)}{2n}$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2n-1} - \frac{1}{2n} \quad (2)$$

$$\text{Let } \langle b_n \rangle = \langle a_{2n} \rangle$$

Consider

$$b_n - b_{n-1} = \left(1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2n-1} - \frac{1}{2n} \right) -$$

$$\left(1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2n-3} - \frac{1}{2n-2} \right)$$

$$= \frac{1}{2n-1} - \frac{1}{2n} > 0$$

Put Your Own Notes

$$\Rightarrow b_n > b_{n-1}$$

$\Rightarrow \langle b_n \rangle$ is monotonically increasing sequence.

$\Rightarrow \langle a_{2n} \rangle$ is monotonically increasing sequence.

Now we show that $\langle a_{2n} \rangle$ is bounded above.

$$\begin{aligned} a_{2n} &= 1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2n-1} - \frac{1}{2n} \\ &= 1 - \left(\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots - \frac{1}{2n-1} + \frac{1}{2n} \right) \\ &= 1 - \left(\left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \dots + \left(\frac{1}{2n-2} - \frac{1}{2n-1} \right) + \frac{1}{2n} \right) \\ &= 1 - \lambda, \quad \lambda > 0 \text{ where } \lambda = \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{2n-2} - \frac{1}{2n-1} \right) + \frac{1}{2n} \end{aligned}$$

$$\Rightarrow a_{2n} \leq 1$$

$\Rightarrow \langle a_{2n} \rangle$ is convergent $\forall n \in \mathbb{N}$.

Similarly we can show that $\langle a_{2n+1} \rangle$ is convergent.

$$\text{Let } \lim_{n \rightarrow \infty} a_{2n} = l_1 \text{ and } \lim_{n \rightarrow \infty} a_{2n+1} = l_2$$

$$\text{Now consider } a_{n+1} - a_n = (-1)^{n+1} \cdot \frac{1}{n}$$

$$\text{Taking } n \rightarrow \infty \text{ we have } l_1 - l_2 = 0 \Rightarrow l_1 = l_2.$$

Since $\langle a_{2n+1} \rangle, \langle a_{2n} \rangle$ forms complementary sequences of $\langle a_n \rangle$ and converges to same limit

$\therefore \langle a_n \rangle$ is convergent.

$$(f) \text{ Let } \gamma_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \log n$$

$$\text{Then } \lim_{n \rightarrow \infty} \gamma_n = \gamma$$

3.5. Advanced Analysis of A Sequence

3.5.1. Limit Superior

Let $\langle a_n \rangle$ be a sequence of real numbers which is bounded above.

$$\text{Define } b_1 = \sup \{a_1, a_2, \dots, a_{n+1}, \dots\}$$

$$b_2 = \sup \{a_2, a_3, a_4, \dots\}$$

$\vdots$

$$b_n = \sup \{a_n, a_{n+1}, a_{n+2}, \dots\}$$

Then $\langle b_n \rangle$ is defined which is monotonically decreasing.

Then limit superior of $\langle a_n \rangle$ is denoted by $\overline{\lim} a_n$ or $\limsup_{n \rightarrow \infty} a_n$ and is defined as $\inf \{b_1, b_2, b_3, b_4, \dots\}$.

If $\langle a_n \rangle$ is not bounded above. Then limit superior of $\langle a_n \rangle$ is defined as $\overline{\lim} a_n = \limsup_{n \rightarrow \infty} a_n = +\infty$.

Example:

(a) Let $\langle a_n \rangle = \langle n \rangle$ Since it is not bounded above $\therefore \overline{\lim} a_n = +\infty$.

(b) Let $\langle a_n \rangle = \langle -n \rangle$

Clearly, $\langle a_n \rangle$ is bounded above

We have, $b_1 = \sup \{-1, -2, -3, \dots\} = -1$

$$b_2 = \sup \{-2, -3, -4, \dots\} = -2$$

$\vdots$

$$b_n = \sup \{-n, -(n+1), \dots\} = -n$$

Now $\inf \{b_1, b_2, \dots\} = -\infty$

$$\therefore \overline{\lim} a_n = -\infty$$

(c) Let $\langle a_n \rangle = \langle (-1)^n \cdot n \rangle$

Since $\langle a_n \rangle$ is not bounded above.

$$\therefore \overline{\lim} \langle a_n \rangle = +\infty.$$

(d) Let $\langle a_n \rangle = \langle (-1)^n \rangle$

Clearly $\langle a_n \rangle$ is bounded sequence.

$$(\because -1 \leq a_n \leq 1 \quad \forall n \in \mathbb{N})$$

We have, $b_1 = \sup \{-1, 1, -1, \dots\} = 1$

$$b_2 = \sup \{1, -1, \dots\} = 1$$

$\vdots$

$$b_n = \sup \{(-1)^n, (-1)^{n+1}, \dots\} = 1$$

Now $\inf \{b_1, b_2, \dots\} = 1$

$$\therefore \overline{\lim} a_n = 1.$$

(e) Let $\langle a_n \rangle = \langle \frac{1}{n} \rangle$

Clear $\langle a_n \rangle$ is bounded sequence.

$$(\because 0 < a_n \leq 1 \quad \forall n \in \mathbb{N})$$

We have, $b_1 = 1$

$$b_2 = \frac{1}{2}$$

$\vdots$

$$b_n = \frac{1}{n}$$

Now $\inf \{b_1, b_2, \dots, b_n, \dots\} = 0$

$$\therefore \overline{\lim} a_n = 0.$$

3.5.2. Limit Inferior

Let $\langle a_n \rangle$ be a sequence of real numbers which is bounded below.

$$\text{Define } c_1 = \inf \{a_1, a_2, \dots, a_n, a_{n+1}, \dots\}$$

$$c_2 = \inf \{a_2, a_3, \dots\}$$

$\vdots$

$$c_n = \inf \{a_n, a_{n+1}, \dots\}$$

Then $c_1 \leq c_2 \leq c_3 \leq \dots$ i.e., $\langle c_n \rangle$ is monotonically increasing sequence.

Now, limit inferior of $\langle a_n \rangle$ denoted by $\liminf_{n \rightarrow \infty} a_n$ or $\liminf \langle a_n \rangle$ is defined to be $\sup \{c_1, c_2, c_3, \dots\}$ and if $\langle a_n \rangle$ is not bounded below. Then we define $\liminf a_n = -\infty$

Examples:

(i) Let $\langle a_n \rangle = \langle n \rangle$, then $\langle a_n \rangle$ is bounded below by 1.

$$c_1 = \inf \{1, 2, 3, \dots\} = 1$$

$$c_2 = \inf \{2, 3, 4, \dots\} = 2$$

$\vdots$

$$c_n = \inf \{n, n+1, \dots\} = n$$

$$\text{and } \sup \{c_1, c_2, c_3, \dots\} = +\infty$$

$$\therefore \liminf a_n = \infty.$$

(ii) Let $\langle a_n \rangle = \langle -n \rangle$

Since $\langle a_n \rangle$ is not bounded below

$$\therefore \liminf a_n = -\infty.$$

(iii) Let $\langle a_n \rangle = \begin{cases} 4 & n=1 \text{ or prime} \\ n & \text{else} \end{cases}$

Clearly $\langle a_n \rangle$ is bounded below by 4.

$$c_1 = \inf \{4, 4, 4, 4, 4, 6, 4, \dots\} = 4$$

$$c_2 = \inf \{4, 4, 4, 4, 6, 4, 8, \dots\} = 4$$

$\vdots$

$$c_n = \inf \{a_n, a_{n+1}, \dots\} = 4$$

$$\Rightarrow \sup \{c_1, c_2, \dots\} = 4$$

$$\therefore \liminf a_n = 4.$$

(iv) $\langle a_n \rangle = \begin{cases} -1 - \frac{1}{n} & , \text{ when } n \text{ is even} \\ 1 + \frac{1}{n} & , \text{ when } n \text{ is odd} \end{cases}$

Then $\overline{\lim} a_n = 1$ and $\underline{\lim} a_n = -1$

Note that $[-1, 1] \cap R(U) = \emptyset$ (where $R(U)$ denote the range set of the sequence $\langle a_n \rangle$)

Note:

1. If α and β are limit inferior and limit superior of a sequence $\langle a_n \rangle$ respectively. Then $[\alpha, \beta]$ may not have any member of the sequence. (above example)
2. If the sequence is bounded. Then
 - a. The Supremum of the set of limit points of the sequence is limit superior.
 - b. The infimum of the set of limit points of the sequence is limit inferior.

Application: If $\langle a_n \rangle = (-1)^n$. Then limit superior and limit inferior of $\langle a_n \rangle$ are 1 and -1 respectively. Since set of limit points of $\langle a_n \rangle$ is $S = \{1, -1\}$ and $\sup S = 1$, $\inf S = -1$.

3.5.3. Convergent Sequence

A sequence $\langle a_n \rangle$ is said to be convergent iff limit superior is equal to the limit inferior and they exist finitely i.e. $\overline{\lim} a_n = \underline{\lim} a_n = l$ ($l \in \mathbb{R}$), then l is called the limit of the sequence $\langle a_n \rangle$.

Example: Let $\langle a_n \rangle = \langle 2012 \rangle \forall n \in \mathbb{N}$, then $\langle a_n \rangle$ is convergent sequence $\therefore \underline{\lim} a_n = \overline{\lim} a_n = 2012$.

3.5.4. Divergent Sequence

A sequence $\langle a_n \rangle$ is said to be divergent if $\underline{\lim} a_n = \overline{\lim} a_n = \infty$ (infinite) Or $\underline{\lim} a_n = \overline{\lim} a_n = -\infty$ (infinite).

Examples:

1. Let $\langle a_n \rangle = \langle n \rangle$
2. Let $\langle a_n \rangle = \langle -n \rangle$

Note:

1. Divergent sequence is unbounded from one side.
2. Unbounded sequence may not be divergent sequence.

Example: Let $\langle a_n \rangle = (-1)^n \cdot n$, then $\langle a_n \rangle$ is unbounded below and unbounded above both but it is not convergent sequence as $\underline{\lim} a_n = -\infty$ and $\overline{\lim} a_n = +\infty$.

3.5.5. Oscillatory Sequence

A sequence $\langle a_n \rangle$ is said to be oscillatory sequence if $\underline{\lim} a_n \neq \overline{\lim} a_n$.

3.5.6. Finitely Oscillatory Sequence

A sequence $\langle a_n \rangle$ is said to oscillate finitely if both $\limsup a_n$ and $\liminf a_n$ exist finitely and $\limsup a_n \neq \liminf a_n$.

Examples:

1. $\langle a_n \rangle = (-1)^n$

2. $\langle a_n \rangle = \begin{cases} -1 - \frac{1}{n} & n \text{ is even} \\ 1 + \frac{1}{n} & n \text{ is odd} \end{cases}$

3.5.7. Infinitely Oscillatory Sequence

A sequence $\langle a_n \rangle$ is said to be oscillate infinitely, if both $\limsup a_n$ and $\liminf a_n$ exist infinitely and $\limsup a_n \neq \liminf a_n$.

Examples: $\langle a_n \rangle = (-1)^n \cdot n$

3.6. Properties on Limit Superior and Limit Inferior

- Let $\langle a_n \rangle$ be a bounded sequence and $l \in \mathbb{R}$. Then $\limsup_{n \rightarrow \infty} a_n = l$ iff for each $\varepsilon > 0$
 - $a_n > l - \varepsilon$ for infinitely many values of n .
 - $a_n < l + \varepsilon$ for all except finitely many values of n .
- Let $\langle a_n \rangle$ be a bounded sequence and $l \in \mathbb{R}$. Then $\liminf_{n \rightarrow \infty} a_n = l$ iff for each $\varepsilon > 0$
 - $a_n < l + \varepsilon$ for infinitely many values of n .
 - $a_n > l - \varepsilon$ for all except finitely many values of n .
- Let $\langle a_n \rangle$ be a bounded sequence and $\limsup_{n \rightarrow \infty} a_n = M$. Then
 - There is a subsequence of $\langle a_n \rangle$ that converges to M .
 - The limit superior of no subsequence of $\langle a_n \rangle$ can exceed M .
 - If a subsequence $\langle a_{n_k} \rangle$ of $\langle a_n \rangle$ converges to number l then $l \leq M$.
- Let $\langle a_n \rangle$ be a bounded sequence and $\liminf_{n \rightarrow \infty} a_n = m$. Then
 - There is a subsequence of $\langle a_n \rangle$ that converges to m .
 - The limit inferior of no subsequence of $\langle a_n \rangle$ can be less than m .
 - If a subsequence $\langle a_{n_k} \rangle$ of $\langle a_n \rangle$ converges to a number l then $l \geq m$.

5. Let $\langle a_n \rangle$ and $\langle b_n \rangle$ are two bounded sequences. Then

(i) If $a_n \leq b_n \quad \forall n \in \mathbb{N} \Rightarrow \limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} b_n$ and

$$\liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} b_n$$

(ii) $\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n$

Equality hold when $\langle a_n \rangle$ is convergent.

(iii) $\liminf_{n \rightarrow \infty} (a_n + b_n) \geq \liminf_{n \rightarrow \infty} a_n + \liminf_{n \rightarrow \infty} b_n$

Equality hold when $\langle a_n \rangle$ is convergent.

(iv) $\limsup_{n \rightarrow \infty} (a_n \cdot b_n) \leq \limsup_{n \rightarrow \infty} a_n \cdot \limsup_{n \rightarrow \infty} b_n$

Equality hold when both $\langle a_n \rangle, \langle b_n \rangle$ are sequences of positive real numbers and $\langle a_n \rangle$ is convergent.

(v) $\liminf_{n \rightarrow \infty} (a_n \cdot b_n) \geq \liminf_{n \rightarrow \infty} a_n \cdot \liminf_{n \rightarrow \infty} b_n$

Equality hold when both $\langle a_n \rangle, \langle b_n \rangle$ are sequences of positive real numbers and $\langle a_n \rangle$ is convergent.

3.7. The Sequence of Natural Numbers

A sequence $\langle a_n \rangle$ is said to be sequence of natural number if its range set contains only natural numbers.

Results Based on Sequence of Natural Numbers

1. Every sequence of natural number has to be bounded below and has infimum in the range set.

Reason: Obviously it is bounded below by 1.

2. If sequence of natural number has limit point p . Then this p has to be natural number & $a_n = p$ for infinite many value of n i.e., there exist subsequence $\langle a_{n_k} \rangle$ which is a constant sequence such that $a_{n_k} = p$ i.e., a sequence $\langle a_n \rangle$ of $\mathbb{N}$ has a limit point $\Leftrightarrow$ it has a constant subsequence.

Reason: Since p is a limit point. So for $\epsilon = \frac{1}{2}$, $(p - \epsilon, p + \epsilon)$ contains infinite member of $\mathbb{N} \Rightarrow a_n = p$ for infinite many values of $n \Rightarrow \exists$ a constant subsequence $\langle a_{n_k} \rangle = \langle p \rangle$ which converges to p .

Conversely: If the sequence $\langle a_n \rangle$ of $\mathbb{N}$ has constant subsequence say $\langle a_{n_k} \rangle = \langle p \rangle$

Then p is a limit point of $\langle a_n \rangle$.

3. Sequence of natural number is convergent iff it is eventually constant.

Reason: Let $\langle a_n \rangle$ be a sequence of natural numbers such that

$$\lim_{n \rightarrow \infty} a_n = l.$$

So, for $\epsilon = \frac{1}{2} \quad \exists m \in \mathbb{N}$ Such that

$$|a_n - l| < \frac{1}{2} \quad \forall n \geq m$$

$$\Rightarrow a_n = l \quad \forall n \geq m.$$

$\Rightarrow \langle a_n \rangle$ is eventually sequence.

Conversely: Suppose that $\langle a_n \rangle$ is eventually constant sequence. Then there exist $m \in \mathbb{N}$ such that $a_n = p \quad \forall n \geq m \quad (p \in \mathbb{N})$

$\therefore$ for any $\epsilon > 0$

$$|a_n - p| < \epsilon \quad \forall n \geq m$$

$\Rightarrow \langle a_n \rangle$ is convergent and converges to p .

4. If a sequence of natural numbers is not divergent then it has constant subsequence.

Reason: Since sequence is not divergent then either it is convergent or Oscillates. If sequence oscillates then it is either Oscillates finitely or Oscillates infinitely. In both the cases, the sequence has limit point and hence it has a constant subsequence.

If sequence is convergent then there exist a constant subsequence.

3.8. Cauchy Sequence

A sequence $\langle a_n \rangle$ is said to be a **Cauchy sequence** if for any $\epsilon > 0$, there exists a positive integer m such that $|a_n - a_m| < \epsilon$, whenever $n \geq m$.

Example: The sequence $\langle a_n \rangle = \left\langle 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots \right\rangle$ is a Cauchy sequence.

Reason: Let $\epsilon > 0$ be given and if $n > m$, consider $|a_n - a_m| =$

$$\left| \frac{1}{n} - \frac{1}{m} \right| = \frac{m-n}{nm} < \frac{1}{n} < \epsilon, \text{ provided } m > \frac{1}{\epsilon}.$$

Let m be a positive integer greater than $1/\epsilon$. Then $|a_n - a_m| < \epsilon \quad \forall n \geq m$

Hence $\left\langle \frac{1}{n} \right\rangle$ is a Cauchy sequence.

Put Your Own Notes

Example: The sequence $\langle a_n \rangle = \left\langle \frac{n}{n+1} \right\rangle$ is a Cauchy sequence.

Reason: Let $\varepsilon > 0$ be given and if $n > m$, Consider

$$|a_n - a_m| = \left| \frac{n}{n+1} - \frac{m}{m+1} \right| = \frac{n-m}{(n+1)(m+1)} < \frac{1}{m}$$

$$\Rightarrow |a_n - a_m| < \frac{1}{m} < \varepsilon, \text{ provided } m > \frac{1}{\varepsilon}.$$

Let m be a positive integer greater than $1/\varepsilon$.

Then $|a_n - a_m| < \varepsilon$, for $n \geq m$.

Hence $\left\langle \frac{n}{n+1} \right\rangle$ is a Cauchy sequence.

Example: The sequence $\langle a_n \rangle = \langle n^2 \rangle$ is not a Cauchy sequence.

Reason: Let $n > m$, consider $|a_n - a_m| = n^2 - m^2 = (n-m)(n+m) > 2m > 1$, whatever m may be.

For $\varepsilon = \frac{1}{2}$, we see that there is no positive integer m such that $|n^2 - m^2| < \varepsilon$ for all $n \geq m$.

3.8.1. Algebra of Cauchy Sequences

Let $\langle a_n \rangle, \langle b_n \rangle$ are two Cauchy sequence. Then

1. $\langle a_n \rangle + \langle b_n \rangle = \langle a_n + b_n \rangle$ is a Cauchy sequence.
2. $\langle a_n \rangle - \langle b_n \rangle = \langle a_n - b_n \rangle$ is a Cauchy sequence.
3. $\langle a_n \rangle \cdot \langle b_n \rangle = \langle a_n \cdot b_n \rangle$ is a Cauchy sequence.
4. If each $a_n \neq 0 \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n \neq 0$ then $\left\langle \frac{1}{a_n} \right\rangle$ is Cauchy sequence.

Remark: If $\lim_{n \rightarrow \infty} a_n = 0$ and each $a_n \neq 0$ where $\langle a_n \rangle$ is a Cauchy sequence.

Then $\left\langle \frac{1}{a_n} \right\rangle$ may not be Cauchy sequence.

Example: Let $\langle a_n \rangle = \left\langle \frac{1}{n} \right\rangle$. Then $\langle a_n \rangle$ is a Cauchy sequence but $\left\langle \frac{1}{a_n} \right\rangle = \langle n \rangle$ is not a Cauchy sequence.

Cauchy's Criterion for Convergence

Statement: A sequence $\langle a_n \rangle$ of real numbers is convergent iff it is a Cauchy sequence.

3.8.2. Cauchy's General Principle of Convergence (CGPC)

A necessary and sufficient condition for a sequence $\langle a_n \rangle$ to be convergent is that to each $\varepsilon > 0$, there corresponds a positive integer m such that

$$|a_{n+p} - a_n| < \varepsilon \quad \forall n \geq m, p \geq 0.$$

Note: This principle is used where the limit l of sequence is not known no guess can be made of the same. This principle which involves only the terms of the sequence is useful for determining whether a sequence convergent or not.

Example:

1. The sequence $\langle a_n \rangle$, where $a_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$ is not convergent.

Reason: Suppose that the given sequence is convergent.

By Cauchy's General Principal of Convergence, for $\varepsilon = \frac{1}{2}$, there exists a

positive integer m such that $|a_n - a_m| < \frac{1}{2} \forall n \geq m$.

In particular for $n = 2m$, $|a_{2m} - a_m| < \frac{1}{2}$ (1)

$$\text{Now } |a_{2m} - a_m| = \left| \left(1 + \frac{1}{2} + \dots + \frac{1}{m} + \frac{1}{m+1} + \dots + \frac{1}{2m} \right) - \left(1 + \frac{1}{2} + \dots + \frac{1}{m} \right) \right|$$

$$= \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{2m}$$

$$> \frac{1}{2m} + \frac{1}{2m} + \dots + \frac{1}{2m} \text{ (} m \text{ terms)} = \frac{1}{2m} \cdot m = \frac{1}{2}$$

$$\left(\because m+1 < m+m \Rightarrow \frac{1}{m+1} > \frac{1}{2m} \text{ etc.} \right)$$

$$\therefore |a_{2m} - a_m| > \frac{1}{2}$$

This contradicts (1).

Hence the given sequence is not convergent.

2. The sequence $\langle a_n \rangle$, where $a_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}$ is a convergent sequence.

Reason: By Cauchy's criterion of convergence, a real sequence converges iff it is a Cauchy sequence. We shall show that the given sequence $\langle a_n \rangle$ is a Cauchy sequence, so that $\langle a_n \rangle$ is convergent.

For $n \geq m$, we have

$$|a_n - a_m| = \left| \left(1 + \frac{1}{1!} + \dots + \frac{1}{m!} + \frac{1}{(m+1)!} + \frac{1}{(m+2)!} + \dots + \frac{1}{n!} \right) - \left(1 + \frac{1}{1!} + \dots + \frac{1}{m!} \right) \right|$$

$$= \frac{1}{(m+1)!} + \frac{1}{(m+2)!} + \dots + \frac{1}{n!}$$

$$< \frac{1}{2^m} + \frac{1}{2^{m+1}} + \dots + \frac{1}{2^{n-1}} \quad \left(\because n! \geq 2^{n-1} \forall n \right)$$

Put Your Own Notes

$$= \frac{1}{2^m} \left[1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{(n-m)-1}} \right]$$

$$= \frac{1}{2^m} \frac{1 - \left(\frac{1}{2}\right)^{n-m}}{1 - \left(\frac{1}{2}\right)} = \frac{1}{2^{m-1}} - \frac{1}{2^{n-1}} \leq \frac{1}{2^{m-1}}$$

$$\left(\because n \geq m \Rightarrow n-1 \geq m-1 \Rightarrow 2^{n-1} \geq 2^{m-1} \Rightarrow \frac{1}{2^{n-1}} \leq \frac{1}{2^{m-1}} \right)$$

For any $\varepsilon > 0$, $|a_n - a_m| < \varepsilon$, if $\frac{1}{2^{m-1}} < \varepsilon$ or $2^{m-1} > \frac{1}{\varepsilon}$

or $(m-1)\log 2 > \log(1/\varepsilon)$ or $m > \left\{ \log(1/\varepsilon)(\log 2)^{-1} + 1 \right\} = N$, (say)

$\therefore |a_n - a_m| < \varepsilon \forall n \geq m$, where m is a positive integer $\geq N$.

Hence $\langle a_n \rangle$ is a Cauchy sequence and so $\langle a_n \rangle$ is convergent.

3. Let $\langle a_n \rangle$ be a sequence of real numbers and for each $n \in \mathbb{N}$, define

$$s_n = a_1 + a_2 + \dots + a_n, t_n = |a_1| + |a_2| + \dots + |a_n| \quad (1)$$

If $\langle t_n \rangle$ is a Cauchy sequence, then so is $\langle s_n \rangle$.

Reason: Since $\langle t_n \rangle$ is a Cauchy sequence, for any $\varepsilon > 0$, there exists a positive integer m such that $|t_n - t_m| < \varepsilon \forall n \geq m$

$$\text{or } \left| \{ |a_1| + \dots + |a_m| + |a_{m+1}| + \dots + |a_n| \} - \{ |a_1| + |a_2| + \dots + |a_m| \} \right| < \varepsilon \forall n \geq m$$

$$\text{or } \| |a_{m+1}| + |a_{m+2}| + \dots + |a_n| \| < \varepsilon \forall n \geq m. \quad (2)$$

Now $|s_n - s_m| = |a_{m+1} + a_{m+2} + \dots + a_n|$, by (1)

$$\leq |a_{m+1}| + |a_{m+2}| + \dots + |a_n|$$

$$< \varepsilon \forall n \geq m, \text{ by (2)}$$

$$\therefore |s_n - s_m| < \varepsilon \forall n \geq m.$$

Hence $\langle s_n \rangle$ is a Cauchy sequence.

Some Important Theorems

1. **Cauchy's First Theorem on limits:** Let $\lim_{n \rightarrow \infty} a_n = 0$, then

$$\lim_{n \rightarrow \infty} \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right) = 0$$

Put Your Own Notes

In General: Let $\lim_{n \rightarrow \infty} a_n = l$. Then $\lim_{n \rightarrow \infty} \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right) = l$

Example: $\lim_{n \rightarrow \infty} \frac{1}{n} \left[1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right] = 0.$

Reason: Let $a_k = \frac{1}{k}$. Then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$

Since $\lim_{n \rightarrow \infty} a_n = 0$, so by Cauchy's first theorem on limits

$$\lim_{n \rightarrow \infty} \frac{1}{n} (a_1 + a_2 + \dots + a_n) = 0.$$

Hence $\lim_{n \rightarrow \infty} \frac{1}{n} \left[1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right] = 0$

Note: Converse of the above theorem need not be true.

Example: $\langle a_n \rangle = \langle -1, 1, -1, 1, \dots \rangle$ is not convergent and
 $a_1 + a_2 = 0, a_1 + a_2 + a_3 = -1, a_1 + a_2 + a_3 + a_4 = 0$, etc.

Now $\frac{a_1 + a_2 + \dots + a_n}{n} = \begin{cases} 0, & \text{if } n \text{ is even} \\ -\frac{1}{n}, & \text{if } n \text{ is odd} \end{cases}$

Hence $\lim_{n \rightarrow \infty} \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right) = 0$, but $\lim_{n \rightarrow \infty} a_n \neq 0.$

2. **Cauchy's Second Theorem on Limits:** If $\langle a_n \rangle$ converges to $l (\neq 0)$ and

$a_n > 0$ then $\lim_{n \rightarrow \infty} (a_1 a_2 \dots a_n)^{1/n} = l$

Example: Let $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$. Then

$$\lim_{n \rightarrow \infty} \left[\frac{2}{1} \left(\frac{3}{2} \right)^2 \left(\frac{4}{3} \right)^3 \dots \left(\frac{n+1}{n} \right)^n \right]^{1/n} = e.$$

Reason: Let $a_n = \left(1 + \frac{1}{n} \right)^n$, then $a_n > 0 \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = e > 0.$

By Cauchy's second theorem on limits, we have

$$\lim_{n \rightarrow \infty} (a_1 a_2 a_3 \dots a_n)^{1/n} = e.$$

Hence $\lim_{n \rightarrow \infty} \left[\frac{2}{1} \left(\frac{3}{2} \right)^2 \left(\frac{4}{3} \right)^3 \dots \left(\frac{n+1}{n} \right)^n \right]^{1/n} = e$, using (1).

3. If all the terms of a sequence $\langle a_n \rangle$ are positive and if $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$

exist. Then $\lim_{n \rightarrow \infty} (a_n)^{1/n}$ also exist and the two limits are equal. i.e.,

$$\lim_{n \rightarrow \infty} (a_n)^{1/n} = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}, \text{ provided the later limit exist.}$$

4. **Cesaro's Theorem:** Let $\lim_{n \rightarrow \infty} a_n = a$ and $\lim_{n \rightarrow \infty} b_n = b$ where $a, b \in \mathbb{R}$

$$\text{, then } \lim_{n \rightarrow \infty} \frac{a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1}{n} = a.b$$

5. **Sandwich Theorem:** Let $\langle a_n \rangle, \langle b_n \rangle, \langle c_n \rangle$ are sequence of real number such that $a_n \leq b_n \leq c_n \quad \forall n \in \mathbb{N}$

$$\text{and } \lim_{n \rightarrow \infty} a_n = l = \lim_{n \rightarrow \infty} c_n \text{ then } \lim_{n \rightarrow \infty} b_n = l.$$

6. If $a_n \geq 0 \quad \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = l$, then $l \geq 0$.

7. If $\langle a_n \rangle, \langle b_n \rangle$ be two sequence such that $a_n \leq b_n \quad \forall n \in \mathbb{N}$ then $\lim a_n \leq \lim b_n$.

8. If $\langle a_n \rangle$ be a sequence of positive real numbers such that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = l$, where $l < 1$ then $\lim_{n \rightarrow \infty} a_n = 0$.

Application:

- (i) Let $\langle a_n \rangle = \langle 2^{-n} \cdot n^2 \rangle$, then $\lim_{n \rightarrow \infty} 2^{(-n)} \cdot n^2 = 0$

$$\text{Reason: We have } a_n = \frac{n^2}{2^n}, \quad a_{n+1} = \frac{(n+1)^2}{2^{n+1}}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{n+1}{n} \right)^2$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} \left(1 + \frac{1}{n} \right)^2 = \frac{1}{2} < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} 2^{(-n)} \cdot n^2 = 0$$

(ii) If $|x| < 1$ and $m \in \mathbb{N}$. Then $\lim_{n \rightarrow \infty} \frac{m(m-1)(m-2)\dots(m-n+1)}{n!} x^n = 0$.

Reason: Let $a_n = \frac{m(m-1)\dots(m-n+1)}{n!} x^n$, so that

$$a_{n+1} = \frac{m(m-1)\dots(m-n+1)(m-n)}{(n+1)!} x^{n+1}$$

$$\text{Now } \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \left\{ \frac{(m-n)}{(n+1)!} \cdot n! \cdot x \right\}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{m-n}{n+1} \right) x = \lim_{n \rightarrow \infty} \frac{\left(\frac{m}{n} \right) - 1}{1 + \left(\frac{1}{n} \right)} \cdot x = -x$$

$$\therefore \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = -x, \text{ where } |-x| = |x| < 1.$$

Hence $\lim_{n \rightarrow \infty} a_n = 0$,

(iii) If $p > 0$, then $\lim_{n \rightarrow \infty} \frac{n^k}{(1+p)^n} = 0$, k being a fixed real number.

Reason: Let $a_n = \frac{n^k}{(1+p)^n}$, so that $a_{n+1} = \frac{(n+1)^k}{(1+p)^{n+1}}$.

$$\therefore \frac{a_{n+1}}{a_n} = \left(\frac{n+1}{n} \right)^k \cdot \frac{1}{1+p} = \left(1 + \frac{1}{n} \right)^k \cdot \frac{1}{1+p}$$

$$\text{Now } \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{1}{(1+p)} \cdot \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^k = \frac{1}{1+p},$$

Where $\frac{1}{1+p} < 1$ ($\because 1+p > 1$ as $p > 0$)

Hence $\lim_{n \rightarrow \infty} a_n = 0$.

9. If $\langle a_n \rangle$ be a sequence of positive real numbers such that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = l > 1$

, then $\langle a_n \rangle$ is not a bounded sequence and hence not convergent

10. Let $\lim_{n \rightarrow \infty} a_n = a$, $\lim_{n \rightarrow \infty} b_n = b$, and $\langle S_n \rangle$ & $\langle T_n \rangle$ are two sequences, where

$S_n = \max\{a_n, b_n\}$ and $T_n = \min\{a_n, b_n\}$. Then the sequences $\langle S_n \rangle$ and $\langle T_n \rangle$ are convergent and $\lim_{n \rightarrow \infty} S_n = \max\{a, b\}$, $\lim_{n \rightarrow \infty} T_n = \min\{a, b\}$.

Reason: For any two real numbers x and y , we know

$$\max\{x, y\} = \frac{1}{2}(x + y) + \frac{1}{2}|x - y| \quad (1)$$

$$\min\{x, y\} = \frac{1}{2}(x + y) - \frac{1}{2}|x - y| \quad (2)$$

Taking $x = a_n$, $y = b_n$ in (1), we obtain

$$S_n = \max\{a_n, b_n\} = \frac{1}{2}(a_n + b_n) + \frac{1}{2}|a_n - b_n|, \text{ using (1).}$$

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} S_n &= \frac{1}{2} \lim_{n \rightarrow \infty} (a_n + b_n) + \frac{1}{2} \lim_{n \rightarrow \infty} |a_n - b_n| \\ &= \frac{1}{2} \left(\lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n \right) + \frac{1}{2} \lim_{n \rightarrow \infty} |c_n|, \quad c_n = a_n - b_n \\ &= \frac{1}{2}(a + b) + \frac{1}{2} \lim_{n \rightarrow \infty} |c_n| \end{aligned} \quad (3)$$

$$\text{We have } \lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} (a_n - b_n) = \lim_{n \rightarrow \infty} a_n - \lim_{n \rightarrow \infty} b_n = a - b$$

$$\therefore \lim_{n \rightarrow \infty} c_n = a - b \Rightarrow \lim_{n \rightarrow \infty} |c_n| = |a - b|,$$

$$\text{From (3), } \lim_{n \rightarrow \infty} S_n = \frac{1}{2}(a + b) + \frac{1}{2}|a - b| = \max\{a, b\}.$$

$$\text{Similarly, } \lim_{n \rightarrow \infty} T_n = \min\{a, b\}$$

11. Consider a sequence $\langle a_n \rangle$, where $a_n = \sqrt{n}(\sqrt{n+1} - \sqrt{n})$. Then $\langle a_n \rangle$ converges.

$$\text{Reason: We have } a_n = \frac{\sqrt{n}(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{(\sqrt{n+1} + \sqrt{n})} \quad (\text{Rationalize})$$

$$\text{or } a_n = \frac{\sqrt{n(n+1-n)}}{\sqrt{n+1} + \sqrt{n}} = \frac{\sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1}$$

Hence $\lim_{n \rightarrow \infty} a_n = \frac{1}{\sqrt{1+0} + 1} = \frac{1}{2}$ and so $\langle a_n \rangle$ converges.

12. Let $\langle a_n \rangle$ be a sequence such that $a_n \rightarrow a$. Then $a_n^2 \rightarrow a^2$.

Reason: Since $\langle a_n \rangle$ is convergent, $\langle a_n \rangle$ is bounded and so there exists

some $k > 0$ such that $|a_n| \leq k \forall n \in \mathbb{N}$.

$$\begin{aligned} \text{Now } |a_n^2 - a^2| &= |(a_n - a)(a_n + a)| = |a_n - a| |a_n + a| \\ &\leq |a_n - a| \{|a_n| + |a|\} \\ &\leq |a_n - a| \{k + |a|\} |a_n - a| \quad \forall n \in \mathbb{N} \end{aligned} \quad (1)$$

Let $\varepsilon > 0$ be given. Since $a_n \rightarrow a$, there exists a positive integer m such that

$$|a_n - a| < \frac{\varepsilon}{k + |a|} \quad \forall n \geq m \quad (2)$$

From (1) and (2), we see that

$$|a_n^2 - a^2| < \varepsilon \quad \forall n \geq m$$

Hence $\lim_{n \rightarrow \infty} a_n^2 = a^2$

However, the converse may not be true.

For example, let $\langle a_n \rangle = \langle (-1)^n \rangle$. Then

$\langle a_n^2 \rangle = \langle 1, 1, 1, \dots \rangle$, which converges to 1.

but $\langle a_n \rangle$ does not converge to 1.

13. Let $\langle a_n \rangle$ be a sequence such that $a_n^2 \rightarrow a^2$, then $|a_n| \rightarrow |a|$ as $n \rightarrow \infty$.

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CHAPTER 4

SERIES OF REAL NUMBERS

4.1. Definition

We know about arithmetic and geometric series etc. A series of n terms is denoted by the expression $u_1 + u_2 + \dots + u_n$ or $\sum_{i=1}^n u_i$. If the series has no last term, then such series is called an infinite series.

4.1.1 Infinite Series:

Let $\langle a_n \rangle$ be a sequence of real numbers, then $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots + a_{n+1} + \dots$ i.e. infinite sum of the members of the sequence, is defined as series of real numbers. It is also denoted by $\sum a_n$.

Example:

(a) $\sum \frac{1}{n}$, here $\langle a_n \rangle = \left\langle \frac{1}{n} \right\rangle \quad \forall n \in \mathbb{N}$

(b) $\sum \sin\left(\frac{1}{n^2}\right)$, here $\langle a_n \rangle = \left\langle \sin\left(\frac{1}{n^2}\right) \right\rangle \quad \forall n \in \mathbb{N}$

4.1.2 Sequence of Partial Sums

Suppose $\sum a_n$ is an infinite series then we define a sequence $\langle S_n \rangle$ as follows:

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$\vdots$$

$$S_n = a_1 + a_2 + \dots + a_n, \text{ and so on.}$$

The sequence $\langle S_n \rangle$ is called the sequence of partial sums of the series $\sum a_n$

4.1.3 Convergent Series

A series $\sum a_n$ is said to be convergent, if the sequence $\langle S_n \rangle$ of partial sums of $\sum a_n$ is convergent and if $\lim_{n \rightarrow \infty} S_n = S$, then S is called the sum of the

series $\sum a_n$ and then we write it as $S = \sum_{n=1}^{\infty} a_n$.

4.1.4 Divergent Series

The series $\sum a_n$ is said to be divergent, if the sequence $\langle S_n \rangle$ of partial sums of $\sum a_n$ is divergent.

4.1.5 Oscillatory Series

The series $\sum a_n$ is said to oscillatory, if the sequence $\langle S_n \rangle$ of partial sums of $\sum a_n$ oscillates.

4.2. Necessary condition for convergence of a series

If the series $\sum a_n$ converge, then $\lim_{n \rightarrow \infty} a_n = 0$.

Remark:

(a) Converse of the theorem need not be true.

Example: Let $\sum a_n = \sum \frac{1}{n}$, then $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ but $\sum \frac{1}{n}$ is not convergent.

(b) If $\lim_{n \rightarrow \infty} a_n \neq 0$ then $\sum a_n$ cannot converge.

Examples:

(a) The series $1+r+r^2+\dots (r>0)$ converges if $r<1$ and diverges if $r \geq 1$.
(Geometric Series)

(b) The series $\sum (-1)^{n-1}$ oscillates.

Solution: We see that

$$S_1 = 1, S_2 = 1-1=0, S_3 = 1-1+1=1,$$

$$S_4 = 1-1+1-1=0, S_5 = 1 \text{ and so on.}$$

$$\therefore \langle S_n \rangle = \langle 1, 0, 1, 0, 1, 0, \dots \rangle, \text{ which oscillates.}$$

Hence the given series oscillates.

(c) Test for convergence of the series :

(i) $\sum_{n=1}^{\infty} (-1)^n n,$

(ii) $\sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{3}\right)$

Solution:

(i) We see that $\sum_{n=1}^{\infty} (-1)^n n = -1+2-3+4-5+6\dots$

$$\therefore S_1 = -1, S_2 = -1+2=1, S_3 = -1+2-3=-2,$$

$$S_4 = -1+2-3+4=2, S_5 = -3, S_6 = 3 \text{ etc.}$$

$$\Rightarrow \langle S_n \rangle = \langle -1, 1, -2, 2, -3, 3, \dots \rangle, \text{ which is not bounded.}$$

Hence $\langle S_n \rangle$ is not convergent and so $\sum (-1)^n n$ is not convergent.

(ii) $\sum_{n=1}^{\infty} \sin \frac{n\pi}{3} = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + 0 - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} + 0 + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + 0 \dots$

The sequence $\langle S_n \rangle$ of partial sums of the above series is

$$\left\langle \frac{\sqrt{3}}{2}, \sqrt{3}, \sqrt{3}, \frac{\sqrt{3}}{2}, 0, 0, \frac{\sqrt{3}}{2}, \sqrt{3}, \dots \right\rangle$$

$$\text{Clearly, } \limsup S_n = \sqrt{3} \text{ and } \liminf S_n = 0.$$

It follows that $\langle S_n \rangle$ is not convergent. Hence the given series is not convergent.

4.3. Telescopic Series

Let $\langle a_n \rangle$ be a sequence of real numbers.

Define $b_n = a_n - a_{n+1}$ and $c_n = a_{n+1} - a_n$

Then $\sum b_n$ and $\sum c_n$ are called telescopic series.

If $\langle S_n \rangle$ denote the sequence of partial sum of the series $\sum b_n$

$$\text{Then } S_n = \sum_{r=1}^n b_r$$

$$= b_1 + b_2 + \dots + b_n$$

$$= a_1 - a_2 + a_2 - a_3 + \dots + a_n - a_{n+1}$$

$$= a_1 - a_{n+1}$$

Similarly if $\langle t_n \rangle$ denote the sequence of partial sum of the series $\sum c_n$

$$\text{Then } t_n = a_{n+1} - a_n$$

Thus $\langle S_n \rangle$ and $\langle t_n \rangle$ are convergent iff $\langle a_n \rangle$ is convergent.

Thus $\sum b_n$ and $\sum c_n$ are convergent iff $\langle a_n \rangle$ is convergent.

Example: The series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$ is convergent.

Solution: We have,

$$\begin{aligned} S_n &= \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} \end{aligned}$$

$$\therefore \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1 - 0 = 1.$$

Hence the given series is convergent.

4.4. Cauchy's General Principal of Convergence

A necessary and sufficient condition for a series $\sum a_n$ to converge is that for each $\varepsilon > 0$, there exists a positive integer m , such that $|a_{m+1} + a_{m+2} + \dots + a_n| < \varepsilon$, for all $n \geq m$.

Pringsheim's Theorem: If a series $\sum u_n$ of positive monotonic decreasing terms converges then not only $u_n \rightarrow 0$ but also $nu_n \rightarrow 0$ as $n \rightarrow \infty$.

Reason: We know that for a convergent series, for any $\varepsilon > 0$, a positive integer N exists such that $|u_{m+1} + u_{m+2} + \dots + u_{m+p}| < \frac{\varepsilon}{2}$, $\forall m \geq N, p \geq 1$

Let us choose $m + p = n > 2N$ and $m = \left\lfloor \frac{n}{2} \right\rfloor$, the greatest integer not greater

$$\text{than } \frac{n}{2} \quad u_{m+1} + u_{m+2} + \dots + u_n < \frac{\varepsilon}{2}$$

But $\sum u_n$ is positive monotonic decreasing so

$$(n-m)u_n < u_{m+1} + u_{m+2} + \dots + u_n < \frac{\varepsilon}{2} \text{ or } \frac{1}{2}nu_n < \frac{1}{2}\varepsilon \text{ i.e. } nu_n < \varepsilon, \forall n \geq N.$$

Hence $\lim nu_n = 0$.

Note: The condition $nu_n \rightarrow 0$ is only a necessary one not sufficient condition for the convergence of the present type of series. If nu_n does not tend to zero then the series $\sum u_n$ is certainly divergent, for example, the harmonic series $\sum \frac{1}{n}$ must diverge because it has positive monotonic decreasing terms and $n \cdot \frac{1}{n}$ does not tend to zero. However, $nu_n \rightarrow 0$ does not imply anything as to the possible convergence of $\sum u_n$ for example, Abel's series $\sum \frac{1}{n \log n}$ diverges although it has positive monotonic decreasing terms and $nu_n \rightarrow 0$.

Example: The series $\sum \frac{n}{n+1} = \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \dots$ is not convergent.

Solution: We have $u_n = \frac{n}{n+1} = \frac{1}{1 + \frac{1}{n}}$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = \frac{1}{1+0} = 1 \neq 0$$

Hence the given series is not convergent.

Example: The series $\sqrt{\frac{1}{4}} + \sqrt{\frac{2}{6}} + \dots + \sqrt{\frac{n}{2(n+1)}} + \dots$ is not convergent.

Solution: We have

$$u_n = \sqrt{\frac{n}{2(n+1)}} = \frac{1}{\sqrt{2}} \sqrt{\frac{1}{1 + \frac{1}{n}}}$$

$$\therefore \lim_{n \rightarrow \infty} u_n = \frac{1}{\sqrt{2}} \neq 0.$$

Hence, the given series does not converge.

Example: Test for convergence of the series $\sum \cos\left(\frac{1}{n}\right)$.

Solution: $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \cos \frac{1}{n} = 1 \neq 0$

Hence the given series does not converge.

Example: The series $\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^{1/n}$ does not converge.

Solution: Let $u_n = \left(\frac{1}{n}\right)^{1/n}$, so that $\log u_n = \frac{1}{n} \log\left(\frac{1}{n}\right)$

$$= \log u_n = \frac{\log 1 - \log n}{n} = \frac{-\log n}{n}$$

$$\therefore \lim_{n \rightarrow \infty} \log u_n = -\lim_{n \rightarrow \infty} \frac{\log n}{n}, \text{ which is } \frac{\infty}{\infty} \text{ form}$$

$$= -\lim_{n \rightarrow \infty} \frac{1/n}{1}$$

(by L'Hospital's Rule)

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$$= -\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

$$\text{Now } \lim_{n \rightarrow \infty} \log u_n = 0 \Rightarrow \log \left(\lim_{n \rightarrow \infty} u_n \right) = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} u_n = e^0 = 1 \neq 0.$$

Hence $\sum u_n$ is not convergent.

4.5. Series of Positive Real Numbers

Series with positive terms are the simplest and the most important type of series one comes across. The simplicity arises mainly from the fact that the sequence of its partial sums is monotonically increasing.

Let $\sum u_n$ be an infinite series of positive terms and $\{S_n\}$ be the sequence of its partial sums so that

$$S_n = u_1 + u_2 + \dots + u_n \geq 0, \forall n$$

$$\therefore S_n - S_{n-1} = u_n \geq 0$$

$$\Rightarrow S_n \geq S_{n-1}, \forall n > 1$$

Thus the sequence $\{S_n\}$ of partial sums of a series of positive terms is a monotonically increasing sequence.

Since a monotonic increasing sequence can either converge, or diverge to ∞ , but cannot oscillate, therefore, there are only two possibilities for a positive term series, it may either converge or diverge to $+\infty$.

Remark:

- A positive term series converges iff the sequence of its partial sum is bounded above.
- The sequence of partial sum of a series with negative terms can be shown to be monotonic decreasing and hence a series with negative terms converges iff the sequence of its partial sum is bounded below.
- It may similarly be seen that a series of negative terms can either converge or diverge to $-\infty$.
- A series $\sum_{n=1}^{\infty} u_n$ whose terms are not necessarily positive may fail to be convergent even if the sequence $\langle S_n \rangle$ is bounded above.

For example consider $u_n = (-1)^n$ so that we have

$$S_n = \begin{cases} -1 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

The sequence $\langle S_n \rangle$, even though bounded above, is obviously not convergent and as such the series is not convergent. The sequence $\langle S_n \rangle$ has two limit points viz. -1 and 0 .

It should be seen that, in general, boundedness of the sequence of partial sums of a series is only a necessary but not a sufficient condition for the convergence of the series $\sum u_n$, and it is only for positive term series that this condition of boundedness of the sequence of its partial sums is

as well a sufficient condition for the convergence of the corresponding series.

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4.5.1. Test for Convergence of Positive Terms Series

1. **First Comparison Test:** If $\sum u_n$ and $\sum v_n$ are positive terms series, $k > 0$ and $\exists m \in \mathbb{N}$ such that $u_n \leq kv_n \quad \forall n \geq m$

(a) $\sum v_n$ converges $\Rightarrow \sum u_n$ converges.

(b) $\sum u_n$ diverges $\Rightarrow \sum v_n$ diverges.

Example: Test for convergence the series $1 + \frac{1}{2^2} + \frac{1}{3^3} + \frac{1}{4^4} + \dots + \frac{1}{n^n} + \dots$

Solution: Clearly, $n^n > 2^n$ for $n > 2$.

$$\therefore \frac{1}{n^n} < \frac{1}{2^n} \text{ for } n > 2$$

Since $\sum \frac{1}{2^n} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots$ is a geometric series with common ratio $\frac{1}{2} < 1$, so $\sum 1/2^n$ is convergent.

Hence by First Comparison Test $\sum \frac{1}{n^n}$ is convergent.

Example: The series $\sum e^{-n^2}$ converges.

Solution: We know $e^x > x$ for $x > 0$. Then

$$e^{n^2} > n^2 \Rightarrow \frac{1}{e^{n^2}} < \frac{1}{n^2} \Rightarrow e^{-n^2} < \frac{1}{n^2} \quad \forall n.$$

Since $\sum \frac{1}{n^2}$ converges, so by First Comparison Test, $\sum e^{-n^2}$ also converges.

Example: Test for convergence the series:

(a) $\sum_{n=2}^{\infty} \frac{1}{\log n},$

(b) $\sum_{n=2}^{\infty} \frac{1}{n^2 \log n}$

Solution:

(a) We know $\log n < n$ for all $n \geq 2$.

$$\therefore \frac{1}{\log n} > \frac{1}{n} \quad \forall n \geq 2 \text{ or } \frac{1}{n} < \frac{1}{\log n} \quad \forall n \geq 2.$$

Since $\sum \frac{1}{n}$ diverges, so by First Comparison Test, $\sum_{n=2}^{\infty} \frac{1}{\log n}$ diverges.

(b) We know $\frac{1}{n^2 \log n} < \frac{1}{n^2} \quad \forall n \geq 3$.

Since $\sum \frac{1}{n^2}$ converges, so by First Comparison Test,

$$\sum_{n=2}^{\infty} \frac{1}{n^2 \log n} \text{ converges.}$$

Example: Test for convergence the series:

(a) $1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} + \dots$

(b) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n!}}$

Solution:

(a) We know that $n! \geq 2^{n-1} \quad \forall n \geq 2$.

$$\therefore \frac{1}{n!} \leq \frac{1}{2^{n-1}} \quad \forall n \geq 2.$$

Now $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots$ being a geometric series with common ratio $\frac{1}{2} < 1$ is convergent. Hence, by First Comparison Test, $\sum \frac{1}{n!}$ is convergent.

(b) We know $\frac{1}{\sqrt{n!}} \leq \frac{1}{2^{\frac{1}{2}(n-1)}} \quad \forall n \geq 2$.

The series on the Right Hand Side being a geometric series with common ratio $\frac{1}{\sqrt{2}} < 1$ is convergent. Hence $\sum \frac{1}{\sqrt{n!}}$ is convergent.

Example: Test for convergence the series $\sum_{n=1}^{\infty} \frac{1}{3^n + x}$, for all positive values of x .

Solution: Clearly $3^n + x > 3^n$, since $x > 0$

$$\Rightarrow \frac{1}{3^n + x} < \frac{1}{3^n} \quad \forall n \in \mathbb{N}$$

Now $\sum \frac{1}{3^n}$ being a geometric series with common ratio $\frac{1}{3} < 1$ is convergent. Hence, by First Comparison Test, the given series is convergent for all $x > 0$.

2. **Second Comparison Test:** If $\sum u_n$ and $\sum v_n$ are two positive term series such that $\frac{u_n}{u_{n+1}} \geq \frac{v_n}{v_{n+1}} \quad \forall n \geq m$ then

(i) $\sum v_n$ converges $\Rightarrow \sum u_n$ converges.

(ii) $\sum u_n$ diverges $\Rightarrow \sum v_n$ diverges.

3. **p Series Test:** The series $\sum \frac{1}{n^p}$ is convergent iff $p > 1$.

4. **Limit form Comparison Test:** Let $\sum u_n$ and $\sum v_n$ be two positive term series such that $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = l$, (l is finite and non-zero). Then $\sum u_n$ and $\sum v_n$ converge or diverge together.

Remark: If $l = 0$ or $l = \infty$, then the conclusion of the above test may not hold good.

Example:

(a) Let $\sum u_n = \sum \frac{1}{n^2}$ and $\sum v_n = \sum \frac{1}{n}$.

Then $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n^2}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Thus $l = 0$ and in this case $\sum u_n$ converges and $\sum v_n$ diverges.

(b) Let $\sum u_n = \sum \frac{1}{n}$ and $\sum v_n = \sum \frac{1}{n^2}$.

Then $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} n = \infty$.

Thus $l = \infty$ and in this case $\sum u_n$ diverges and $\sum v_n$ converges.

5. If $\sum a_n^2$ & $\sum b_n^2$ are convergent series of positive real numbers. Then $\sum \sqrt{a_n b_n}$ is convergent.

Proof: As $(a_n - b_n)^2 \geq 0 \Rightarrow a_n^2 + b_n^2 - 2a_n b_n \geq 0$

$\Rightarrow \sum a_n^2 + \sum b_n^2 \geq \sum 2\sqrt{a_n b_n}$

$\Rightarrow \frac{\sum a_n^2 + \sum b_n^2}{2} \geq \sum \sqrt{a_n b_n}$

By comparison test, $\sum \sqrt{a_n b_n}$ is convergent. Hence proved.

6. If $\sum a_n$ is convergent. Then $\sum \frac{a_n}{n}$ is convergent.

Proof: As $\left(a_n - \frac{1}{n}\right)^2 \geq 0$

$\Rightarrow a_n^2 + \frac{1}{n^2} - \frac{2a_n}{n} \geq 0$

$\Rightarrow \sum a_n^2 + \sum \frac{1}{n^2} \geq \sum \frac{2a_n}{n}$

Since $\sum a_n^2$ and $\sum \frac{1}{n^2}$ are both convergent $\Rightarrow \sum a_n^2 + \sum \frac{1}{n^2}$ is convergent.

Then by comparison test $\sum \frac{2a_n}{n}$ is convergent

$\Rightarrow \sum \frac{a_n}{n}$ is convergent. Hence proved.

Example: Test each of the following series for convergence:

(a) $\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \dots$

(b) $\frac{1}{\sqrt{1.2}} + \frac{1}{\sqrt{2.3}} + \frac{1}{\sqrt{3.4}} + \dots$

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Solution:

(a) $\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \dots$

The n th term of this series is $u_n = \frac{2n-1}{n(n+1)(n+2)}$.

For large values of n , $u_n \sim \frac{2n}{n.n.n} = \frac{2}{n^2}$.

Let $v_n = \frac{1}{n^2}$. Then $\frac{u_n}{v_n} = \frac{(2n-1)n^2}{n(n+1)(n+2)}$
 $= \left(\frac{n}{n+1}\right) \left(\frac{2n-1}{n+2}\right) = \frac{1}{1+(1/n)} \cdot \frac{2-(1/n)}{1+(2/n)}$.

Now $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = 2 \neq 0$ and finite.

So, by limit form comparison test $\sum u_n$ and $\sum v_n$ converge or diverge together.

Since $\sum v_n = \sum \frac{1}{n^2}$ converges, so $\sum u_n$ converges.

(b) $\frac{1}{\sqrt{1.2}} + \frac{1}{\sqrt{2.3}} + \frac{1}{\sqrt{3.4}} + \dots$

The n th term is $u_n = \frac{1}{\sqrt{n(n+1)}} \sim \frac{1}{\sqrt{n.n}} = \frac{1}{n}$.

Let $v_n = \frac{1}{n}$. Then $\frac{u_n}{v_n} = \frac{\sqrt{n}}{\sqrt{n+1}} = \frac{1}{\sqrt{1+(1/n)}}$.

Now $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = 1 \neq 0$ and finite.

So, $\sum u_n$ and $\sum v_n$ converge or diverge together by limit form comparison test

Since $\sum v_n = \sum \frac{1}{n}$ diverges, so $\sum u_n$ diverges.

Example: Test for convergence the series

$\frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \frac{1}{\sqrt{4}+\sqrt{5}} + \dots$

Solution: We have $u_n = \frac{1}{\sqrt{n+1}+\sqrt{n+2}} \sim \frac{1}{\sqrt{n}+\sqrt{n}} = \frac{1}{2\sqrt{n}}$.

Let $v_n = \frac{1}{\sqrt{n}}$. Then $\frac{u_n}{v_n} = \frac{\sqrt{n}}{\sqrt{n+1}+\sqrt{n+2}}$.

Now, $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+(1/n)}+\sqrt{1+(2/n)}} = \frac{1}{\sqrt{2}} \neq 0$ and finite

So $\sum u_n$ and $\sum v_n$ converge or diverge together by limit form comparison test,

Since $\sum v_n = \sum \frac{1}{\sqrt{n}}$ diverge, so $\sum u_n$ diverges.

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Example: Test for convergence the series $\sum u_n = \sum \frac{\sqrt{2n-1}}{(2n+2)(2n+4)}$

Solution: We have, $u_n = \frac{\sqrt{2n-1}}{(2n+2)(2n+4)} \sim \frac{\sqrt{2n}}{2n \cdot 2n}$.

Let $v_n = \frac{\sqrt{n}}{n \cdot n} = \frac{1}{n^{3/2}}$.

Then $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \sqrt{\frac{2n-1}{n}} \left(\frac{n}{2n+2} \right) \left(\frac{n}{2n+4} \right)$

$$= \lim_{n \rightarrow \infty} \sqrt{2 - \frac{1}{n}} \cdot \frac{1}{\left(2 + \frac{2}{n}\right) \left(2 + \frac{4}{n}\right)}$$

$$= \frac{\sqrt{2}}{2 \cdot 2} = \frac{1}{2\sqrt{2}} \neq 0 \text{ and finite}$$

So $\sum u_n$ and $\sum v_n$ converge or diverge together. (By limit form comparison test)

Since $\sum v_n = \sum \frac{1}{n^{3/2}}$ converges, so the given series converges.

Example: Test for converges the series $\frac{1}{3.7} + \frac{1}{4.9} + \frac{1}{5.11} + \frac{1}{6.13} + \dots$

Solution: We have $u_n = \frac{1}{(n+2)(2n+5)} \sim \frac{1}{n \cdot 2n} = \frac{1}{2n^2}$.

Let $v_n = \frac{1}{n^2}$.

Then $\frac{u_n}{v_n} = \frac{n}{n+2} \cdot \frac{n}{2n+5} = \frac{1}{1 + (2/n)} \cdot \frac{1}{2 + (5/n)}$.

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \frac{1}{2} \neq 0 \text{ and finite.}$$

So, by limit form comparison test $\sum u_n$ and $\sum v_n$ converge or diverge together.

Since $\sum v_n = \sum 1/n^2$ converges, so the given series $\sum u_n$ also converges.

Example: Test for convergence the series whose n th term is

$$u_n = \frac{\sqrt{n+1} - \sqrt{n-1}}{n}$$

Solution: We have $u_n = \frac{\sqrt{n+1} - \sqrt{n-1}}{n} = \frac{(\sqrt{n+1} - \sqrt{n-1})(\sqrt{n+1} + \sqrt{n-1})}{n(\sqrt{n+1} + \sqrt{n-1})}$

(Rationalize)

$$= \frac{2}{n(\sqrt{n+1} + \sqrt{n-1})} \sim \frac{2}{n(\sqrt{n} + \sqrt{n})} = \frac{1}{n\sqrt{n}}$$

Let $v_n = \frac{1}{n\sqrt{n}} = \frac{1}{n^{3/2}}$, so that $\sum v_n$ converges.

$$\text{Now } \lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{2\sqrt{n}}{(\sqrt{n+1} + \sqrt{n-1})}$$

$$\lim_{n \rightarrow \infty} \frac{2}{\sqrt{1+(1/n)} + \sqrt{1-(1/n)}} = 1 \neq 0 \text{ and finite}$$

So $\sum u_n$ and $\sum v_n$ converge or diverge together by limit form comparison test.

Since $\sum v_n$ converges, so the given series $\sum u_n$ also converges.

Example: The series $\sum \frac{1}{\sqrt{n} + \sqrt{n+1}}$ diverges.

Solution: Take $v_n = \frac{1}{\sqrt{n}}$, then $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \frac{1}{2} \neq 0$ and finite.

By limit form comparison test, the given series diverges.

Example: Test for convergence the series $\sum_{n=1}^{\infty} (\sqrt{n^3+1} - \sqrt{n^3})$.

$$\text{Solution: } u_n = \sqrt{n^3+1} - \sqrt{n^3} = \frac{(\sqrt{n^3+1} - \sqrt{n^3})(\sqrt{n^3+1} + \sqrt{n^3})}{\sqrt{n^3+1} + \sqrt{n^3}}$$

$$= \frac{(n^3+1) - n^3}{\sqrt{n^3+1} + \sqrt{n^3}} = \frac{1}{\sqrt{n^3+1} + \sqrt{n^3}}$$

Let $v_n = \frac{1}{\sqrt{n^3}} = \frac{1}{n^{3/2}}$, so that $\sum v_n$ converges.

$$\text{Now } \lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n^3}}{\sqrt{n^3+1} + \sqrt{n^3}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+(1/n)^3} + 1} = \frac{1}{2} \neq 0 \text{ and finite.}$$

So, by limit form comparison test $\sum u_n$ and $\sum v_n$ converge or diverge together.

Since, $\sum v_n$ converges, so the given series $\sum u_n$ converges.

Example: Test for convergence the series whose n th term is

$$u_n = \left\{ (n^3+1)^{1/3} - n \right\}.$$

$$\text{Solution: Let } u_n = (n^3+1)^{1/3} - n = n \left[\left(1 + \frac{1}{n^3} \right)^{1/3} - 1 \right]$$

$$= n \left[\left(1 + \frac{1}{3n^3} - \frac{1}{9n^6} + \dots \right) - 1 \right] = \frac{1}{3n^2} - \frac{1}{9n^5} + \dots$$

Let $v_n = \frac{1}{n^2}$, so that $\sum v_n$ converges.

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Now, $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \frac{1}{3} \neq 0$ and finite.

So, by limit form comparison test $\sum u_n$ and $\sum v_n$ converge or diverge together.

Since $\sum v_n$ converges, so $\sum u_n$ converges.

Example: Test for convergence the series $\sum u_n = \sum \frac{n+1}{n^p}$, $p \in \mathbb{R}$

Solution: Let $u_n = \frac{n+1}{n^p}$ and $v_n = \frac{n}{n^{p-1}}$.

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n+1}{n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = 1 \neq 0 \text{ and finite.}$$

So, by limit form comparison test $\sum u_n$ and $\sum v_n$ converge or diverge together by limit form comparison test.

Now, $\sum v_n = \sum \frac{1}{n^{p-1}}$ converges, if $p-1 > 1$ i.e., $p > 2$ and diverges if $p-1 \leq 1$ i.e., $p \leq 2$.

Hence $\sum \frac{n+1}{n^p}$ converges if $p > 2$ and diverges if $p \leq 2$.

Example: Test the convergence of the series $\sum \frac{1}{n^{1+(1/n)}}$.

Solution: We have $u_n = \frac{1}{n^{1+(1/n)}}$. Let $v_n = \frac{1}{n}$.

$$\text{Then } \lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{n^{1/n}} = 1.$$

Thus the two series $\sum u_n$ and $\sum v_n$ converge or diverge together by limit form comparison test.

Since $\sum v_n = \sum 1/n$ diverges, so the given series $\sum u_n$ diverges.

Example: Test for convergence the series $\frac{\sqrt{2}-1}{3^3-1} + \frac{\sqrt{3}-1}{4^3-1} + \frac{\sqrt{4}-1}{5^3-1} + \dots$

Solution: The n th term of the given series is $u_n = \frac{\sqrt{n+1}-1}{(n+2)^3-1}$.

$$\text{Let } v_n = \frac{\sqrt{n}}{n^3} = \frac{1}{n^{5/2}}$$

$$\text{Now, } \lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \left(\sqrt{\frac{n+1}{n}} - \frac{1}{\sqrt{n}} \right) \lim_{n \rightarrow \infty} \frac{1}{\left[\left(\frac{n+2}{n} \right)^3 - \frac{1}{n^3} \right]}$$

$$= \lim_{n \rightarrow \infty} \sqrt{1 + \left(\frac{1}{n} \right)} \cdot \lim_{n \rightarrow \infty} \frac{1}{\left[1 + \left(\frac{2}{n} \right)^3 \right]} = 1.$$

So $\sum u_n$ and $\sum v_n$ converge or diverge together.

Since $\sum v_n = \sum \frac{1}{n^{5/2}}$ converges, so the given series converges.

7. If $\sum a_n$ is a convergent series of positive terms then $\sum \frac{a_n}{1+a_n}$ is convergent.

Reason: Since $\sum a_n$ is convergent so $\lim_{n \rightarrow \infty} a_n = 0$.

Let $b_n = \frac{a_n}{1+a_n} \forall n$. Then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} (1+a_n) = 1+0=1$

Hence, $\sum a_n$ and $\sum b_n$ converge or diverge together by limit form comparison test. Since $\sum a_n$ is convergent, so $\sum b_n = \sum \frac{a_n}{1+a_n}$ is convergent.

Remark: If $\sum a_n$ is convergent $\Rightarrow \sum \frac{a_n}{1-a_n}$ is convergent ($a_n \geq 0$ and $a_n \neq 1 \forall n \in \mathbb{N}$)

Take $b_n = \frac{a_n}{1-a_n}$ then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} (1-a_n) = 1 \neq 0$ so, by limit form comparison test, $\sum a_n$ & $\sum b_n$ converges & diverges together. Since $\sum a_n$ is convergent, so $\sum \frac{1}{1-a_n}$ is convergent.

8. **D' Alembert's Ratio Test :** Let $\sum a_n$ be a positive term series such

that $\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = l$. Then

- (i) $\sum a_n$ converges if $l > 1$
- (ii) $\sum a_n$ diverges if $l < 1$
- (iii) Test fails if $l = 1$

Example: Test for convergence the series $\frac{1}{2} + \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} + \frac{1}{4 \cdot 2^4} + \dots$

Solution: We have $u_n = \frac{1}{n \cdot 2^n}, u_{n+1} = \frac{1}{(n+1)2^{n+1}}$.

and $\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \lim_{n \rightarrow \infty} \frac{2(n+1)}{n} = 2 \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = 2 > 1$.

By Ratio Test, the given series converges.

Example: Test for convergence the series $\left(\frac{1}{3}\right)^2 + \left(\frac{1 \cdot 2}{3 \cdot 5}\right)^2 + \left(\frac{1 \cdot 2 \cdot 3}{3 \cdot 5 \cdot 7}\right)^2 + \dots$

Solution: We have

$u_n = \left[\frac{1 \cdot 2 \cdot 3 \dots n}{3 \cdot 5 \cdot 7 \dots (2n+1)} \right]^2, u_{n+1} = \left[\frac{1 \cdot 2 \cdot 3 \dots n(n+1)}{3 \cdot 5 \cdot 7 \dots (2n+1)(2n+3)} \right]^2$

and $\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \lim_{n \rightarrow \infty} \left[\frac{2n+3}{n+1} \right]^2 = \lim_{n \rightarrow \infty} \left(\frac{2 + \frac{3}{n}}{1 + \frac{1}{n}} \right)^2 = 4 > 1$.

By Ratio Test, the given series converges.

Example: Test for convergence the series $\sum_{n=1}^{\infty} \frac{1.2.3 \dots n}{7.10 \dots (3n+4)}$

Solution: We have $u_n = \frac{1.2.3 \dots n}{7.10 \dots (3n+4)}$, $u_{n+1} = \frac{1.2.3 \dots n(n+1)}{7.10 \dots (3n+4)\{3(n+1)+4\}}$

$$\text{and } \lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \lim_{n \rightarrow \infty} \frac{3n+7}{n+1} = \lim_{n \rightarrow \infty} \frac{3+\frac{7}{n}}{1+\frac{1}{n}} = 3 > 1$$

By Ratio Test, the given series converges.

Example: Test for convergence the series $\frac{1^2.2^2}{1!} + \frac{2^2.3^2}{2!} + \frac{3^2.4^2}{3!} + \dots$

Solution: We have $u_n = \frac{n^2(n+1)^2}{n!}$, $u_{n+1} = \frac{(n+1)^2(n+2)^2}{(n+1)!}$

$$\text{and } \lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \lim_{n \rightarrow \infty} \frac{(n+1)!}{n!} \left(\frac{n}{n+2} \right)^2$$

$$= \lim_{n \rightarrow \infty} (n+1) \cdot \lim_{n \rightarrow \infty} \left(\frac{1}{1+\frac{2}{n}} \right)^2 = \infty$$

Hence the series converges.

Example: Test for convergence the series $\sum_{n=1}^{\infty} \frac{r^n}{n!}$, where r is any positive number.

Solution: We have $u_n = \frac{r^n}{n!}$, $u_{n+1} = \frac{r^{n+1}}{(n+1)!}$

$$\text{Now } \frac{u_n}{u_{n+1}} = \frac{1}{r} \frac{(n+1)!}{n!},$$

$$\text{and } \lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \frac{1}{r} \lim_{n \rightarrow \infty} (n+1) = \infty \quad [\because r > 0]$$

By Ratio Test, the given series converges.

Example: Test for convergence the series whose n th term is $\frac{r^n}{n^n}$, $r > 0$

Solution: We have $u_n = \frac{r^n}{n^n}$, $u_{n+1} = \frac{r^{n+1}}{(n+1)^{n+1}}$

$$\frac{u_n}{u_{n+1}} = \frac{1}{r} \frac{(n+1)^{n+1}}{n^n} = \frac{1}{r} \left(\frac{n+1}{n} \right)^n (n+1)$$

$$= \frac{1}{r} \left(1 + \frac{1}{n} \right)^n (n+1).$$

$$\text{Now } \lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \frac{1}{r} \cdot e \cdot \infty = \infty$$

By Ratio Test, the given series converges.

Example: Test for convergence the series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.

Solution: We have $u_n = \frac{n!}{n^n}$, $u_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}}$.

$$\begin{aligned} \text{Now } \frac{u_n}{u_{n+1}} &= \frac{n!}{(n+1)!} \cdot \frac{(n+1)^{n+1}}{n^n} \\ &= \frac{1}{(n+1)} \left(\frac{n+1}{n} \right)^n (n+1) = \left(1 + \frac{1}{n} \right)^n. \end{aligned}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e = 2.718 > 1.$$

By Ratio Test, the given series converges.

Example: Test for convergence the series $x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ for all positive values of x .

Solution: $u_n = \frac{x^{2n-1}}{(2n-1)!}$, $u_{n+1} = \frac{x^{2n+1}}{(2n+1)!}$.

$$\begin{aligned} \text{Now } \frac{u_n}{u_{n+1}} &= \frac{1}{x^2} \frac{(2n+1)!}{(2n-1)!} \\ &= \frac{1}{x^2} \frac{(2n+1) \cdot 2n \cdot (2n-1)!}{(2n-1)!} \end{aligned}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \lim_{n \rightarrow \infty} \frac{2n(2n+1)}{x^2} = \infty.$$

Hence, by Ratio Test, $\sum u_n$ converges.

9. **Rabbe's Test:** Let $\sum a_n$ be a positive term series such that

$$\lim_{n \rightarrow \infty} n \left(\frac{a_n}{a_{n+1}} - 1 \right) = l. \text{ Then}$$

- (i) $\sum a_n$ converges if $l > 1$
- (ii) $\sum a_n$ diverges if $l < 1$
- (iii) Test fails if $l = 1$

Example: Test for convergence the series $\sum_{n=1}^{\infty} \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} \cdot \frac{1}{n}$.

Solution: We have $u_n = \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} \cdot \frac{1}{n}$,

$$u_{n+1} = \frac{1.3.5 \dots (2n-1)(2n+1)}{2.4.6 \dots 2n(2n+2)} \cdot \frac{1}{n+1}$$

$$\frac{u_n}{u_{n+1}} = \frac{(2n+2)(n+1)}{(2n+1)n} = \left(1 + \frac{1}{n} \right) \frac{(1+1/n)}{(1+1/2n)}$$

Now $\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = 1$ means that the Ratio test fails.

So, we apply Raabe's Test. We have

$$n \left(\frac{u_n}{u_{n+1}} - 1 \right) = n \left[\frac{(n+1)(2n+2)}{n(2n+1)} - 1 \right] = \frac{3n+2}{2n+1} = \frac{3 + \frac{2}{n}}{2 + \frac{1}{n}}$$

$$\text{and } \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \frac{3}{2} > 1.$$

Hence, by Raabe's test, the given series converges.

Example: Test for convergence the series $\sum_{n=1}^{\infty} \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} \cdot \frac{4n+1}{2n+3}$,

Solution: We have $u_n = \frac{1.3 \dots (2n-1)(4n+1)}{2.4 \dots 2n(2n+3)}$, $u_{n+1} = \frac{1.3 \dots (2n+1)(4n+5)}{2.4 \dots (2n+2)(2n+5)}$

$$\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1} \cdot \frac{2n+5}{4n+5} \cdot \frac{2n+3}{4n+1}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = \frac{2.2.2}{2.4.4} = \frac{1}{4} < 1.$$

By Raabe's test, $\sum u_n$ diverges.

10. **Cauchy's nth Root Test :** Let $\sum a_n$ be a positive term series such that

$$\lim_{n \rightarrow \infty} (a_n)^{1/n} = l. \text{ Then}$$

(i) $\sum a_n$ converges if $l < 1$

(ii) $\sum a_n$ diverges if $l > 1$

(iii) Test fails if $l = 1$

Example: Test for convergence the series whose n^{th} term is

$$u_n = \frac{n^{n^2}}{(n+1)^{n^2}}$$

Solution: We have $u_n = \left[\frac{n}{n+1} \right]^{n^2}$

$$\Rightarrow (u_n)^{1/n} = \left(\frac{n}{n+1} \right)^n = \frac{1}{(1+1/n)^n}$$

$$\text{and } \lim_{n \rightarrow \infty} (u_n)^{1/n} = \frac{1}{\lim_{n \rightarrow \infty} (1+1/n)^n} = \frac{1}{e} < 1. (e = 2.718).$$

Hence, by Cauchy's root test, the given series is convergent.

Example: The series $\sum_{n=1}^{\infty} (n^{1/n} - 1)^n$ converges.

Solution: Let $u_n = (n^{1/n} - 1)^n$, so that $(u_n)^{1/n} = n^{1/n} - 1$.

$$\text{Now } \lim_{n \rightarrow \infty} (u_n)^{1/n} = \lim_{n \rightarrow \infty} n^{1/n} - 1 = 1 - 1 = 0 < 1.$$

Hence, by Cauchy's root test, the given series converges.

Example: Test for the convergence the series whose n^{th} term is

$$u_n = \left(1 + \frac{1}{\sqrt{n}}\right)^{-n^{3/2}}$$

Solution: We have $u_n = \frac{1}{\left(1 + \frac{1}{\sqrt{n}}\right)^{n^{3/2}}}$

Then $(u_n)^{1/n} = \frac{1}{\lim_{n \rightarrow \infty} \left(1 + \frac{1}{\sqrt{n}}\right)^{\sqrt{n}}} = \frac{1}{e} < 1$ ($\because e = 2.7182$)

Hence, by Cauchy's root test, the given series converges.

Example: Test the convergence of the series

$$\left(\frac{2^2}{1^2} - \frac{2}{1}\right)^{-1} + \left(\frac{3^3}{2^3} - \frac{3}{2}\right)^{-2} + \left(\frac{4^4}{3^4} - \frac{4}{3}\right)^{-3} + \dots$$

Solution: The n^{th} term of the given series is $u_n = \left[\left(\frac{n+1}{n}\right)^{n+1} - \frac{n+1}{n}\right]^{-n}$

Now, $(u_n)^{1/n} = \left[\left(\frac{n+1}{n}\right)^{n+1} - \frac{n+1}{n}\right]^{-1}$

$$= \left(\frac{n+1}{n}\right)^{-1} \left[\left(\frac{n+1}{n}\right)^n - 1\right]^{-1}$$

$$= \left(1 + \frac{1}{n}\right)^{-1} \left[\left(1 + \frac{1}{n}\right)^n - 1\right]^{-1}$$

$$\lim_{n \rightarrow \infty} (u_n)^{1/n} = 1 \cdot (e-1)^{-1} = \frac{1}{e-1} < 1 \quad (e = 2.718).$$

Hence, by Cauchy's root test, the given series converges.

Example: Test for convergence the series whose n^{th} term is $2^{-n-(-1)^n}$.

Solution: Let $u_n = 2^{-n-(-1)^n} = 2^{-\{n+(-1)^n\}}$

and $(u_n)^{1/n} = \frac{1}{\left[2^{n+(-1)^n}\right]^{1/n}} = \frac{1}{2^n \cdot 2^{\frac{(-1)^n}{n}}}$

Now $\lim_{n \rightarrow \infty} (u_n)^{1/n} = \frac{1}{2} \cdot \frac{1}{\lim_{n \rightarrow \infty} 2^{\frac{(-1)^n}{n}}} = \frac{1}{2} < 1$. $\because \lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$

Hence, by Cauchy's root test, the given series converges.

Some Important Definitions

1. The infinite integral $\int_1^{\infty} f(x) dx$ is said to be convergent (divergent),

if $\lim_{l \rightarrow \infty} \int_1^l f(x) dx$ is finite (infinite).

2. Let $f(x)$ be a real valued function with domain $[1, \infty[$. Then the function $f(x)$ is said to be **non-negative**, if $f(x) \geq 0 \forall x \geq 1$.
3. The function $f(x)$ is said to be **monotonically decreasing**, if $x \leq y \Rightarrow f(x) \geq f(y) \forall x, y \in [1, \infty[$.

Example: Let $f(x) = 1/x^2$. Then $f(x)$ is non-negative and monotonically decreasing $\forall x \geq 1$.

11. **Cauchy's Integral Test:** If $u(x)$ is a non-negative, monotonically decreasing and integrable function such that $u(n) = u_n \forall n \in \mathbb{N}$, then the series $\sum_{n=1}^{\infty} u_n$ is convergent if and only if $\int_1^{\infty} u(x) dx$ is convergent.

Example: The series $\sum_{n=1}^{\infty} \frac{1}{n^p} (p > 0)$ is convergent if $p > 1$ and divergent if $p \leq 1$.

Example: The series $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ is convergent if $p > 1$ and divergent if $0 < p \leq 1$.

Example: examine the convergence of the following series $\sum_{n=1}^{\infty} \frac{1}{n^2 + n}$.

Solution: Let $u(x) = \frac{1}{x^2 + x}$, so that $u(n) = u_n \forall n \in \mathbb{N}$.

For $x \geq 1$, $u(x)$ is non-negative, monotonically decreasing and integrable function. Now

$$\begin{aligned} \int_1^t \frac{dx}{x^2 + x} &= \int_1^t \frac{dx}{x(x+1)} = \int_1^t \left(\frac{1}{x} - \frac{1}{x+1} \right) dx \\ &= \left| \log x - \log(x+1) \right|_1^t = \left| \log \frac{x}{1+x} \right|_1^t \\ &= \log \frac{t}{1+t} - \log \frac{1}{2} = \log \frac{1}{1+(1/t)} - \log \frac{1}{2}. \end{aligned}$$

$$\therefore \lim_{t \rightarrow \infty} \int_1^t \frac{dx}{x^2 + x} = \log 1 + \log 2 = \log 2, \text{ which is finite.}$$

Thus $\int_1^{\infty} u(x) dx$ is a convergent and so by Cauchy's Integral Test, the given series is convergent.

Example: The series $\sum_{n=3}^{\infty} \frac{1}{n \log n (\log \log n)^p}$, ($p > 0$) converges $p > 1$ and diverges if $0 < p \leq 1$.

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4.6. Alternating Series

A series of the form $u_1 - u_2 + u_3 - \dots$ where $u_n > 0 \forall n \in \mathbb{N}$ is called an alternating series and is denoted by $\sum_{n=1}^{\infty} (-1)^{n-1} u_n$.

Example:

$$(a) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

$$(b) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$$

$$(c) \sum_{n=2}^{\infty} \frac{(-1)^n}{\log n} = \frac{1}{\log 2} - \frac{1}{\log 3} + \frac{1}{\log 4} - \frac{1}{\log 5} + \dots$$

4.6.1 Leibnitz's Test for Alternating Series: If an alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} u_n$ satisfies

$$(i) u_{n+1} \leq u_n \forall n$$

$$(ii) \lim_{n \rightarrow \infty} u_n = 0$$

Then, the series $\sum_{n=1}^{\infty} (-1)^{n-1} u_n$ converges.

Remark: The alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} u_n$ will not be convergent if either $u_{n+1} \nless u_n, \forall n$ or $\lim_{n \rightarrow \infty} u_n \neq 0$.

Example: The series $1 - \frac{3}{2} + \frac{4}{3} - \frac{5}{4} + \dots$ is not convergent, as

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right) = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) = 1 \neq 0.$$

Example: Test for convergence the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

Solution: We see that $\frac{1}{2} < 1, \frac{1}{3} < \frac{1}{2}, \frac{1}{4} < \frac{1}{3}$ and so on. Thus

$$(a) u_{n+1} < u_n \forall n$$

$$(b) \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Hence, by Leibnitz's Test, the given series is convergent.

4.7. Absolute Convergence

If $\sum a_n$ is a series of real numbers such that $\sum |a_n|$ is a convergent series. Then $\sum a_n$ is called an absolutely convergent series.

Example:

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \text{ is absolute convergent series.}$$

$$(b) \sum_{n=1}^{\infty} \sin\left(\frac{1}{n^2}\right) \text{ is absolute convergent series}$$

$$(c) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \text{ is not absolutely convergent series.}$$

Result: Let $\sum u_n$ be absolutely convergent series. Then $\sum u_n$ is also convergent. The converse of this is not true i.e., a convergent series may not be absolutely convergent.

Example: Consider the series $\sum u_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

We have seen that the given series is convergent, by Leibnitz's Test but $\sum |u_n| = 1 + \frac{1}{2} + \frac{1}{3} + \dots = \sum \frac{1}{n}$ is not convergent i.e., $\sum u_n$ is not absolutely convergent.

4.7.1 Conditional Convergence

A series $\sum u_n$ is said to be conditionally convergent, if

- (i) $\sum u_n$ is convergent, and
- (ii) $\sum u_n$ is not absolutely convergent.

Example: Test for convergence and absolute convergence the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^p} = 1 - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} + \dots \text{ for } p > 0.$$

Solution: Since $p > 0$ and $(n+1)^p > n^p$, so

$$\frac{1}{(n+1)^p} < \frac{1}{n^p} \Rightarrow u_{n+1} < u_n \quad \forall n$$

Also, $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$, since $p > 0$.

By Leibnitz's Test, $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^p}$ is convergent.

Now, $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n^p} \right| = \sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leq 1$.

Hence the given series is absolutely convergent if $p > 1$ and conditionally convergent if $0 < p \leq 1$.

Example: Test for convergence and absolute convergence the series

$$\frac{1}{\sqrt{1}} - \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{5}} - \frac{1}{\sqrt{7}} + \dots$$

Solution: We have $u_{n+1} < u_n$ for each n and

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2n-1}} = 0.$$

Thus $\sum u_n$ is convergent, by Leibnitz's test.

Now we show $\sum u_n$ is not absolutely convergent. We have $|u_n| = \frac{1}{\sqrt{2n-1}}$.

$$\text{Let } v_n = \frac{1}{\sqrt{n}} \Rightarrow \lim_{n \rightarrow \infty} \frac{|u_n|}{v_n} = \lim_{n \rightarrow \infty} \sqrt{\frac{n}{2n-1}} = \frac{1}{\sqrt{2}} \neq 0 \text{ and finite}$$

Since $\sum v_n$ is divergent, so $\sum |u_n|$ is divergent. (by limit form comparison test)

Hence $\sum u_n$ is not absolutely convergent.

Example: Test for convergence the series $\sum_{n=1}^{\infty} (-1)^n \frac{\sin n\alpha}{n^3}$, α being real number.

Solution: We have $|u_n| = \frac{|\sin n\alpha|}{n^3} \leq \frac{1}{n^3} \quad \forall n$. $(\because \sum 1/n^3 \text{ is convergent})$

Thus $\sum u_n$ is absolutely convergent for all real values of α and so $\sum u_n$ is convergent for all real values of α .

Example: Discuss the convergence of the series $\sum_{n=1}^{\infty} \frac{\sin nx + \cos nx}{n^{3/2}}$.

Solution: We have

$$|u_n| = \left| \frac{\sin nx + \cos nx}{n^{3/2}} \right| \leq \frac{1}{n^{3/2}} + \frac{1}{n^{3/2}} = \frac{2}{n^{3/2}} \quad \left\{ \because |\sin nx| \leq 1, |\cos nx| \leq 1 \quad \forall x \right\}$$

Since $\sum \frac{1}{n^{3/2}}$ is convergent, so by First Comparison Test, $\sum |u_n|$ is convergent i.e., $\sum u_n$ is absolutely convergent.

Hence $\sum u_n$ is convergent.

Example: Test for convergence and absolute convergence the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\log(n+1)} = \frac{1}{\log 2} - \frac{1}{\log 3} + \frac{1}{\log 4} - \dots$$

Solution: We have $u_n = \frac{1}{\log(n+1)} \quad \forall n \in \mathbb{N}$

$$\text{Clearly, } \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{\log(n+1)} = 0$$

Since $\log x$ is an increasing function for all $x > 0$, therefore

$$\log(n+2) > \log(n+1) \quad (\because n+2 > n+1)$$

$$\Rightarrow \frac{1}{\log(n+2)} < \frac{1}{\log(n+1)} \quad \forall n \Rightarrow u_{n+1} < u_n \quad \forall n$$

Thus the two conditions of Leibnitz Test are satisfied and so the given series is convergent.

Now we test the absolute convergence of the given series.

We have $\sum_{n=2}^{\infty} |u_n| = \sum_{n=2}^{\infty} \frac{1}{\log n}$, which is divergent.

Hence $\sum u_n$ is not absolutely convergent i.e., $\sum u_n$ is conditionally convergent.

Example: The series $\frac{\log 2}{2^2} - \frac{\log 3}{3^2} + \frac{\log 4}{4^2} - \dots$ converges.

Solution: We have $u_n = \frac{\log(n+1)}{(n+1)^2} \quad \forall n \in \mathbb{N}$.

$$\text{Since, } \lim_{n \rightarrow \infty} \frac{\log(n+1)}{(n+1)^2} = 0.$$

$$\Rightarrow \lim_{n \rightarrow \infty} u_n = 0$$

Now we shall prove that $u_{n+1} \leq u_n \forall n$.

$$\text{Let } f(x) = \frac{\log x}{x^2} \text{ so that } f'(x) = \frac{x^2(1/x) - 2x \log x}{x^4}$$

$$\Rightarrow f'(x) = \frac{1 - 2 \log x}{x^3} < 0 \quad \forall x > e^{1/2} \left(\because x > e^{1/2} \Leftrightarrow \log x > \frac{1}{2} \Leftrightarrow 1 - 2 \log x < 0 \right)$$

$$\Rightarrow f(x) \text{ is decreasing function } \forall x > e^{1/2}$$

$$\Rightarrow f(n+2) \leq f(n+1) \quad \forall n \quad \left(\because n+2 > n+1 > e^{1/2} \quad \forall n \right)$$

$$\Rightarrow \frac{\log(n+2)}{(n+2)^2} \leq \frac{\log(n+1)}{(n+1)^2} \quad \forall n$$

$$\Rightarrow u_{n+1} \leq u_n \quad \forall n.$$

Thus both the conditions of Leibnitz test are satisfied and so the given series is convergent.

Put Your Own Notes

ASSIGNMENT SHEET - 1

- Let F be the set of all functions from $[0, 1]$ to $[0, 1]$ itself. If $\text{card}(F) = f$ then
 - F is similar to $[0, 1]$
 - f is less than the cardinality of $[0, 1]$
 - $f > c$ where cardinality of $[0, 1]$ is c
 - F is countable
- If $f: A \rightarrow B$ is a one-one map and A is countable. Then which is correct
 - B is countable.
 - B is uncountable
 - There exists a subset of B which is countable
 - None of these.
- Let A be an infinite set of disjoint open sub intervals of $(0, 1)$. Let B be the power set of A . Then-
 - Cardinality of A & B are equal
 - A similar to $(0, 1)$
 - B similar to $(0, 1)$
 - A & B both are uncountable
- Match the following list-I with list-II and choose the correct:

List-I

 - Countable
 - Uncountable
 - Empty set

List-II

 - Set of transcendental elements of $\mathbb{R}$ over $\mathbb{Q}$
 - Set of all function from $\mathbb{Z}_2 = \{0, 1\}$ to $\mathbb{N}$
 - $\{m \in \mathbb{N} : \sqrt{n+1} - \sqrt{n} \text{ is rational}\}$

Code:

	A	B	C
(a.)	2	3	1
(b.)	1	2	3
(c.)	3	2	1
(d.)	2	1	3
- Let A be the set of lines passing through the origin and slope is integral multiple of $\frac{\pi}{12}$. Then
 - A is similar to $\mathbb{R}$
 - A is countably infinite
 - A is similar to the set of months in a year
 - A is similar to the power set of $\mathbb{R}$.
- Consider the following statements:
 - The set of all finite subsets of the natural numbers is countable
 - The set of all polynomials with integer coefficients is countable

Choose the correct answer:

 - Only 1 is true
 - Only 2 is true
 - Both 1 and 2 are true
 - Both are false
- Let A & B are infinite sets. Let f is a map from A to B such that the collection of pre images of any non-empty subset of B is non empty. Then choose the incorrect
 - If A is countable then B is countable
 - Such map f is always onto
 - A & B are similar
 - B may be countable even if A is not countable
- If f be a function with domain A and range B then which of following is correct.
 - B countable $\Rightarrow A$ countable
 - A countable $\Rightarrow B$ countable
 - A uncountable $\Rightarrow B$ uncountable
 - All of above
- Which of the following is correct?
 - The set of rational numbers in any interval of finite length is countable.
 - The set of irrational numbers in any interval of finite length is countable.
 - Every subset of uncountable set is uncountable.
 - All of above.

10. Read the following statements

1. S is a countable set
2. There exists a surjection of $\mathbb{N}$ onto S.
3. There exists an injection of S into $\mathbb{N}$

Codes:

- (a.) 1 implies 2 & 3 but not conversely
~~(b.) 2 & 3 imply 1 but not conversely~~
 (c.) 1 implies either of 2 & 3 but not both.
~~(d.) All the three statements are equivalent.~~

11. Which set is countable

- (a.) The set of all polynomials with real coefficients
 (b.) The set of all subsets of a countably infinite set
 (c.) The set $A - B$ where A is uncountable but B is countable
~~(d.) The set of all finite subsets of $\mathbb{N}$; (the set of natural number)~~

12. If F be set of all function defined on

$I_n = \{1, 2, 3, \dots, n\}; n \in \mathbb{N}$ with range $B \subseteq I^+$
 (set of positive integer) Then

- ~~(a.) F is countable.~~
 (b.) F is uncountable
~~(c.) F is infinite~~
~~(d.) F is countable if B is finite~~

13. Select the correct statements

- ~~1. Every countable set is similar to $\mathbb{N}$~~
 2. The set of all disjoint intervals is not similar to the set of real numbers
 3. The power set of $\mathbb{N}$ is similar to the set of real numbers.

Codes:

- (a.) 1 and 2 only
~~(b.) 2 and 3 only~~
~~(c.) 1 and 3 only~~
 (d.) All of these

14. Let for each pair of natural numbers (m, n) , $P_{n,m}$ be the set of polynomials of degree n with integral coefficients s.t. $|a_0| + |a_1| + \dots + |a_n| = m$; where a_i 's are coefficients then

- (a.) $P_{m,n}$ is countably infinite
 (b.) $P_{m,n}$ is finite for some m, n only
 (c.) $P = \bigcup \{P_{mn} : m, n \in \mathbb{N}\}$ is set of all algebraic number.
~~(d.) P_{mn} is finite $\forall m, n \in \mathbb{N}$~~

15. $S = \{a \in \mathbb{R} : 'a' \text{ is a recurring decimal number}\}$

$T = \{b \in \mathbb{R} : b = \sqrt{\frac{p}{q}} \text{ for some distinct primes } p \& q\}$ then

- (a.) S is countable but T is not
 (b.) T is countable but S is not
~~(c.) S & T both are countable sets~~
 (d.) S & T are uncountable sets

16. Consider the following statements:

1. Every infinite set is equivalent to at least one of its proper subset.
2. If a set is equivalent to one of its proper subset then it is infinite set

Code:

- (a.) 1 is correct and 2 is incorrect
 (b.) 2 is correct and 1 is incorrect
~~(c.) Both are correct~~
 (d.) Both incorrect

17. Let $A_1, A_2, \dots, A_n$ sets, where n be a fixed natural number. Consider following statements:

1. If $A = \bigcap_{i=1}^n A_i$ countably infinite, then there exist at least one A_i for $i = 1, 2, \dots, n$ which is countable
2. If $A = A_1 \times A_2 \times \dots \times A_n$ is countably infinite. Then each A_i for $i = 1, 2, \dots, n$ is countable

Code:

- (a.) 1 is correct and 2 is incorrect
~~(b.) 2 is correct and 1 is incorrect~~
 (c.) Both are correct
 (d.) Neither 1 nor 2 are correct

18. How many statements is/are false?

1. Cardinality of $[0, 1] \times [0, 1]$ is the same as cardinality of $\mathbb{R}$
 2. Cardinality of $\mathbb{R}$ is the same as the cardinality of irrationals
 3. Cardinality of $\mathbb{R}$ is the same as the cardinality of $\mathbb{C}$
- ~~(a.) Zero~~
 (b.) One
 (c.) Two
 (d.) Three

19. Let A be an uncountable subset of $\mathbb{R}$ and B be a proper infinite subset of $\mathbb{N}$. Define $f: B \rightarrow A$ such that f is one-one, then
- (a) A and $A - B$ are similar
(b) A and $A - f(B)$ are similar
(c) $\mathbb{R}$ and $\mathbb{R} - f(B)$ are similar
(d) All of the above are correct
20. Let P_n be the set of all polynomials of degree n with integral coefficients. The P_n is
- (a.) A finite set with n^n elements
(b.) A finite set
(c.) A countable set
(d.) An uncountable set
21. Consider the following statements.
1. S be set of all straight line in a plane each of which passes through at least two different rational co-ordinate
2. If $S = \{x: x \text{ be a rational point of } \mathbb{R}^2\}$ (a point $x = (x_1, x_2) \in \mathbb{R}^2$ called rational point if each $x_1, x_2 \in \mathbb{Q}$)
- Then
- (a) 1 and 2 both countable.
(b) 1 countable and 2 uncountable
(c.) 2 countable and 1 uncountable
(d.) 1 and 2 both uncountable
22. Consider the following statements.
1. If every subset of a set is countable then set is countable.
2. If every proper subset of a set is countable then set is countable
- Then
- (a.) 1 correct 2 may or may not correct
(b.) 2 correct 1 may or may not correct
(c.) Both may or may not correct
(d.) Both correct
23. If $F = \{A_1, A_2, \dots\}$ is a countable collection of countable sets let $G = \{B_1, B_2, \dots\}$ where $B_1 = A_1$ and for $n > 1$,
- $$B_n = A_n - \bigcup_{k=1}^{n-1} A_k.$$
- Then form the following statements which is/are true
1. G is a collection of disjoint sets
2. $\bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} B_k$
3. $\bigcup_{k=1}^{\infty} B_k$ is countable
- Code:
- (a.) Only 1
(b.) 1 and 3
(c.) 2 and 3
(d.) All three
24. Which of the following is/are true?
- (a.) The set $\{e^x: x \in \mathbb{R}\}$ is a countable set
(b.) The set $\{\log x: x > 0\}$ is a countable set
(c.) $\{\sin x: -\pi/2 \leq x \leq \pi/2\}$ is a countable set
(d.) The set of all solutions of the equation $z^n = 1; n = 1, 2, \dots$, where z is a complex number, is a countable set.
25. If $A = \{(x, y): y = e^x, x \in \mathbb{R}\}$ and $B = \{(x, y): y = e^{-x}, x \in \mathbb{R}\}$ then
- (a.) $A \cap B = \phi$
(b.) $A \cap B \neq \phi$
(c.) $A \cup B = \mathbb{R}$
(d.) None of these
26. Let X denote the two-point set $\{0, 1\}$ and write $X_j = \{0, 1\}$ for every $j = 1, 2, 3, \dots$. Let $Y = \prod_{j=1}^{\infty} X_j$. Which of the following is/are correct
- (a.) Y is countable set
(b.) cardinality $Y = \text{cardinality of } [0, 1]$
(c.) $\bigcup_{n=1}^{\infty} \left(\prod_{j=1}^n X_j \right)$ is countable
(d.) Y is uncountable

27. Which of the following subset of $\mathbb{R}^2$ is/are NOT countable set?

(a) $\{(a, b) \in \mathbb{R}^2 \mid a \leq b\}$

(b) $\{(a, b) \in \mathbb{R}^2 \mid a + b \in \mathbb{Q}\}$

(c) $\{(a, b) \in \mathbb{R}^2 \mid ab \in \mathbb{Z}\}$

(d) $\{(a, b) \in \mathbb{R}^2 \mid a, b \in \mathbb{Q}\}$

28. For each $J=1,2,3,\dots$ let A_J be a finite set containing at least two distinct element, then

(a) $\bigcup_{j=1}^{\infty} A_j$ is countable set

(b) $\bigcup_{j=1}^{\infty} A_j$ is uncountable

(c) $\prod_{j=1}^{\infty} A_j$ is uncountable

(d) $\prod_{j=1}^{\infty} A_j$ is countable

29. Which of the following sets of functions are uncountable? ($\mathbb{N}$ stands for the set of natural numbers.)

(a) $\{f \mid f: \mathbb{N} \rightarrow \{1, 2\}\}$

(b) $\{f \mid f: \{1, 2\} \rightarrow \mathbb{N}\}$

(c) $\{f \mid f: \{1, 2\} \rightarrow \mathbb{N}, f(1) \leq f(2)\}$

(d) $\{f \mid f: \mathbb{N} \rightarrow \{1, 2\}, f(1) \leq f(2)\}$

30. Let $A = (0, 1)$ & B is the sets of all disjoint open subintervals of A . then

(a) A & B are similar

(b) There is no onto map from A to B hence A & B are not similar.

(c) There is no onto map from B to A hence they are not similar.

(d) There is no one-one map from A to B , hence they are not similar.

ASSIGNMENT SHEET - 2

- Let G be the set of all irrational numbers. The interior and the closure of G are denoted by G° and $\bar{G}$, respectively. Then
 - $G^\circ = \phi$, $\bar{G} = G$
 - $G^\circ = \mathbb{R}$, $\bar{G} = \mathbb{R}$
 - $G^\circ = \phi$, $\bar{G} = \mathbb{R}$
 - $G^\circ = G$, $\bar{G} = \mathbb{R}$
- The set $\{x \in \mathbb{R} : x^2 > x\}$ is same as
 - the interval $(0,0)$
 - the complement of the interval $(0,1)$
 - the complement of the interval $[0,1]$
 - the interval $[0,1]$.
- The set of all boundary points of $\mathbb{Q}$ in $\mathbb{R}$ is
 - $\mathbb{R}$
 - $\mathbb{R} \setminus \mathbb{Q}$
 - $\mathbb{Q}$
 - Empty set
- Lim superior $\left\{ \frac{(-1)^n}{2^n} : n = 1, 2, \dots \right\}$ is
 - 0
 - $\frac{1}{4}$
 - 1
 - 1
- If $Y = \left\{ \frac{x}{1+|x|} : x \in \mathbb{R} \right\}$, then the set of all limit points of Y is
 - $(-1, 1)$
 - $(-1, 1]$
 - $[0, 1]$
 - $[-1, 1]$
- Which of these subsets of the plane is a bounded set?
 - $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 10\}$
 - $\left\{ (x, y) \in \mathbb{R}^2 : 1 < \frac{x^2}{4} + \frac{y^2}{9} \right\}$
 - $\{(x, y) \in \mathbb{R}^2 : x = y\}$
 - $\{(x, y) \in \mathbb{R}^2 : y^2 < 4x\}$
- The set $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$
 - A bounded closed set
 - A bounded open set
 - An unbounded open set
 - An unbounded closed set
- The set $\left\{ \frac{1}{n} \sin \frac{1}{n} : n \in \mathbb{N} \right\}$ has
 - One limit point and it is 0
 - One limit point and it is 1
 - One limit point and it is -1
 - Three limit points and these are -1, 0 and 1.
- The set $U = \left\{ x \in \mathbb{R} : \sin x = \frac{1}{2} \right\}$ is
 - open
 - closed
 - both open and closed
 - neither open nor closed
- Let S be the set of all numbers of the form $\sum_{k=0}^{\infty} \frac{a_k}{3^k}$; $a_k = 0$ or 2 . Which of the following is true?
 - S is countable
 - S is dense in $[0, 1]$
 - S is closed in $[0, 1]$
 - $S = [0, 1]$
- Consider the following subsets of $\mathbb{R}$:
 $E = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}$, $F = \left\{ \frac{1}{1-x} : 0 \leq x < 1 \right\}$ Then
 - Both E and F are closed
 - E is closed and F is NOT closed
 - E is NOT closed and F is closed
 - Neither E nor F is closed.
- How many subsets A of $\{1, 2, 3, 4\}$ are there that $(A \cup \{1, 2\}) - (A \cap \{1, 2\})$ has exactly one point?
 - 1
 - 2
 - 3
 - 4

13. Let $A = \{m + n\sqrt{2} : m, n \in \mathbb{Z}\}$, where $\mathbb{Z}$ stands for the set of all integers
- A is dense in $\mathbb{R}$
 - A has only countable many limit point in $\mathbb{R}$
 - A has no limit point in $\mathbb{R}$
 - Only irrational numbers can be limit points of A
14. Let $X = \left\{\frac{1}{n} : n \in \mathbb{Z}, n \geq 1\right\}$ and let $\bar{X}$ be its closure. Then
- $\bar{X} \setminus X$ is a single point
 - $\bar{X} \setminus X$ is open in $\mathbb{R}$
 - $\bar{X} \setminus X$ is infinite but not open in $\mathbb{R}$
 - $\bar{X} \setminus X = \emptyset$
15. Which of the following sets satisfy the condition that for every positive integer n there is some a in A such that $a < n$?
- $A = \{-1, 5\}$
 - A is empty set
 - $A = \mathbb{N}$
 - $A = \{x \in \mathbb{R} \mid x > 10\}$
16. Let S be an infinite subset of $\mathbb{R}$ such that $S \cap \mathbb{Q} = \emptyset$. Which of the following statements is true?
- S must have a limit point which belongs to $\mathbb{Q}$
 - S must have a limit point which belongs to $\mathbb{R} \setminus \mathbb{Q}$
 - S cannot be a closed set in $\mathbb{R}$
 - $\mathbb{R} \setminus S$ must have a limit point which belongs to S .
17. Let $A \neq \mathbb{R}$ be a dense subset of $\mathbb{R}$. If $U \subseteq \mathbb{R}$ is a non-empty open subset then
- $U \subseteq \overline{A \cap U}$
 - $\overline{A \cap U} = \emptyset$
 - $\overline{A \cap U} \subseteq U$
 - $\overline{A \cap U} = A \cap \bar{U}$
18. If $S = \left\{\frac{1}{2m} + \frac{1}{3n} : m, n \in \mathbb{N}\right\}$ then the derived sets S' and S'' has the property
- S' & S'' both are finite
 - S' is finite & S'' is empty
 - S' is infinite & S'' is singleton
 - S' & S'' both are infinite
19. Let $A = \left\{1 + \frac{1}{n} : n \in \mathbb{N}\right\}$ and $B = \left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\}$ then derived set of $(A \cup B)$
- Has cardinality 2
 - Has cardinality 1
 - Is empty set
 - Is countably infinite set
20. Let $S = \{x\sqrt{2} : x \in \mathbb{Q}\}$ then the derived set of S is
- Empty set
 - Singleton set
 - Equivalent to the set of natural numbers
 - Uncountable set
21. Let $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \neq 1\}$ then,
- S is closed in $\mathbb{R}^2$
 - S is open in $\mathbb{R}^2$
 - S is neither closed nor open
 - S is both closed and open.
22. In $\mathbb{R}$, let $F_n = \left(\frac{-1}{n}, \frac{1}{n}\right), n = 1, 2, 3, \dots$. Then $\bigcap_{n=1}^{\infty} F_n$ is
- Empty
 - Open but not closed
 - Closed but not open
 - Both open and closed
23. If S be finite intersection of family of all nbd of a point x then,
- S is infinite set.
 - S singleton
 - S finite but need not to be singleton
 - Can't say

24. Let A be a non empty subset of $\mathbb{R}$ having a supremum. Let $B = \{x \in \mathbb{R} : (x+1)^2 \in A\}$ then
- $\sup B = 2 \sup A - 1$
 - $\sup B = (\sup A + 1) / 2$
 - $\sup B = \sup A$
 - B does not have supremum
25. Let A be any subset of $\mathbb{R}$ such that $A \cap A' = \emptyset$ where A' is set of all limit point of A . Then,
- A is uncountable
 - A is finite
 - A is countable
 - None of these
26. Consider the set $S = \{x + iy : x \text{ and } y \text{ are real and } x, y \in (0, 1)\}$. Then
- S is uncountable and unbounded
 - S is countable and bounded
 - S is countable and unbounded
 - S is uncountable and bounded
27. The supremum of the set
- $$S = \left\{ 1, 1 + \frac{1}{2}, 1 + \frac{1}{2} + \frac{1}{2^2}, \dots, 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \right\}$$
- 1
 - 2
 - 3
 - $\frac{1}{2}$
28. Let $A = \left\{ x : x = \frac{4n+3}{n}; n \in \mathbb{N} \right\}$ then $\sup A$ & $\inf A$ are respectively
- 7 & 4
 - 0 & 4
 - 3 & 4
 - None of these
29. Select the incorrect statement about the set
- $$A = \left\{ 1, 1 + \frac{1}{3}, 1 + \frac{1}{3} + \frac{1}{3^2}, \dots, 1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}} \right\}$$
- The set A is bounded.
 - The $\inf A$ is 1
 - The set A has its largest elements
 - None of these.
30. $S = \left[(-1)^n \left(\frac{1}{4} - \frac{4}{n} \right) : n \in \mathbb{N} \right]$ Then
- $\sup S = \frac{1}{4}$ and $\inf S = \frac{1}{4}$
 - $\sup S = \frac{15}{4}$ and $\inf S = \frac{-7}{4}$
 - $\sup S = \frac{15}{4}$ and $\inf S = -\frac{1}{4}$
 - None of these
31. S be subset of $\mathbb{R}$ and $\inf S = \sup S$, Then
- S empty
 - S singleton
 - S finite but may not be singleton
 - Can't say
32. $S = \left\{ \left(1 - \frac{1}{n} \right) \sin \frac{n\pi}{2} : n \in \mathbb{N} \right\}$. Then correct statement.
- S has smallest element but not greatest element
 - S has greatest element but not smallest element
 - S has both greatest & least element
 - S bounded but have not smallest and greatest element both
33. Which of the following is not a nbd of each of its point?
- Set $\mathbb{Q}$ of rational numbers
 - Set $\mathbb{Q}^c$ of irrational number
 - Set $\mathbb{Z}$ of integers
 - None of these
34. Let S be an uncountable set and T be a set of those real number x s.t. $(x - \delta, x + \delta) \cap S$ is uncountable then which of the statements is/are correct
- T is countable
 - $S - T$ is countable
 - $S \cap T$ is uncountable
 - All are correct

35. Let E_i are subsets of $\mathbb{R}$ s.t. $E_i \cap E_j = \emptyset$. Then select the correct statement:
 (a.) Each E_i is countable
 (b.) $\mathbb{R} - \cup E_i$ is uncountable for $i = 1$ to n
 (c.) $\mathbb{R} - \cup E_i$ is uncountable $\forall i \in \mathbb{N}$
 (d.) None of these.
36. Choose the incorrect statement:
 (a.) $S \subset T \Leftrightarrow S' \subset T'$
 (b.) $x \in S' \Leftrightarrow x \in (S \cup \{x\})'$ where S is any subset of $\mathbb{R}$
 (c.) If $S = \left\{ \frac{1}{2m} + \frac{1}{2n}; m, n \in \mathbb{N} \right\}$, then $(S')' = \{0\}$
 (d.) $(1, 3) = \cup \left[1 + \frac{1}{n}, 3 - \frac{1}{n} \right] \forall n \in \mathbb{N}$
37. Consider the following statements and choose correct
 (a.) Every infinite and bounded set must have a limit point
 (b.) Any finite set cannot have a limit point
 (c.) Any infinite but unbounded set can't have a limit point
 (d.) None of these
38. Let $E \subset \mathbb{R}, E \neq \emptyset$. Let (1), (2) and (3) denote the following conditions:
 E is infinite
 E is bounded
 E is closed
 (a.) 1 is necessary for E to have a limit point
 (b.) 1 and 2 together are sufficient for E to have a limit point
 (c.) 1 and 3 together are sufficient for E to have a limit point
 (d.) 3 is sufficient for every limit point of E to belong to E
39. Which of the following subsets of $\mathbb{R}$ is closed?
 (a.) $[0, 1] \cup [2, 3] \cup [4, 5]$
 (b.) $[0, 1]$
 (c.) $(1, \infty)$
 (d.) The set of rational numbers in $[0, 1]$
40. Which set is/are not open?
 (a.) $\bigcap_{n=1}^{\infty} G_n$, where $G_n =]-\frac{1}{n}, \frac{1}{n}[$
 (b.) The set $\mathbb{Q}$ of rational number
 (c.) The union of arbitrary family of open set
 (d.) All of above

ASSIGNMENT SHEET - 3

- The sequence $a_n = \sqrt{A^2 n^2 + n + 1} - n$ is
(a.) convergent for all real values of A
(b.) divergent for all real values of A
(c.) convergent for exactly one real value of A
(d.) Convergent for exactly two real values of A
- The least upper bound of the sequence $\left\langle 1 - \frac{1}{n} \right\rangle$ is
(a.) 0
(b.) -1
(c.) 1
(d.) None of these
- Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers defined as $a_1 = 1$ and for $n \geq 1$,
 $a_{n+1} = a_n + (-1)^n 2^{-n}$, $b_n = \frac{2a_{n+1} - a_n}{a_n}$. Then
(a.) $\{a_n\}$ converges to zero and $\{b_n\}$ is a Cauchy sequence
(b.) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is a Cauchy sequence
(c.) $\{a_n\}$ converges to zero and $\{b_n\}$ is not a convergent sequence
(d.) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is not a convergent sequence
- Which of the following sequence $\langle a_n \rangle$ is monotonic:
(a.) $\langle a_n \rangle = 1 - \frac{(-1)^n}{n}$
(b.) $a_{n+2} = \frac{1}{2}(a_n + a_{n+1})$, where $a_1 < a_2$ and a_1, a_2 are given
(c.) $a_{n+1} = \sqrt{\frac{ab^2 + a_n^2}{a+1}} \forall n$, where $a > 0$, $0 < a_1 < b$ and $a_1 = a$
(d.) None of the above
- Define a sequence s_n by $s_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}$. Then the limit of s_n as n tends to infinity
(a.) is 0
(b.) is 1
(c.) is ∞
(d.) doesn't exist
- The sequence $\langle S_n \rangle$, where $S_n = 1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1}$ is
(a.) Convergent
(b.) Monotonically decreasing
(c.) Not Cauchy
(d.) None of these
- Let $\{x_n\}$ be a real sequence. If sequence of even terms of $\{x_n\}$ converges to 1 and sequence of odd terms converges to -1. Then the sequence $\{x_n\}$ will
(a.) Converge to 0
(b.) Converge to 1
(c.) Converge to -1
(d.) None of these
- Let $S_k = \sum_{n=2}^k \frac{(-1)^n}{n \ln n}$. Then the sequence $\{S_k\}$
(a.) converges to a finite number
(b.) diverges to ∞
(c.) diverges to $-\infty$
(d.) oscillates
- Consider a mapping f from the set of natural numbers $\mathbb{N}$ to the set of integers $\mathbb{Z}$ defined by $f(n) = \begin{cases} \frac{n-1}{2}, & \text{when } n \text{ is odd} \\ -\frac{n}{2}, & \text{when } n \text{ is even} \end{cases}$
then, f is
(a.) One-one but not onto
(b.) Onto but not one-one
(c.) Bijective
(d.) Neither one-one nor onto

10. Let a_1, a_2 be positive real numbers and let $a_{n+1} = \frac{1}{2}(a_n + a_{n-1})$ for $n \geq 2$. Then the sequences $\{a_{2n}\}$ and $\{a_{2n-1}\}$
- are not bounded.
 - both diverge to ∞
 - both converge to the same limit.
 - both converge but to different limits.
11. Consider the sequence
4, 0, 4, 1, 0, 4, 1, 1, 0, 4, 1, 1, 1, 0,
Then
- converges to $4\frac{1}{9}$
 - is divergent
 - is unbounded
 - is not convergent and has supremum $= 4\frac{1}{9}$
12. Let $x_n = 2^{2n} \left(1 - \cos\left(\frac{1}{2^n}\right) \right)$ for all $n \in \mathbb{N}$. Then the sequence $\{x_n\}$
- does NOT converge
 - converges to 0
 - converges to $\frac{1}{2}$
 - converges to $\frac{1}{4}$
13. Let $p(x)$ be a polynomial in the real variable x of degree 5. Then $\lim_{n \rightarrow \infty} \frac{p(n)}{2^n}$ is
- 5
 - 1
 - 0
 - ∞
14. If $0 < c < d$, then the sequence $a_n = (c^n + d^n)^{1/n}$
- is bounded and monotonically decreasing
 - is bounded and monotonically increasing
 - is monotonically increasing, but unbounded for $1 < c < d$
 - is monotonically decreasing, but unbounded for $1 < c < d$
15. The limit superior and the limit inferior of the following sequence $\left\langle (-1)^n \left(1 + \frac{1}{n} \right) \right\rangle$ are
- 1, -1
 - 2, -1
 - 2, 1
 - 1, 1
16. Let $\{a_n\}, \{b_n\}$ and $\{c_n\}$ be sequences of real numbers such that $b_n = a_{2n}$ and $c_n = a_{2n+1}$. Then $\{a_n\}$ is convergent
- implies $\{b_n\}$ is convergent but $\{c_n\}$ need not be convergent
 - implies $\{c_n\}$ is convergent but $\{b_n\}$ need not be convergent
 - implies both $\{b_n\}$ and $\{c_n\}$ are convergent
 - if both $\{b_n\}$ and $\{c_n\}$ are convergent
17. For each positive integer n , let a_n be the number of points of intersection of the graph $y = \sin x$ with the line $y = \frac{x}{n}$. The sequence $\{a_n\}$ is
- decreasing
 - constant
 - converging to zero
 - diverging to infinity
18. The value of $\lim_{x \rightarrow \infty} \left(\frac{a_1^{1/x} + a_2^{1/x} + \dots + a_n^{1/x}}{n} \right)^{nx}$, $a_i > 0, \forall i$, is
- $\frac{a_1 + a_2 + \dots + a_n}{n}$
 - $e^{a_1 + a_2 + \dots + a_n}$
 - $a_1 + a_2 + \dots + a_n$
 - $a_1 a_2, \dots, a_n$
19. Let $\{x_n\}$ be a convergent sequence and $\{y_n\}$ be a monotonic sequence. Then $\{x_n \cdot y_n\}$
- Always converges
 - Converges if $\{x_n\}$ is monotonic
 - Converges if $\{x_n y_n\}$ is monotonic
 - Converges if $\{y_n\}$ is bounded

20. Consider the sequence $\{l_n\}$ $n \in \mathbb{N}$ with $l_n = \frac{1}{n+1} + \dots + \frac{1}{2n}$. This sequence
- is increasing and bounded
 - increases to ∞
 - decreases to 0
 - decreases to a positive number
21. The sequence $\{S_n\}$, where $S_n = 1 + \frac{1}{4^n} + \frac{1}{7} + \dots + \frac{1}{3n-2}$
- A Cauchy sequence
 - A convergent sequence
 - Cannot converge
 - None of these
22. The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n^2 + kn}}$ is
- $2(\sqrt{2}-1)$
 - $2\sqrt{2}-1$
 - $2-\sqrt{2}$
 - $\frac{1}{2}(\sqrt{2}-1)$
23. Consider the sequence $\{a_n\}$ of real numbers where $a_1 > 1$ and $a_{n+1} = 2 - \frac{2}{a_n}$, $n \geq 1$. Then the sequence $\{a_n\}$ is
- bounded but not monotone.
 - not bounded but monotone
 - both bounded and monotone.
 - neither bounded nor monotone.
24. Let $\langle a_n \rangle$ be a sequence of real numbers define b_n and c_n as $b_n = n^{\text{th}}$ rational terms in $\langle a_n \rangle$, $c_n = n^{\text{th}}$ irrational terms in $\langle a_n \rangle$. Then
- $\langle b_n \rangle$ and $\langle c_n \rangle$ both are subsequence, of $\langle a_n \rangle$
 - Both $\langle b_n \rangle$ and $\langle c_n \rangle$ may fail to be subsequence, of $\langle a_n \rangle$
 - At least one of $\langle b_n \rangle$ and $\langle c_n \rangle$ is always a subsequence of $\langle a_n \rangle$
 - None of the above
25. The largest term in sequence $x_n = \frac{1000^n}{n!}$, $n = 1, 2, 3, \dots$
- is x_{999}
 - is x_{1001}
 - is x_1
 - does not exist
26. Let the sequence $\{x_n\}_{n \in \mathbb{N}}$ of real numbers converges to a non zero real number a and let $y_n = a - x_n$. Then $\max_{n \in \mathbb{N}} \{x_n, y_n\}$ converges to
- a always
 - 0 always
 - $\max\{a, 0\}$
 - $\min\{a, 0\}$
27. Let $\{x_n\}$ be an unbounded sequence of non-zero real numbers. Then,
- $\{x_n\}$ must have a convergent subsequence
 - $\{x_n\}$ cannot have a convergent subsequence
 - $\left\{\frac{1}{x_n}\right\}$ must have a convergent subsequence
 - $\left\{\frac{1}{x_n}\right\}$ cannot have a convergent subsequence.
28. Let $\{a_n\}$ be a sequence of real numbers. Let $b_n = a_n + a_{n+1}$, for $n = 1, 2, \dots$. Which of the following is always true:
- If $\{b_n\}$ converges, then $\{a_n\}$ converges.
 - If $\{a_n\}$ converges, then $\{b_n\}$ diverges.
 - If $\{a_n\}$ converges, then $\{b_n\}$ converges.
 - $\{a_n\}$ is a subsequence of $\{b_n\}$.
29. $x_{n+1} = \frac{-3}{4}x_n$, $x_0 = 1$. The sequence $\{x_n\}$
- diverges
 - x_n is monotonically increasing and converges to 0
 - x_n is monotonically decreasing and converges to 0
 - none of the above

30. Let $\{x_n\}$ be sequence of real numbers such that $\lim_{n \rightarrow \infty} (x_{n+1} - x_n) = c$, where c is a positive real number. Then the sequence $\left\{\frac{x_n}{n}\right\}$
- is NOT bounded
 - is bounded but NOT convergent
 - converges to c
 - converges to 0
31. If $\langle a_n \rangle$ is a convergent sequence then $\lim_{n \rightarrow \infty} n(a_{n+1} - a_n) =$
- ∞
 - 1
 - e
 - 0
32. Suppose $a > 0$. Consider the sequence $a_n = n\{\sqrt[n]{ea} - \sqrt[n]{a}\}$, $n \geq 1$. Then
- $\lim_{n \rightarrow \infty} a_n$ does not exist
 - $\lim_{n \rightarrow \infty} \overline{a_n} = e$
 - $\lim_{n \rightarrow \infty} a_n = 0$
 - None of the above.
33. There is bijection from the set of natural number to
- The set of all sequences with entries as 0 & 1 only
 - The set of all transcendental number
 - The set of all unit circles in the Cartesian plane
 - None of these
34. $\lim_{n \rightarrow \infty} \left[\left(\frac{2}{1}\right) \left(\frac{3}{2}\right)^2 \left(\frac{4}{3}\right)^3 \dots \left(\frac{n+1}{n}\right)^n \right]^{1/n} =$
- 1
 - 1
 - $-e$
 - e
35. Given a_1, a_2 & $\forall n \in \mathbb{N}$, $a_n = \left(1 - \frac{1}{n}\right)a_{n-1} + \frac{1}{n}a_{n-2}$. Then $\lim_{n \rightarrow \infty} a_n =$
- $2(a_2 - a_1) + a_1 e^{-1}$
 - $2(a_1 - a_2)e^{-1} + a_2$
 - $2(a_1 - a_2)e^{-1} + a_1$
 - $2(a_2 - a_1)e^{-1} + a_1$
36. Which of the following is/are correct?
- $n \log \left(1 + \frac{1}{n+1}\right) \rightarrow 1$ as $n \rightarrow \infty$
 - $(n+1) \log \left(1 + \frac{1}{n+1}\right) \rightarrow 1$ as $n \rightarrow \infty$
 - $n^2 \log \left(1 + \frac{1}{n}\right) \rightarrow 1$ as $n \rightarrow \infty$
 - $n \log \left(1 + \frac{1}{n^2}\right) \rightarrow 1$ as $n \rightarrow \infty$
37. If $\{x_n\}$ and $\{y_n\}$ are sequences of real numbers, which of the following is/are true?
- $\limsup_n (x_n + y_n) \leq \limsup_n x_n + \limsup_n y_n$
 - $\limsup_n (x_n + y_n) \geq \limsup_n x_n + \limsup_n y_n$
 - $\liminf_n (x_n + y_n) \leq \liminf_n x_n + \liminf_n y_n$
 - $\liminf_n (x_n + y_n) \geq \liminf_n x_n + \liminf_n y_n$
38. If $\langle a_n \rangle$ is a sequence of real numbers such that for some $l \in \mathbb{R}$, define a set $K = \{n \in \mathbb{N} : a_n \notin (l - \varepsilon, l + \varepsilon), \forall \varepsilon > 0\}$ if K is bounded, then choose the incorrect
- l is a limit point of $\langle a_n \rangle$ but $\langle a_n \rangle$ may fail to be convergent
 - $\langle a_n \rangle$ is monotonic.
 - $\langle a_n \rangle$ is a Cauchy sequence
 - $\langle a_n \rangle$ is bounded but need not be convergent
39. Let $\langle a_n \rangle$ and $\langle b_n \rangle$ are two sequences. Then choose the incorrect
- If $\langle a_n + b_n \rangle$ and $\langle a_n \cdot b_n \rangle$ are both convergent. Then $\langle a_n \rangle$ and $\langle b_n \rangle$ are convergent
 - If $\langle a_n - b_n \rangle$ and $\left\langle \frac{a_n}{b_n} \right\rangle$ are both convergent. Then $\langle a_n \rangle$ and $\langle b_n \rangle$ are convergent
 - If $\langle a_n + b_n \rangle$ and $\langle a_n - b_n \rangle$ are both convergent. Then $\langle a_n \rangle$ and $\langle b_n \rangle$ are convergent
 - If $\langle a_n + b_n \rangle$ and $\langle a_n \cdot b_n \rangle$ are both convergent. Then $\langle a_n \cdot b_n \rangle$ convergent

40. Which of the following are null sequences
- $\left\{\frac{n}{\alpha^n}\right\} : \alpha > 1$
 - $\left\{\frac{\log n}{n}\right\}$
 - $\{\alpha^n\} : |\alpha| < 1$
 - $\left(1 + \frac{3}{n}\right)^n : n \in \mathbb{N}$
41. Choose the correct statements
- Every bounded sequence has a limit point
 - There exist an onto map from any set to its power set
 - Set of polynomials over $\mathbb{R}$ is a countable set
 - Every sequence has a monotonic subsequence
42. Which of following is/are Incorrect statement
- Every sequence contains a monotonic subsequence
 - Every limit point of the sequence is the limit of the range set of sequence
 - If a sequence has a unique limit point then it must be limit of the sequence
 - The set of disjoint intervals in $\mathbb{R}$ is an uncountable set.
43. Let $\{x_n\}$ and $\{y_n\}$ be two sequences of real numbers such that $x_n \leq y_n \leq x_{n+2}$, $n = 1, 2, 3, \dots$
- $\{y_n\}$ is a bounded sequence.
 - $\{x_n\}$ is an increasing sequence.
 - $\{x_n\}$ and $\{y_n\}$ converge together.
 - $\{y_n\}$ is an increasing sequence.
44. Let $\langle a_n \rangle$ be a sequence of real numbers defined as $a_n = \begin{cases} -1 - \frac{1}{n} & n \text{ is even} \\ 1 + \frac{1}{n} & n \text{ is odd} \end{cases}$
- Then which of the following is correct:
- $\langle a_n \rangle$ is oscillatory
 - If $\liminf \langle a_n \rangle = a$ and $\limsup \langle a_n \rangle = b$ then $a_n \notin (a, b)$ for $\forall n \in \mathbb{N}$
 - $S = \{\lim a_{n_k} : \langle a_{n_k} \rangle \text{ is a convergent subsequence of } \langle a_n \rangle\}$ is finite
 - None of these
45. Which of the following is not possible about $\langle a_n \rangle$
- Σa_n is convergent but $\langle a_n \rangle$ is non convergent
 - $\langle |a_n| \rangle$ is convergent but $\langle a_n \rangle$ is non convergent
 - $\langle a_n \rangle$ is convergent but $\langle |a_n| \rangle$ is non convergent
 - None of these
46. For the sequence $\{x_n\}$ where $x_n = 1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1}$, which of following is/are incorrect statements
- $\{x_n\}$ is a Cauchy sequence.
 - $\{x_n\}$ is a convergent sequence
 - $\{x_n\}$ is a bounded sequence
 - None of these
47. A Cauchy sequence in $\mathbb{Q}$ which does not have a limit in $\mathbb{Q}$ is
- $\left\{1, 2 + \frac{1}{2}, 3 + \frac{1}{3}, 4 + \frac{1}{4}, \dots\right\}$
 - $\left\{1, 1 + \frac{1}{1!}, 1 + \frac{1}{1!} + \frac{1}{2!}, 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!}, \dots\right\}$
 - $\left\{1, \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \dots\right\}$
 - $\left\{1, -1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^2}, \dots\right\}$
48. If $\{a_n\}$ is a sequence of real numbers with $\limsup_{n \rightarrow \infty} a_n = 1$, then
- Set of values taken by $\{a_n\}$ is not a finite set
 - $\{a_n\}$ is convergent
 - $\{a_n\}$ is bounded
 - $\{a_n\}$ cannot diverge to $+\infty$

49. If u_n is a sequence of positive terms then

(a.) $\lim_{n \rightarrow \infty} \frac{u_{n-1}}{u_n} \geq \overline{\lim}_{n \rightarrow \infty} \frac{u_{n-1}}{u_n}$

(b.) $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} \geq \overline{\lim}_{n \rightarrow \infty} u_n^{1/n}$

(c.) $\lim_{n \rightarrow \infty} u_n^{1/n} \leq \overline{\lim}_{n \rightarrow \infty} \frac{u_{n+1}}{u_n}$

(d.) $\overline{\lim}_{n \rightarrow \infty} u_n^{1/n} \leq \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n}$

50. Let $\{a_n\}$, $\{b_n\}$ & $\{c_n\}$ be sequences of real numbers such that $b_n = a_{2n}$ & $c_n = a_{2n-1}$. Then $\{a_n\}$ is convergent

(a.) implies $\{b_n\}$ is convergent but $\{c_n\}$ need not be convergent

(b.) implies $\{c_n\}$ is convergent but $\{b_n\}$ need not be convergent

(c.) implies both $\{b_n\}$ and $\{c_n\}$ are convergent

(d.) implies both $\{b_n\}$ and $\{c_n\}$ are divergent

ASSIGNMENT SHEET-4

1. If $b_n = \begin{cases} \frac{1}{\sqrt{n}} & \text{if } n \text{ is odd} \\ \frac{1}{n} & \text{if } n \text{ is even} \end{cases}$

Then

- (a.) Both $\langle b_n \rangle$ and $\sum b_n$ are convergent
- (b.) Both $\langle b_n \rangle$ and $\sum b_n$ are divergent
- (c.) $\langle b_n \rangle$ is convergent but $\sum b_n$ is not
- (d.) $\sum b_n$ is convergent but $\langle a_n \rangle$ is convergent

2. Consider the sequence $\left\{ \frac{(-1)^n}{n} \right\}$ and the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$. Then

- (a.) the sequence converges but not the series.
- (b.) the series converges but not the sequence.
- (c.) neither the series nor the sequence converges.
- (d.) both the series and the sequence converge.

3. Which of the following conditions does NOT ensure the convergence of a real sequence $\{a_n\}$?

- (a.) $|a_n - a_{n+1}| \rightarrow 0$ as $n \rightarrow \infty$
- (b.) $\sum_{n=1}^{\infty} |a_n - a_{n+1}|$ is convergent
- (c.) $\sum_{n=1}^{\infty} na_n$ is convergent
- (d.) The sequence $\{a_{2n}\}$, $\{a_{2n+1}\}$ and $\{a_{3n}\}$ are convergent.

4. The series $\sum \left\{ \frac{1}{2^{n-1}} + (-1)^{n-1} \right\}$ is

- (a.) Convergent
- (b.) divergent
- (c.) oscillating infinitely
- (d.) oscillating finitely

5. The sum of the series $\frac{1}{1!} + \frac{1+2}{2!} + \frac{1+2+3}{3!} + \dots$ equals

- (a.) e
- (b.) $\frac{e}{2}$
- (c.) $\frac{3e}{2}$
- (d.) $1 + \frac{e}{2}$

6. Let $\{a_n\}$ be an increasing sequence of positive real numbers such that the series $\sum_{k=1}^{\infty} a_k$ is divergent. Let $S_n = \sum_{k=1}^n a_k$ for $n=1, 2, \dots$ and $t_n = \sum_{k=2}^n \frac{a_k}{S_{k-1}S_k}$ for $n=2, 3, \dots$. Then $\lim_{n \rightarrow \infty} t_n$ is equal to

- (a.) $\frac{1}{a_1}$
- (b.) 0
- (c.) $\frac{1}{(a_1 + a_2)}$
- (d.) $a_1 + a_2$

7. In each of the following cases, which of the series is absolutely convergent.

- (a.) $\sum_{n=1}^{\infty} (-1)^n \frac{1}{2n+3}$
- (b.) $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+2}$
- (c.) $\sum_{n=1}^{\infty} (-1)^n \frac{n \log n}{e^n}$
- (d.) None of these

8. Let $\{x_n\}$ be a sequence of real numbers so that $\sum_{n=1}^{\infty} |x_n - x| = c$, with c finite. Then

- (a.) $\{x_n\}$ may not be bounded
- (b.) $\{x_n\}$ must converge to x
- (c.) $\{x_n\}$ must converge to $x+c$
- (d.) $\{x_n\}$ is bounded but not necessarily convergent.

9. Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series, where

$$a_n = \frac{(-1)^n n}{2^n}, b_n = \frac{(-1)^n}{\log(n+1)} \text{ for all } n \in \mathbb{N}.$$

Then

- (a.) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent

- (b.) $\sum_{n=1}^{\infty} a_n$ is absolutely convergent but $\sum_{n=1}^{\infty} b_n$ is conditionally convergent

- (c.) $\sum_{n=1}^{\infty} a_n$ is conditionally convergent but $\sum_{n=1}^{\infty} b_n$ is absolutely convergent

- (d.) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are conditionally convergent

10. We are given a convergent series $\sum_{n=1}^{\infty} a_n$, where $a_n \geq 0$ for each n . Which of the following correctly describes the behavior of the series $\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$, $1 \leq p \leq 2$?

- (a.) Diverges when $p=1$, but converges for $p \in (1, 2]$

- (b.) Converges for every $p \in [1, 2]$

- (c.) Diverges when $p \in \left[1, \frac{5}{4}\right]$, but converges for $p \in \left[\frac{5}{4}, 2\right]$

- (d.) Diverges for every $p \in [1, 2]$

11. Consider the series $\sum \frac{1}{\sqrt{n}}$ and $\sum \frac{1}{n^{3/2}}$. Then

- (a.) both the series converge to the same value.

- (b.) both the series converge to different values.

- (c.) both the series are divergent.

- (d.) first series is divergent and second series is convergent.

12. Let $\{x_n\}$ be a sequence of positive real numbers. A sufficient condition for $\{x_n\}$ to have no convergent subsequence is

(a.) $|x_{n+2} - x_{n+1}| > |x_{n+1} - x_n| \forall n \in \mathbb{N}$

(b.) $\forall i, j \in \mathbb{N}$, the set $\left\{n \in \mathbb{N} : |x_i - x_n| < \frac{1}{j}\right\}$ is finite

(c.) $\sum_{k=1}^{\infty} x_{n_k} = \infty$ for every increasing sequence $\{n_k\}$ of natural numbers.

- (d.) None of the above.

13. If the terms of this oscillating series are grouped pairwise $3 - \frac{2^2+1}{2} + \frac{2^3+1}{2^2} - \frac{2^4+1}{2^3} + \dots$ then the resulting series becomes

- (a.) Convergent

- (b.) Divergent

- (c.) Oscillate finitely

- (d.) Oscillate infinitely

14. Which of the following series is divergent?

(a.) $\sum_{n=1}^{\infty} \sqrt{n} \sin \frac{1}{n}$

(b.) $\sum_{n=1}^{\infty} \sqrt{n} \sin^{-1} \left(\frac{1}{n^2} \right)$

(c.) $\sum_{n=1}^{\infty} n \tan \left(\frac{1}{n^3} \right)$

(d.) $\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{1}{n^2} \right)$

15. The largest interval in which the series $\sum_{n=1}^{\infty} x^n$ converges

- (a.) For x with $-1 < x < 1$

- (b.) For x with $-\frac{1}{2} < x < \frac{1}{2}$

- (c.) For x with $-2 < x < 2$

- (d.) For x with $-\frac{1}{2} \leq x \leq \frac{1}{2}$

16. The series

$$\left(\frac{2^2}{1^2} - \frac{2}{1}\right)^{-1} + \left(\frac{3^3}{2^3} - \frac{3}{2}\right)^{-2} + \left(\frac{4^4}{3^4} - \frac{4}{3}\right)^{-3} + \dots \text{ is}$$

- (a.) convergent
- (b.) divergent
- (c.) oscillating finitely
- (d.) oscillating infinitely

17. The series $x + \frac{2^2 x^2}{2!} + \frac{3^3 x^3}{3!} + \frac{4^4 x^4}{4!} + \dots$ is convergent if

- (a.) $0 < x < \frac{1}{e}$
- (b.) $x > \frac{1}{e}$
- (c.) $\frac{2}{e} < x < \frac{3}{e}$
- (d.) $\frac{3}{e} < x < \frac{4}{e}$

18. Consider the following statements:

1. If $\sum_{n=1}^{\infty} u_n$ is a series of positive terms, then the convergence of $\sum_{n=1}^{\infty} (-1)^n u_n$ implies the convergence of $\sum_{n=1}^{\infty} u_n$.
2. If $\sum_{n=1}^{\infty} u_n$ is a series of positive terms then the convergence of $\sum_{n=1}^{\infty} u_n$ implies the convergence of $\sum_{n=1}^{\infty} (-1)^n u_n$.
3. The convergence of $\sum_{n=1}^{\infty} u_n$ ($u_n > 0$) implies the convergence of $\sum_{n=1}^{\infty} u_n^2$.

Select the correct answer using the codes given below

- (a.) 2 and 3
- (b.) 1 and 3
- (c.) 1 and 2
- (d.) 1, 2 and 3

19. Let the series $\sum_{n=1}^{\infty} u_n$ has bounded partial series, then the series $\sum_{n=1}^{\infty} u_n v_n$ is convergent if the sequence $\{v_n\}$ is

- (a.) Monotonic and bounded
- (b.) Monotonic and converges to zero
- (c.) Bounded and converges to zero
- (d.) Bounded and monotonic which converges to zero.

20. If p is a real number, then the series $\frac{1}{1^p} + \frac{1}{3^p} + \frac{1}{5^p} + \frac{1}{7^p} + \dots$ to ∞ is convergent for

- (a.) $p > 0$
- (b.) $p > 1$
- (c.) $p > 2$
- (d.) $p < 1$

21. The series $\frac{1}{1.4} + \frac{1}{2.5} + \frac{1}{3.6} + \dots$ to ∞ is

- (a.) Convergent
- (b.) Divergent
- (c.) Oscillating
- (d.) Converges conditionally

22. $\sum_{n=1}^{\infty} \frac{1}{(n^2 + 1)^k}$ is convergent for

- (a.) $k < \frac{1}{2}$
- (b.) $k > \frac{1}{2}$
- (c.) $k = \frac{1}{2}$
- (d.) All values of k

23. Let $\{a_n\}$ is a non-negative real number sequence s.t. $\sum_{n=1}^{\infty} a_n$ is convergent. If p is a real no. s.t. $\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$ diverges then

- (a.) $p \in (0, 1)$
- (b.) $p \in [0, 1]$
- (c.) $p \in [0, 1)$
- (d.) $p \in (0, 1]$

24. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{3n+k}$ is

(a.) $\log \frac{4}{3}$

(b.) $\log \frac{3}{4}$

(c.) $\log \frac{3}{2}$

(d.) $\log \frac{5}{4}$

25. $\sum \frac{n^n}{(n+a)^n} a : a > 0$

(a.) Converges for all $a > 1$

(b.) Converges for $a > 0$

(c.) Converges for $a \geq 0$

(d.) Div for all a

26. Which of the following series are convergent?

(a.) $\sum_{n=1}^{\infty} \frac{\sqrt{2n^2+3}}{5n^3+7}$

(b.) $\sum_{n=1}^{\infty} \frac{(n+1)^n}{n^{\frac{n+3}{2}}}$

(c.) $\sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{1}{n}\right)$

(d.) None of these

27. Consider the sequence of rational number

$\{q_k\}_{k \geq 1}$ where $q_k = \sum_{n=1}^k \frac{1}{10^{n^2}}$ i.e., the sequence

is $q_1 = .1, q_2 = .1001, q_3 = .100100001$ etc.

Which of the following is true?

(a.) This sequence is bounded and convergent in $\mathbb{Q}$.

(b.) This sequence is not bounded.

(c.) This sequence is bounded, but not a Cauchy sequence.

(d.) This sequence is bounded and Cauchy but not convergent in $\mathbb{Q}$

28. Let $\{a_0, a_1, a_2, \dots\}$ be a sequence of real numbers. For any $k \geq 1$, let $s_n = \sum_{k=0}^n a_{-k}$

Which of the following statements are correct?

(a.) If $\lim_{n \rightarrow \infty} s_n$ exists, then $\sum_{m=0}^{\infty} a_m$ exists.

(b.) If $\lim_{n \rightarrow \infty} s_n$ exists, then $\sum_{m=0}^{\infty} a_m$ need not exist.

(c.) If $\sum_{m=0}^{\infty} a_m$ exists, then $\lim_{n \rightarrow \infty} s_n$ exists.

(d.) If $\sum_{m=0}^{\infty} a_m$ exists, then $\lim_{n \rightarrow \infty} s_n$ need not exist.

29. If $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then which of the following is NOT true?

(a.) $\sum_{m=n}^{\infty} a_m \rightarrow 0$ as $n \rightarrow \infty$.

(b.) $\sum_{n=1}^{\infty} a_n \sin n$ is convergent.

(c.) $\sum_{n=1}^{\infty} e^{an}$ is divergent.

(d.) $\sum_{n=1}^{\infty} a_n^2$ is divergent.

30. Using the fact that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, $\sum_{n=1}^{\infty} \frac{1}{(2n+1)^2}$ equals

(a.) $\frac{\pi^2}{12}$

(b.) $\frac{\pi^2}{12} - 1$

(c.) $\frac{\pi^2}{8}$

(d.) $\frac{\pi^2}{8} - 1$

31. Pick out the series which are absolutely convergent:
- $\sum_{n=1}^{\infty} (-1)^n \frac{\cos n\alpha}{n^2}$ where $\alpha \in \mathbb{R}$ is a fixed real number.
 - $\sum_{n=1}^{\infty} (-1)^n \frac{n \log n}{e^n}$
 - $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+2}$
 - None of these
32. It is given that the series $\sum_{n=1}^{\infty} a_n$ is convergent, but not absolutely convergent and $\sum_{n=1}^{\infty} a_n = 0$. Denote by s_k the partial sum $\sum_{n=1}^k a_n, k = 1, 2, \dots$. Then
- $s_k = 0$ for infinitely many k
 - $s_k > 0$ for infinitely many k , and $s_k < 0$ for infinitely many k
 - It is possible that $s_k > 0$ for all k
 - It is possible that $s_k > 0$ for all but a finite number of values of k
33. If $\sum a_n$ is series of positive & negative term and $\sum q_n, \sum p_n$ are series of negative & positive terms respectively and if $\sum a_n$ is conditionally convergent then
- $\sum p_n$ is convergent but $\sum q_n$ not
 - $\sum p_n$ is divergent but $\sum q_n$ not
 - $\sum p_n$ & $\sum q_n$ both are convergent
 - $\sum p_n$ & $\sum q_n$ both are divergent
34. Choose the correct answer, where $a_n \geq 0$
- If $\sum a_n$ is convergent then $\sum a_n^2$ is also convergent
 - If $\sum a_n$ is convergent then $\sum \sqrt{a_n}$ is also convergent
 - If $\sum a_n$ is convergent then $\sum \frac{a_n}{1+a_n}$ is also convergent
 - If $\sum a_n$ is convergent then $\sum \frac{a_n}{1-a_n}$ (provided $a_n \neq 1$) is also convergent.
35. Let $\sum_{n=1}^{\infty} x_n$ be a series of real numbers. Which of the following is true?
- If $\sum_{n=1}^{\infty} x_n$ is divergent, then $\langle x_n \rangle$ does not converge to 0
 - If $\sum_{n=1}^{\infty} x_n$ is convergent, then $\sum_{n=1}^{\infty} x_n$ is absolutely convergent
 - If $\sum_{n=1}^{\infty} x_n$ is convergent, then $x_n^2 \rightarrow 0$ as $n \rightarrow \infty$
 - If $x_n \rightarrow 0$, then $\sum_{n=1}^{\infty} x_n$ is convergent
36. If $\sum a_n$ be a convergent series and $a_n > 0, \forall n \in \mathbb{N}$ then the correct statement:
- $\sum \sqrt{a_n a_{n+1}}$ is convergent
 - $\sum a_n^2$ is convergent
 - $\sum \frac{\sqrt{a_n}}{n}$ is convergent
 - None of the above
37. Let $\langle a_n \rangle$ and $\langle b_n \rangle$ be two sequences of real number such that $a_n = b_n - b_{n+1}$ for $n \in \mathbb{N}$. Then
- Convergence of $\sum a_n$ implies the convergence of $\langle b_n \rangle$ not conversely
 - Convergence of $\sum a_n$ implies the convergence of $\sum b_n$ but not conversely
 - Convergence of $\sum a_n$ implies and implied by the convergence of $\langle b_n \rangle$
 - Convergence of $\sum b_n$ implies and implied by the convergence of $\langle a_n \rangle$
38. The alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} u_n$ converges if $\{u_n\}$ is a/an
- Decreasing sequence which converges to zero
 - Increasing sequence which converges to zero
 - Monotone sequence which converges to zero
 - Strictly monotone sequence which converges to zero

39. Which one of the following series is convergent?

(a.) $\frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \dots$ to ∞

(b.) $1\frac{1}{2} - 1\frac{1}{3} + 1\frac{1}{4} - 1\frac{1}{5} + \dots$ to ∞

(c.) $\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \dots$ to ∞

(d.) $x + x^2 + x^3 + x^4 + \dots$ to a ∞ , where $|x| < 1$

40. Select the incorrect statements

(a.) Finite set has infinite number of limit points

(b.) The sequence $1 + r + r^2 + \dots + r^n$, where $-1 \leq r < 1$ and $n \in \mathbb{N}$ is bounded above but not bounded below.

(c.) If $u_n > 0 \forall n$ then $\sum u_n$ & $\sum \frac{1}{u_n}$ converge or diverge together

(d.) Set of real number except the integers which are multiples of 2 is uncountable set